



ELECTRONIC PRINCIPLES AND DEVICES

Department of Electronics and Communication
Engineering

[https://youtube.com/playlist?
list=PLwjK_iyK4LLDoFG8FeiKAr3ISt
RkPSxqq&si=Oj8z6xY-OWftpPZy](https://youtube.com/playlist?list=PLwjK_iyK4LLDoFG8FeiKAr3IStRkPSxqq&si=Oj8z6xY-OWftpPZy)

Understand all the needed theory from these yt vids

ELECTRONIC PRINCIPLES AND DEVICES



TRANSISTORS

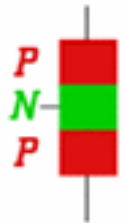
BJT (Bipolar Junction Transistor)

Department of Electronics and Communication.

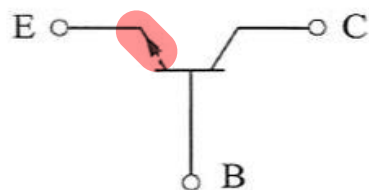
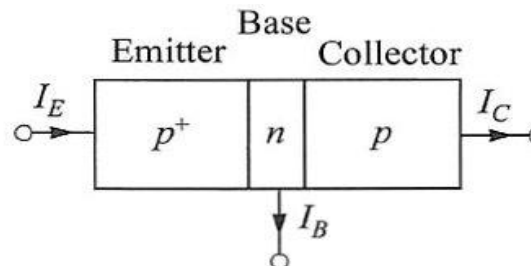
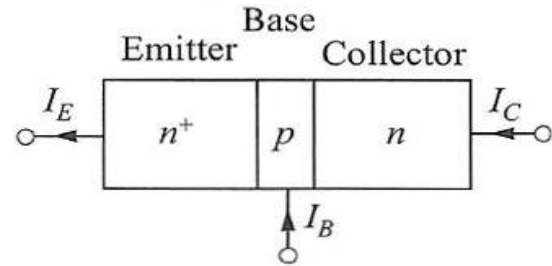
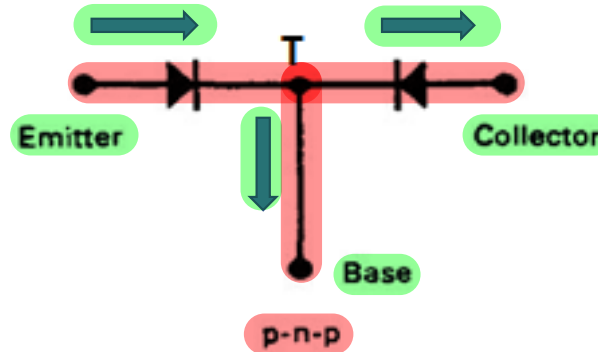
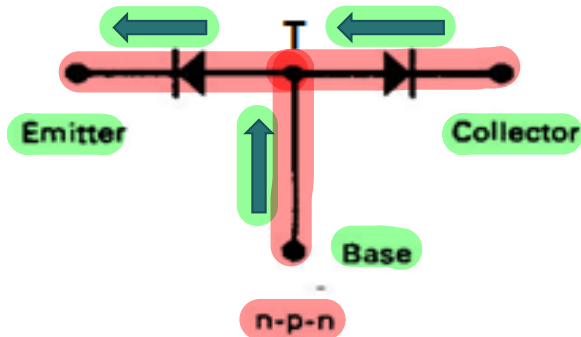
➤ A **bipolar junction transistor (BJT)** is a **three terminal device** consisting of either a p-type semiconductor or n-type semiconductor sandwiched between opposite types. It has **two p-n junctions** connected back-to-back.

➤ In a BJT, the operation depends on the active participation of both the **majority carriers**, and the **minority carriers**; hence the name “**bipolar**”.

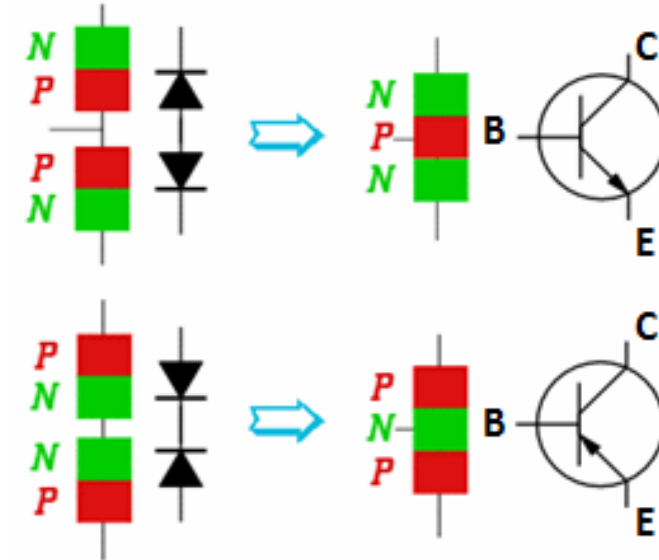
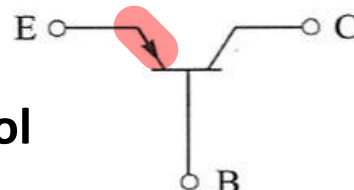
➤ The combination of two terms **transfer + resistor** results in **TRANSISTOR** as the current is transferred from a low to a high-resistance circuit.



➤ The three terminals of a Bipolar junction transistor are Emitter, Base & Collector.

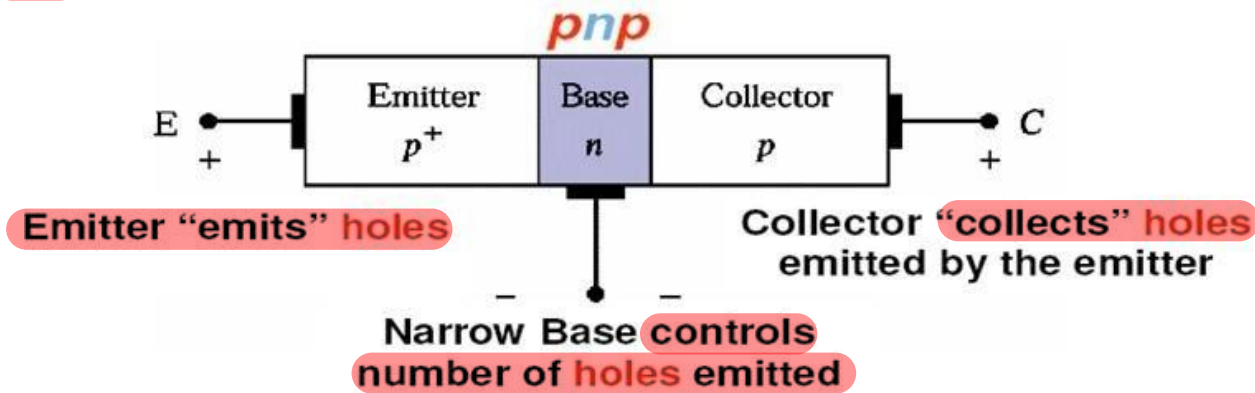


Transistor Symbol

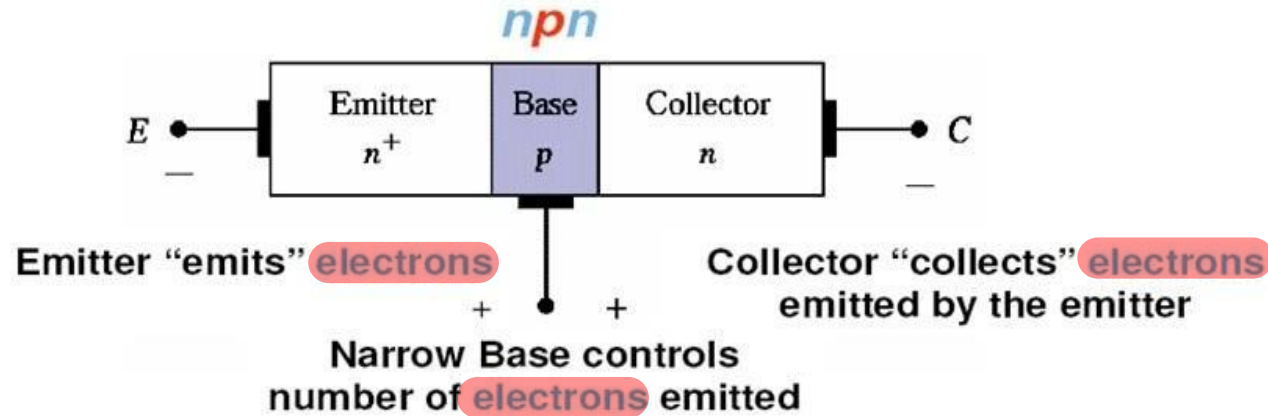


- The **Emitter (E)** is the portion of the transistor that **supplies** charge carriers (either electrons or holes) and it is a **heavily doped** region. The area is **moderate** in size.
- The **Base (B)** is the **middle** portion of the transistor that forms **two PN junction** between the emitter and the collector and it is a **lightly doped** region. It is the **smallest** in width of all the three regions.
- The **Collector (C)** is the portion of the transistor that **collects** charge carriers (either electrons or holes) and it is a **moderately doped** region. It is the **widest** of all the three regions as **maximum heat is dissipated in this region.**

1. **pn**p transistor:



2. **np**n transistor:



Electronic Principles and Devices

Transistor Configurations



Based on the terminal common to the input and output side of the transistor, following are the transistor configurations:

1)Common Base (CB) configuration

2)Common Emitter (CE) configuration

3)Common Collector (CC) configuration

A Transistor has two p-n junctions.

1) Emitter-base (EB) Junction

2) Collector-base (CB) Junction

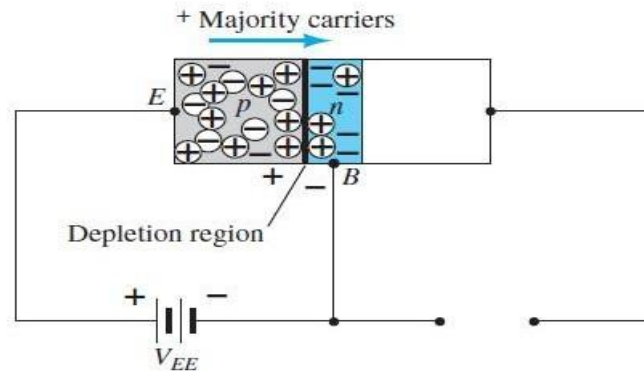
➤ Each junction can either be **forward or reverse biased** independently.

➤ Depending on the bias we have the following regions of operation of BJT

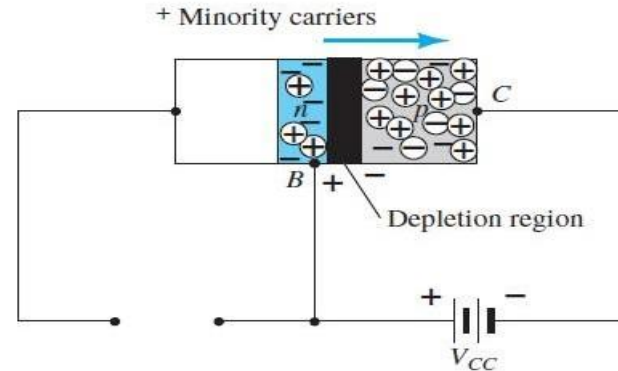
Regions of Operation	Emitter base Junction	Collector-base Junction	Application
Active region(<i>commonly used</i>)	Forward biased	Reverse biased	Transistor works as an Amplifier
Saturation region	Forward biased	Forward biased	Transistor works as a switch (ON)– Digital Application
Cut-off region	Reverse biased	Reverse biased	Transistor works as a switch (OFF)– Digital Application
Inverse active region(<i>rarely used</i>)	Reverse biased	Forward biased	Transistor works as an Attenuator

The transistor operated in **Common Base** configuration, with the external sources, V_{EE} and V_{CC} , connected as shown:

- The emitter – base junction is **forward biased**
- The collector – base junction is **reverse biased**
- As the EB junction is forward biased the current flow is because of majority charge carriers.
- Similarly as the CB junction is reverse biased the current flow is because of minority charge carriers (Hence wider depletion region)



Forward Bias



Reverse Bias

1. Active Region (Commonly Used)

- Condition:
 - Base-Emitter Junction: Forward-biased
 - Base-Collector Junction: Reverse-biased
- Characteristics:
 - The transistor acts as an **amplifier**.
 - A small base current controls a much larger collector current.
 - The collector current (I_C) is proportional to the base current (I_B) and the current gain (β):

$$I_C = \beta I_B$$

- Application: Amplification circuits (e.g., audio amplifiers).

2. Saturation Region

- Condition:
 - Base-Emitter Junction: Forward-biased
 - Base-Collector Junction: Forward-biased
- Characteristics:
 - The transistor is fully "on" and acts like a **closed switch**.
 - Both junctions conduct heavily.
 - The collector-emitter voltage (V_{CE}) is minimal (close to 0.2V in silicon BJTs).
- Application: Switching applications (e.g., turning a load on/off).

3. Cut-off Region

- Condition:
 - Base-Emitter Junction: Reverse-biased
 - Base-Collector Junction: Reverse-biased
- Characteristics:
 - The transistor is fully "off" and behaves like an **open switch**.
 - Both base current (I_B) and collector current (I_C) are nearly zero.
 - Only leakage currents (negligible) flow.
- Application: Switching applications (e.g., turning a load off).

Inverse Active Region (Rarely Used)

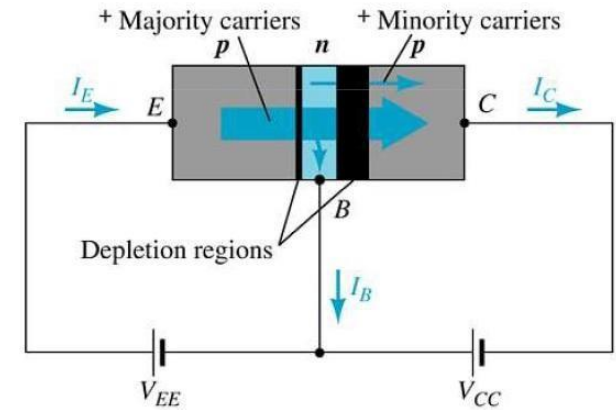
- Condition:
 - Base-Emitter Junction: Reverse-biased
 - Base-Collector Junction: Forward-biased
- Characteristics:
 - The roles of the emitter and collector are **swapped**.
 - The collector acts as the emitter, and the emitter acts as the collector.
 - The transistor's current gain (β) is much lower in this region due to poor design optimization for reverse operation.
- Application: Rarely used in practical circuits but can appear in some specialized designs or reverse-engineering applications.

Visualization of Transistor Regions:

Imagine a **tap (faucet)** controlling water flow:

Region	Tap Analogy	Operation	Applicatio
Active Region	Tap partially open, flow is adjustable	Amplifies signals, small base current controls larger collector current.	Amplifiers, signal control.
Saturation Region	Tap fully open, maximum flow	Fully "on", acts like a closed switch.	Switching (turning devices on
Cut-off Region	Tap fully closed, no flow	Fully "off", acts like an open switch.	Switching (turning devices off
Inverse Active Region	Tap reversed, water flows poorly backward	Rarely used; roles of emitter and collector are swapped.	Specialized reverse-engineerin cases.

- A large number of majority carriers will diffuse across the forward biased p-n junction into the n -type material.
- Since the n -type material is very thin and lightly doped, a very small number of these (holes) will recombine with the electrons in base region and flow through the base terminal as base current (is typically in the order of microamperes).
- Thus, a larger number of these majority carriers from the emitter region will diffuse across the reverse-biased junction into the p -type material connected to the collector terminal.



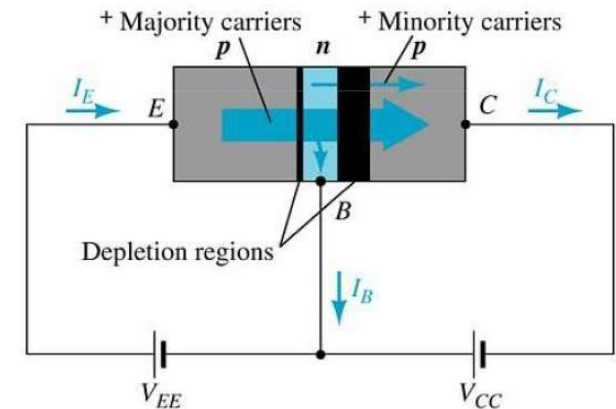
Currents in a BJT

- Applying KCL to the transistor, we get

$$I_E = I_C + I_B$$

- The collector current, comprises two components—the majority and the minority carriers.
- The minority-current is called as leakage current (I_{CO} - I_C with emitter terminal Open).
- Therefore the total collector current is given as

$$I_C = I_{C_{\text{majority}}} + I_{CO_{\text{minority}}}$$



Currents in a BJT

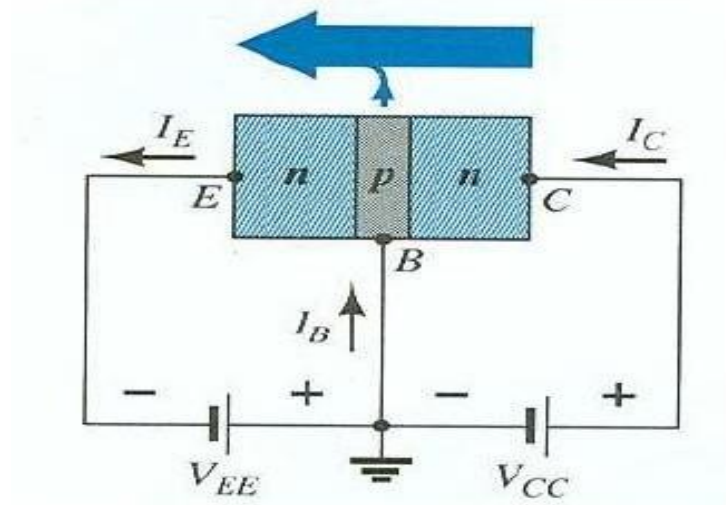
Currents in a transistor

- Emitter current is the sum of the collector and base currents:

$$I_E = I_C + I_B$$

- The collector current is comprised of two currents:

$$I_C = I_C (\text{Majority}) + I_{CO} (\text{Minority})$$

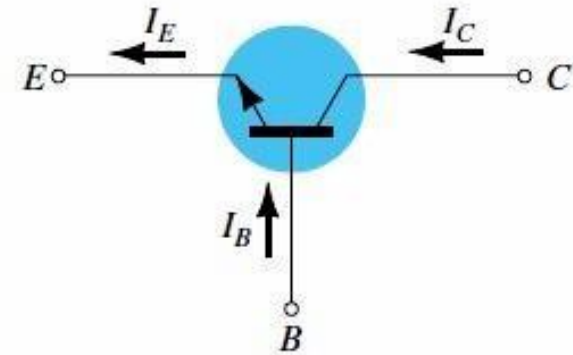
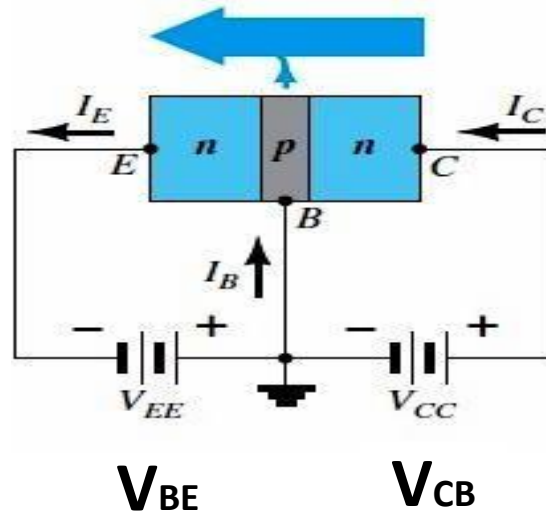


Current directions in a BJT

Electronic Principles and Devices

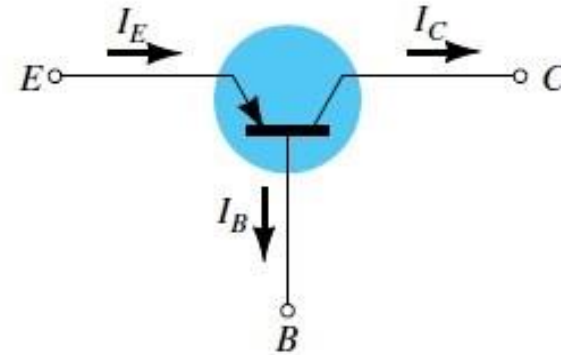
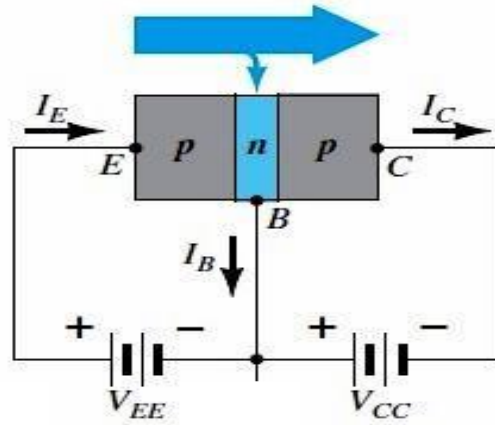
Common Base Configuration

- The **arrow** in the diagram indicates the direction of the emitter current.
- Easy way to remember matching the letters ***npn*** of the transistor type and with the appropriate letters of the phrases “***not pointing in***”
- The ***NPN*** transistor requires two voltage sources **V_{EE}** and **V_{CC}** to bias the two junctions respectively.



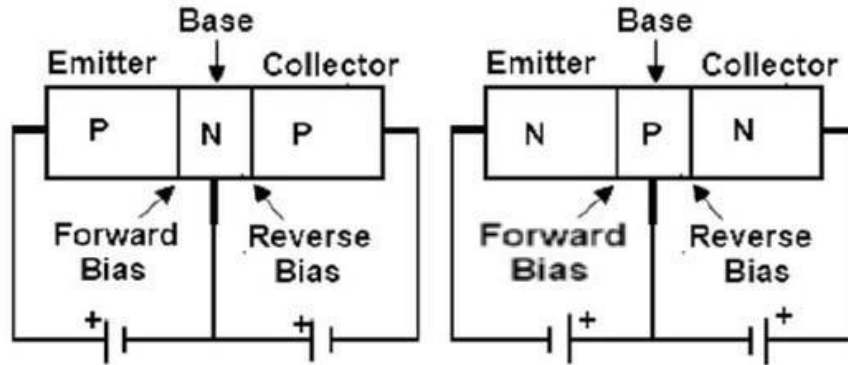
➤ The **arrow** in the diagram, indicates the direction of the emitter current and matching the letters **npn** of the transistor type and with the appropriate letters of the phrases “**p**ointing **i**n **p**ermanently”

➤ The **PNP** BJT requires two voltage sources **V_{EE}** and **V_{CC}** to bias the two junctions respectively.



Electronic Principles and Devices

Common Base Configuration



The biasing of **PNP** and **NPN** transistors to operate in active region

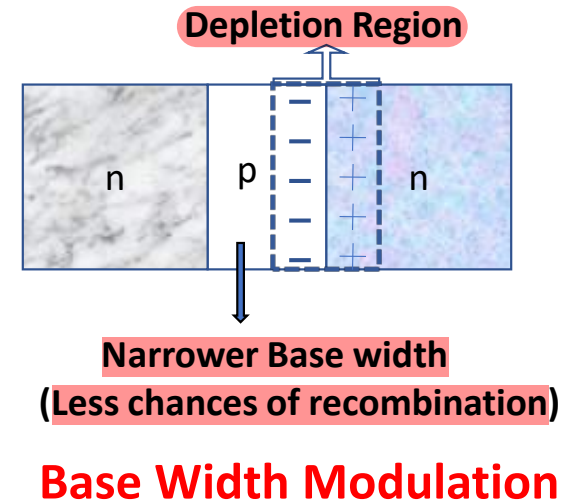
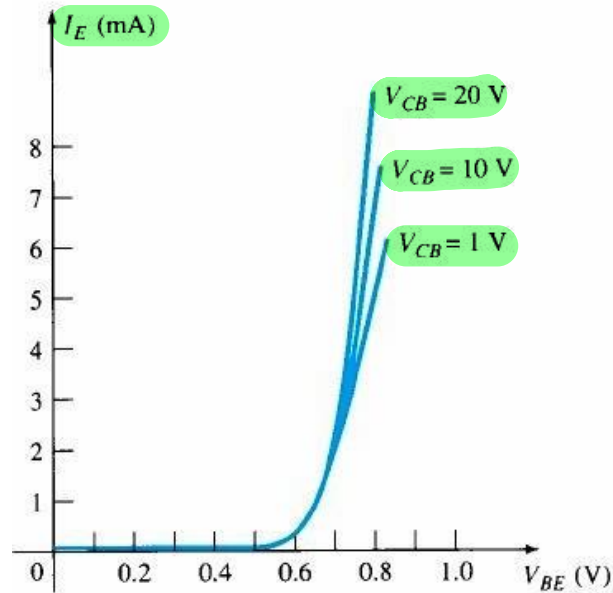
➤ Since the analogy of transistor is two diodes connected back-to-back. One diode is **forward biased** and the other is **reverse biased** for the device to operate in **active region**.

➤ Hence, biasing the **emitter base junction (EB Jn)** in forward bias condition and **collector base junction (CB Jn)** in reverse bias condition, gives the input characteristic of the device.

Electronic Principles and Devices

CB - Input Characteristics

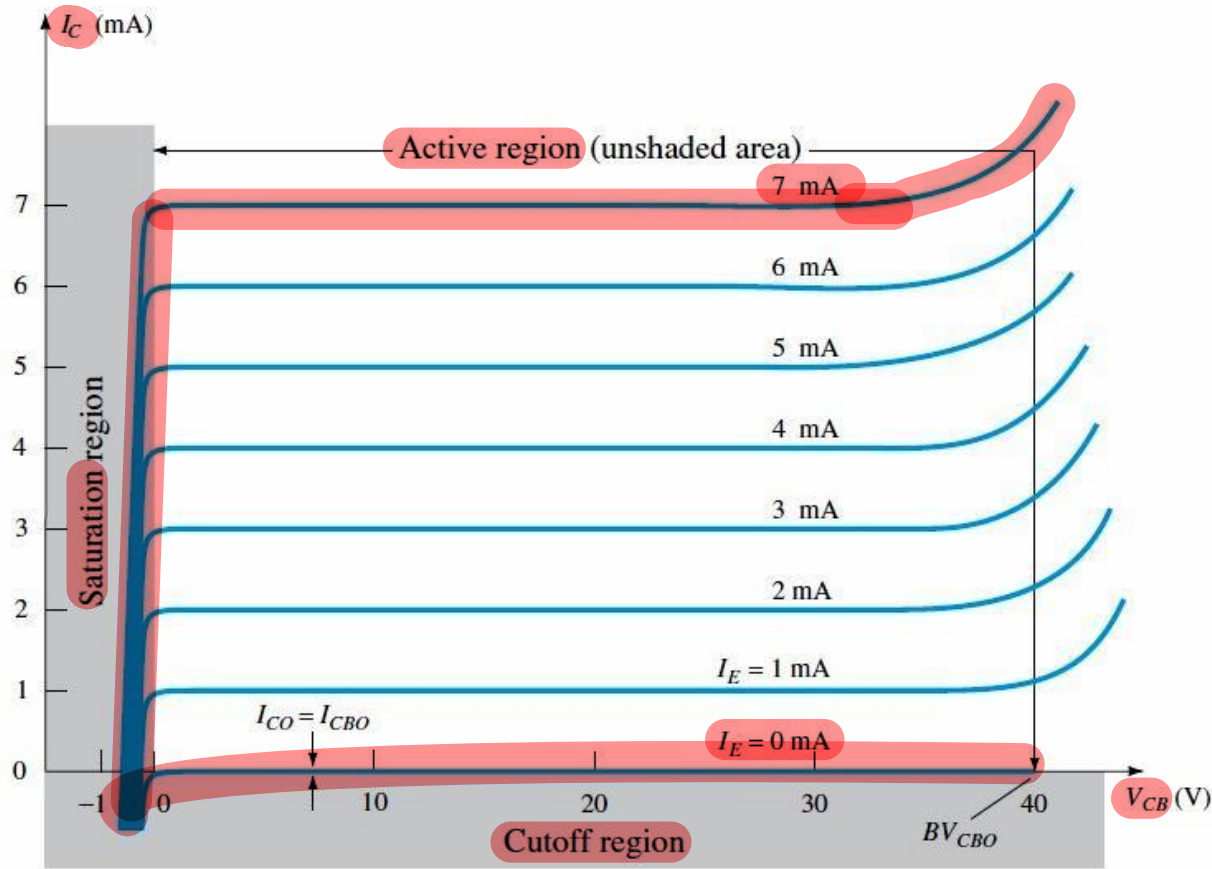
For the transistor in CB configuration, when the output voltage V_{CB} is increased, it causes an **increase** in the emitter current and hence the graph shifts inwards or towards the left.



The input characteristic of BJT (Si) for varying output voltage V_{CB}

Base Width Modulation or Early Effect :

- A greater reverse bias across the collector- base junction increases the collector- base depletion width. There is a variation in the width of the base in a bipolar transistor due to a variation in the applied base-to-collector voltage.
- Base-narrowing has two consequences that affect the current:
 - There is a lesser chance for recombination within the "smaller" base region.
 - The charge gradient is increased across the base, and consequently, the current of minority carriers injected across the collector-base junction increases.
- The Early effect is observed as an increase in the collector or "output" current with increasing collector-emitter voltage.

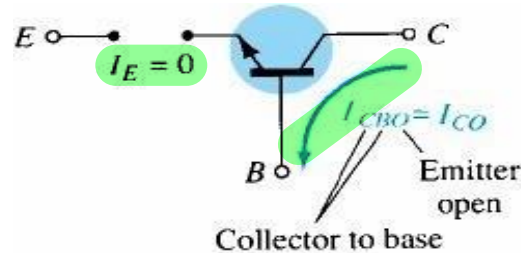


The output characteristic of BJT (Si) for varying input current I_E

Electronic Principles and Devices

CB - Output Characteristics

- **Active region:** The operating range of the amplifier. It is noticed that I_E is approximately equal to I_C ($I_E \approx I_C$). BJT is a current controlled device in the active region and is independent of the output voltage V_{CB}
- **Cut-off region:** The region where the collector current I_C is approximately equal to 0 ($I_C < I_{CBO}$). The device is basically **OFF** i.e., there's no input voltage, but only reverse – bias output voltage and negligible reverse saturation current



Reverse saturation current

- **Saturation region:** The region to the left of $V_{CB} = 0V$. Note the exponential increase in collector current I_C as V_{CB} approaches 0V. In this region, I_C depends on V_{CB} but is independent of I_E

Electronic Principles and Devices

CB Configuration - Approximations

- **Emitter and collector current:** $I_E \approx I_C$.
- **Base – emitter voltage (V_{BE}):** $V_{BE} = 0.7V$ (Si).

Alpha (α): Ratio of I_C to I_E .

$$\alpha_{dc} = \frac{I_C}{I_E}$$

$$I_C = \alpha I_E + I_{CBO}$$

Ideally: $\alpha = 1$

In reality: α is between 0.9 and 0.998

For AC mode:

$$\alpha_{ac} = \left. \frac{\Delta I_C}{\Delta I_E} \right|_{V_{CB} = \text{constant}}$$

α - Current amplification factor or current gain in CB config.

	DC gains	AC gains
Voltage	$A_V = \frac{V_{\text{output}}}{V_{\text{input}}}$	$A_V = \frac{\Delta V_{\text{output}}}{\Delta V_{\text{input}}}$

What is I_{CBO} ?

- I_{CBO} stands for **Collector-Base current with the emitter Open**.
- It is a leakage current that flows from the collector to the base of a transistor when no external current is supplied to the emitter terminal. This current occurs because of the minority charge carriers present in the collector and base regions.

Why does I_{CBO} exist?

- Transistors are made of semiconductor materials that inherently allow small leakage currents due to the thermal generation of electron-hole pairs.
- In the **collector-base junction**, which is reverse-biased under normal operation, a small number of thermally generated minority carriers cross the junction, resulting in this leakage current.

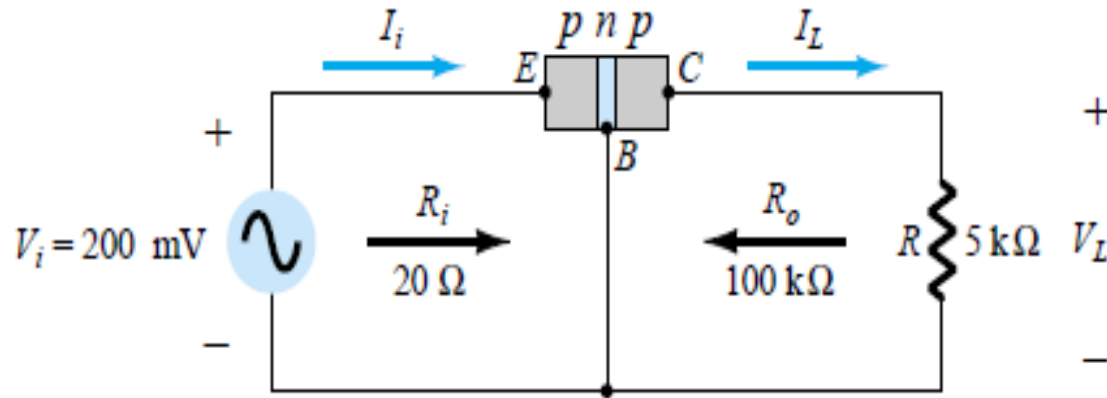
What is I_{CEO} ?

- I_{CEO} stands for **Collector-Emitter current with the base Open**.
- It represents the leakage current flowing from the collector to the emitter in a transistor when no external connection or current is applied to the base terminal.

Why does I_{CEO} exist?

- The collector-emitter path is indirectly influenced by the collector-base leakage current (I_{CBO}), which is amplified by the transistor's current gain (β).
- In a transistor, the collector-base junction is typically reverse-biased, and the emitter-base junction is forward-biased. Even when the base is open, minority carriers in the collector region can cause a leakage current to flow to the emitter.

Common Base Amplifying action



$$I_i = \frac{V_i}{R_i} = \frac{200 \text{ mV}}{20 \Omega} = 10 \text{ mA}$$

If we assume for the moment that $\alpha_{ac} = 1$ ($I_c = I_e$),

$$I_L = I_i = 10 \text{ mA}$$

and

$$\begin{aligned} V_L &= I_L R \\ &= (10 \text{ mA})(5 \text{ k}\Omega) \\ &= 50 \text{ V} \end{aligned}$$

The voltage amplification is

$$A_v = \frac{V_L}{V_i} = \frac{50 \text{ V}}{200 \text{ mV}} = 250$$

1. A common base transistor amplifier has an input resistance of $20\ \Omega$ and output resistance of $100\ \text{k}\Omega$. The collector load is $1\ \text{k}\Omega$. If a signal of $500\ \text{mV}$ is applied between emitter and base, find the voltage amplification. Assume α_{ac} to be nearly one.

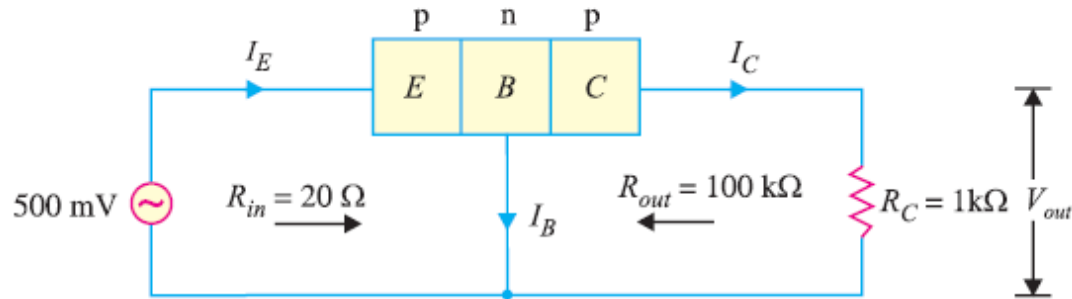


Fig. 1

Input current, $I_E = \frac{\text{Signal}}{R_{in}} = \frac{500\ \text{mV}}{20\ \Omega} = 25\ \text{mA}$. Since α_{ac} is nearly 1, output current, $I_C = I_E = 25\ \text{mA}$.

Output voltage, $V_{out} = I_C R_C = 25\ \text{mA} \times 1\ \text{k}\Omega = 25\ \text{V}$

$\therefore$ Voltage amplification, $A_v = \frac{V_{out}}{\text{signal}} = \frac{25\ \text{V}}{500\ \text{mV}} = 50$

2. In a common base connection, $I_E = 1\text{mA}$, $I_C = 0.95\text{mA}$. Calculate the value of I_B .

Using the relation,

$$I_E = I_B + I_C$$
$$1 = I_B + 0.95$$
$$I_B = 1 - 0.95 = \mathbf{0.05\text{ mA}}$$

3. In a common base connection, current amplification factor is 0.9. If the emitter current is 1mA, determine the value of base current.

Here, $\alpha = 0.9$, $I_E = 1\text{ mA}$

Now

$$\alpha = \frac{I_C}{I_E}$$

or

$$I_C = \alpha I_E = 0.9 \times 1 = 0.9\text{ mA}$$

Also

$$I_E = I_B + I_C$$

$\therefore$ Base current, $I_B = I_E - I_C = 1 - 0.9 = \mathbf{0.1\text{ mA}}$

4. In a common base connection, $I_C = 0.95$ mA and $I_B = 0.05$ mA. Find the value of α .

$$\text{We know } I_E = I_B + I_C = 0.05 + 0.95 = 1 \text{ mA}$$

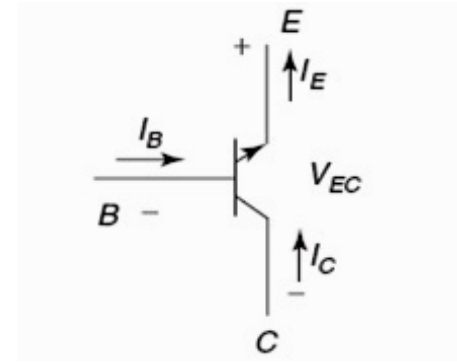
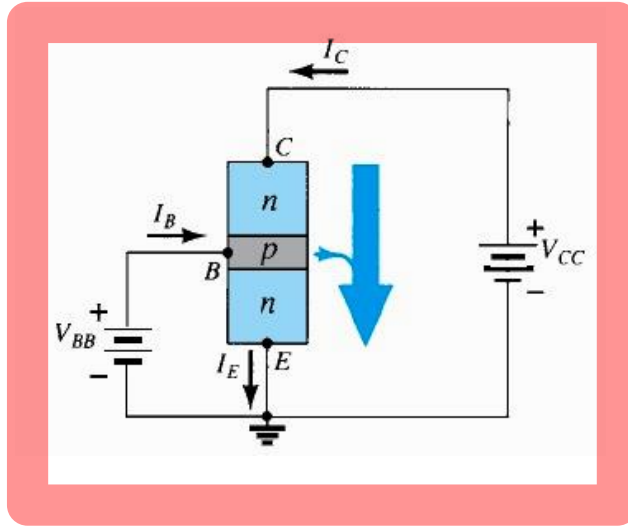
$$\therefore \text{ Current amplification factor, } \alpha = \frac{I_C}{I_E} = \frac{0.95}{1} = \mathbf{0.95}$$

5. In a common base connection, the emitter current is 1mA. If the emitter circuit is open, the collector current is 50 μ A. Find the total collector current. Given that $\alpha = 0.92$.

$$\text{Here, } I_E = 1 \text{ mA, } \alpha = 0.92, \quad I_{CBO} = 50 \mu\text{A}$$

$$\begin{aligned} \therefore \text{ Total collector current, } I_C &= \alpha I_E + I_{CBO} = 0.92 \times 1 + 50 \times 10^{-3} \\ &= 0.92 + 0.05 = \mathbf{0.97 \text{ mA}} \end{aligned}$$

- The **arrow** in the diagram, indicates the direction of the emitter current.
- The **NPN** BJT requires two voltage sources V_{BB} and V_{CC} to bias the two junctions respectively.



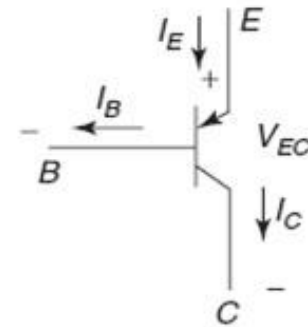
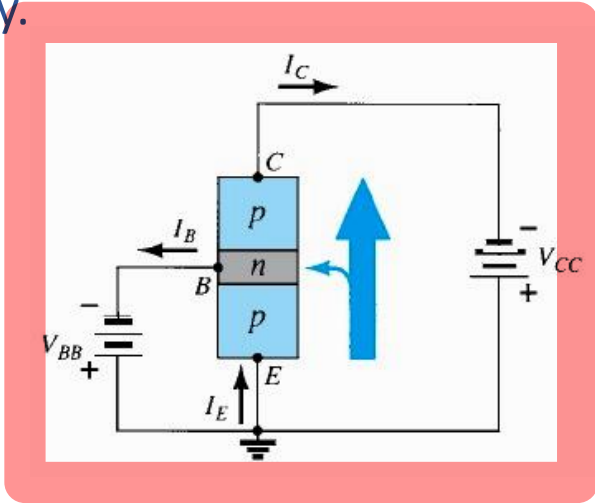
An **NPN** transistor used in Common Emitter (CE) configuration

Electronic Principles and Devices

Common Emitter Configuration

➤ The **arrow** in the diagram, indicates the direction of the emitter current and we can observe that the **direction** of the current in this type of transistor is **opposite** to that of a NPN transistor.

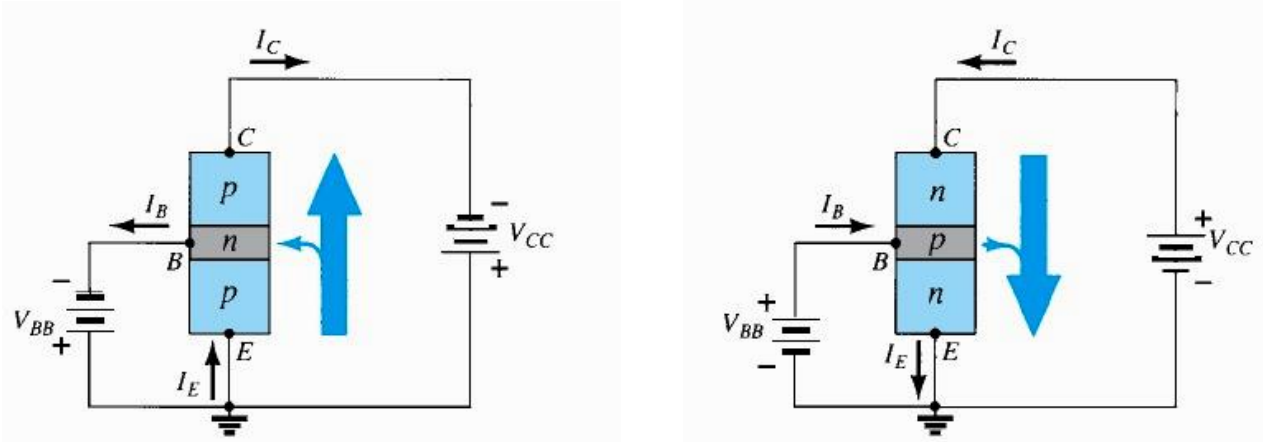
➤ The **PNP** BJT requires two voltage sources V_{BB} and V_{CC} to bias the two junctions respectively.



A **PNP** transistor used in Common Emitter (CE) configuration

Electronic Principles and Devices

Common Emitter Configuration



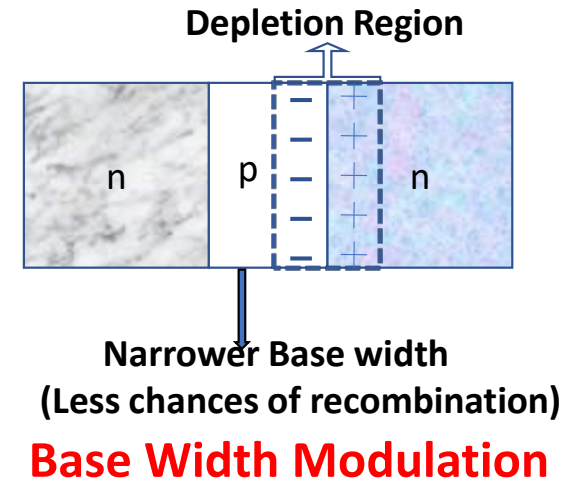
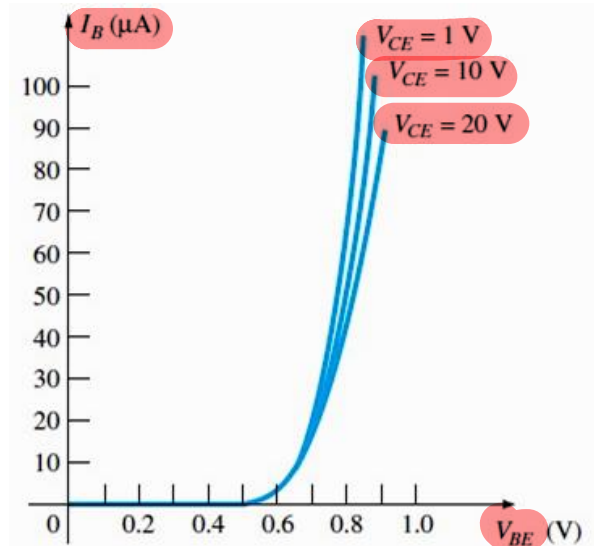
The biasing of **PNP** and **NPN** transistor to operate in active region

- The analogy of a BJT is two diodes connected back-to-back. One diode is **forward biased** and the other is **reverse biased** for the device to operate in **active region**.
- Hence, biasing the **emitter base junction (EB Jn)** in forward bias condition and **collector base junction (CB Jn)** in reverse bias condition, gives the input characteristic of the device.

Electronic Principles and Devices

CE Configuration - Input Characteristics

- For the transistor in CE configuration, I_B is the input current (usually in μA) and V_{BE} is the input voltage. A plot of I_B v/s V_{BE} is called the input characteristics.
- The characteristics resemble the forward bias characteristics of a diode.
- When the output voltage V_{CE} is increased, this high voltage causes a decrease in the current I_B through the device and hence the graph shifts towards the right.



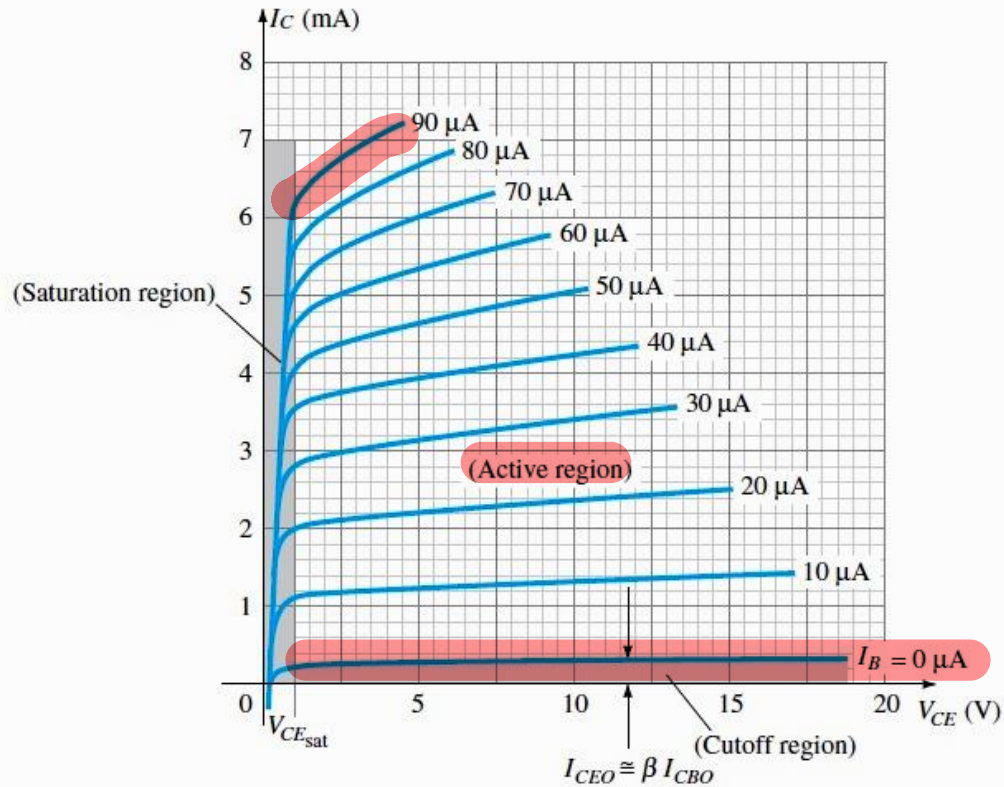
The input characteristic of BJT (Si) for varying output voltage V_{CE}

Base Width Modulation or Early Effect :

- A greater reverse bias across the collector- emitter junction increases the base-emitter depletion width. There is a variation in the width of the base in a bipolar transistor due to a variation in the applied collector- to-emitter voltage.
- Base-narrowing has two consequences that affect the current:
 - There is a lesser chance for recombination within the "smaller" base region & so I_B decreases further .
 - The charge gradient is increased across the base, and consequently, the current of minority carriers injected across the collector-base junction increases.
- The Early effect is observed as an increase in the collector or "output" current with increasing collector-emitter voltage.

Electronic Principles and Devices

CE Configuration - Output Characteristics



The output characteristic of BJT (Si) for varying input current I_B

Electronic Principles and Devices

CE Configuration - Output Characteristics

- **Active region:** The B-E junction is forward biased. Collector voltage is greater than the base voltage which reverse biases the C-B junction.

$$V_{CE} > V_{BE}$$

$$I_C = \beta \cdot I_B$$

$$V_{CE} \uparrow, I_C \uparrow$$

- **Cut-off region:** Both junctions are *reverse biased*. No base current. Only a small reverse leakage current between C and E. For practical purposes, current is assumed to be zero.

- **Saturation region:** The E-B and C-B junctions are forward biased. I_C varies with V_{CE} and is *independent* of I_B and β .

$$I_C = \alpha I_E + I_{CB0}$$

Understand this
derivation from yt

$$= \alpha (I_C + I_B) + I_{CB0}$$

$$(\because I_E = I_C + I_B)$$

$$I_C = \frac{\alpha I_B + I_{CB0}}{1 - \alpha}$$

Let $\beta = \frac{\alpha}{(1 - \alpha)}$

$$[\because \frac{1}{(1 - \alpha)} = 1 + \beta]$$

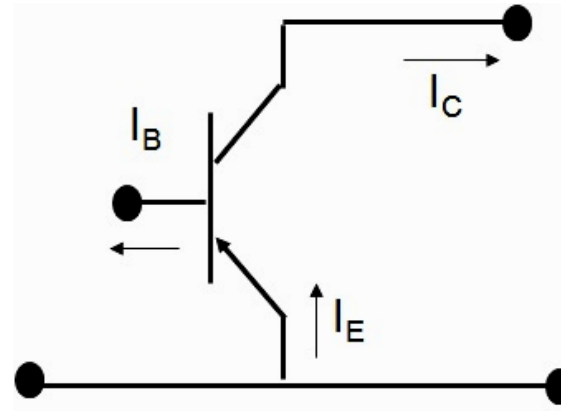
$$I_C = \beta I_B + (1 + \beta) I_{CB0}$$

$$I_C = \beta I_B + I_{CE0}$$

$$I_C = \beta I_B$$

Neglecting leakage current I_{CE0}

$$50 \leq \beta \leq 500$$



$$(\because I_{CE0} = (1 + \beta) I_{CB0})$$

$$I_{CE0} \approx \beta I_{CB0}$$

or

$$I_{CE0} = \frac{I_{CB0}}{1 - \alpha} \Big|_{I_B = 0 \mu A}$$

A relationship can be developed between α and β using the basic relationships introduced thus far. Using $\beta = I_C / I_B$ we have $I_B = I_C / \beta$, and from $\alpha = I_C / I_E$ we have $I_E = I_C / \alpha$. Substituting into

$$I_E = I_C + I_B$$

$$I_C / \alpha = I_C + I_C / \beta$$

and dividing both sides of the equation by I_C will result in

$$\frac{1}{\alpha} = 1 + \frac{1}{\beta}$$

or

$$\beta = \alpha \beta + \alpha = (\beta + 1) \alpha$$

so that

$$\alpha = \frac{\beta}{1 + \beta} \quad \text{or} \quad \beta = \frac{\alpha}{(1 - \alpha)}$$

$$\begin{aligned} I_C &= \beta I_B \\ I_E &= I_C + I_B \\ &= \beta I_B + I_B \\ I_E &= (\beta + 1) I_B \end{aligned}$$

1. Find the value of β if (i) $\alpha = 0.9$ (ii) $\alpha = 0.98$ (iii) $\alpha = 0.99$.

(i) $\alpha = 0.9$

$$\beta = \frac{\alpha}{1-\alpha} = \frac{0.9}{1-0.9} = 9$$

(ii) $\alpha = 0.98$

$$\beta = \frac{\alpha}{1-\alpha} = \frac{0.98}{1-0.98} = 49$$

(iii) $\alpha = 0.99$

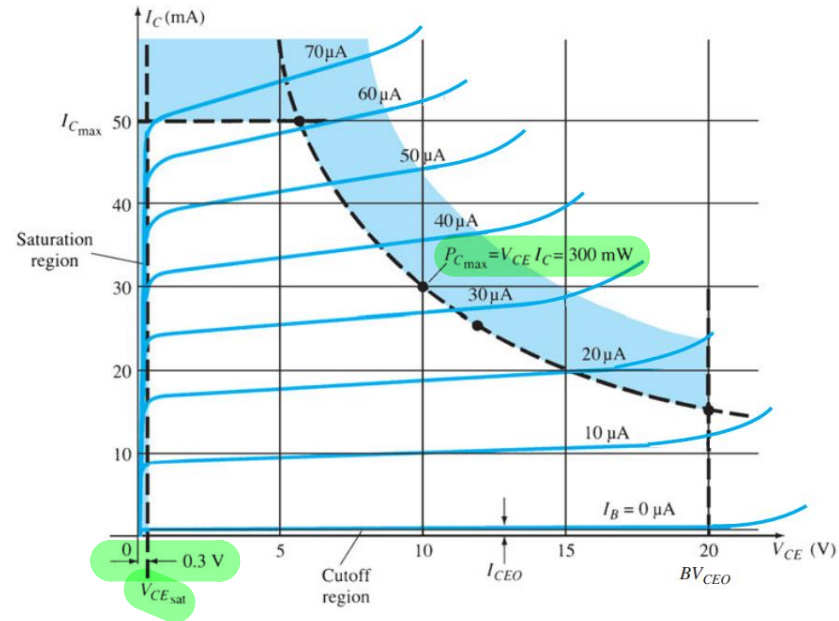
$$\beta = \frac{\alpha}{1-\alpha} = \frac{0.99}{1-0.99} = 99$$

The limit of operation ensures that the transistor works within its maximum rating with minimum signal distortion. It is specified in the data sheet.

V_{CEsat} : Specifies the minimum V_{CE} that can be applied without transistor going to saturation region.

The maximum power dissipation is given as

$$P_{Cmax} = V_{CE} I_C$$



Defining the linear (undistorted) region of operation for a transistor.

Electronic Principles and Devices

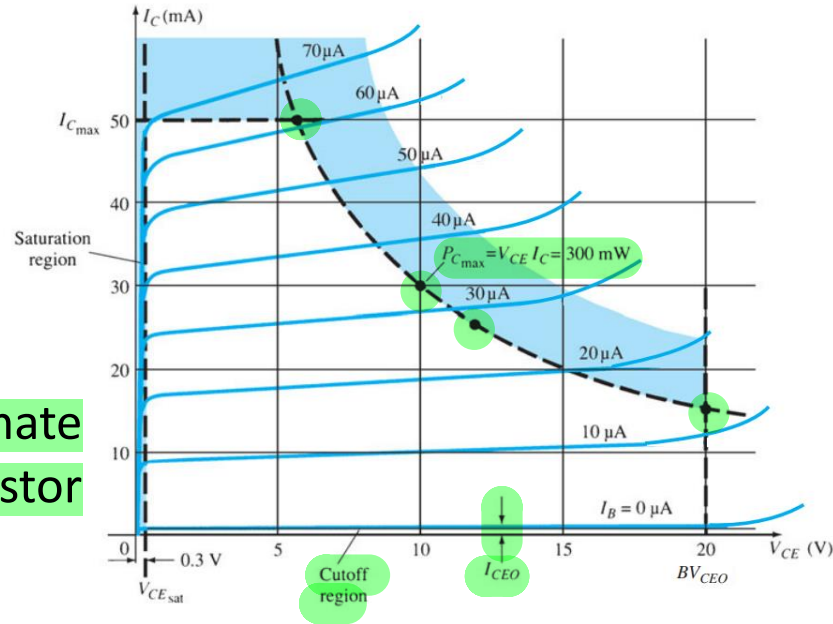
Limits of Operation

For example if,

$$P_{C_{\max}} = V_{CE} I_C = 300 \text{ mW}$$

- At $I_C = I_{C_{\max}}$ i.e, 50 mA, $V_{CE} = 6\text{V}$
- At $V_{CE} = V_{CE_{\max}}$ i.e, 20V, $I_C = 15\text{mA}$
- At $I_C = I_{C_{\max}}/2$, $V_{CE} = 12\text{V}$

These three points give a rough estimate of the power curve and the transistor must be operated with in this region.



Defining the linear (undistorted) region of operation for a transistor.

The cutoff region is the region below $I_C = I_{CEO}$ ($= \beta I_{CBO}$). This region must also be avoided to have minimum distortion.

If the characteristic curve/specification sheet is not available, one must ensure the following conditions are satisfied to ensure minimum distortion.

$$\begin{aligned} I_{CEO} &\leq I_C \leq I_{C_{max}} \\ V_{CE_{sat}} &\leq V_{CE} \leq V_{CE_{max}} \\ V_{CE} I_C &\leq P_{C_{max}} \end{aligned}$$

For CB Characteristics, $P_{C_{max}}$ is defined as

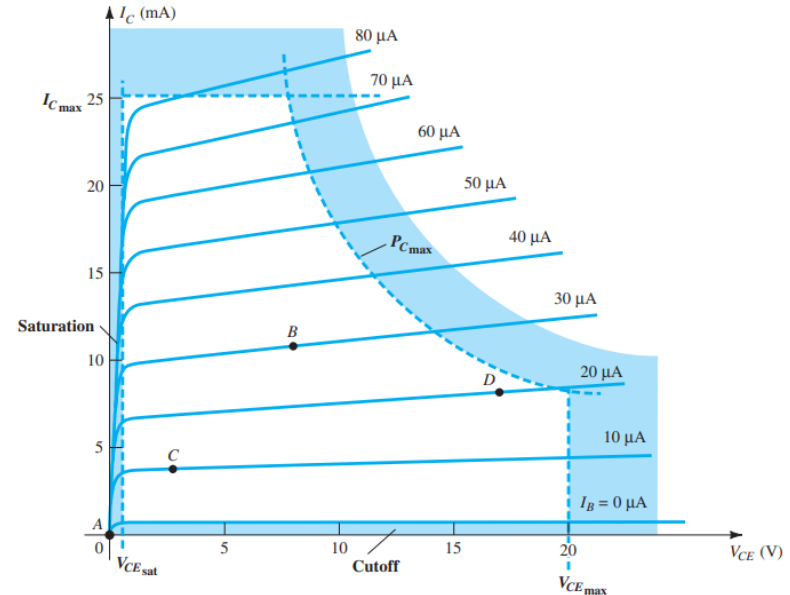
$$P_{C_{max}} = V_{CB} I_C$$

Biasing refers to application of DC voltages to establish a fixed level of current and voltage in a transistor.

For a transistor amplifier, the resulting DC current and voltage on the characteristic curve establish an **Operating Point**. It gives the DC values of I_B , I_C and V_{CE} in the absence of AC input

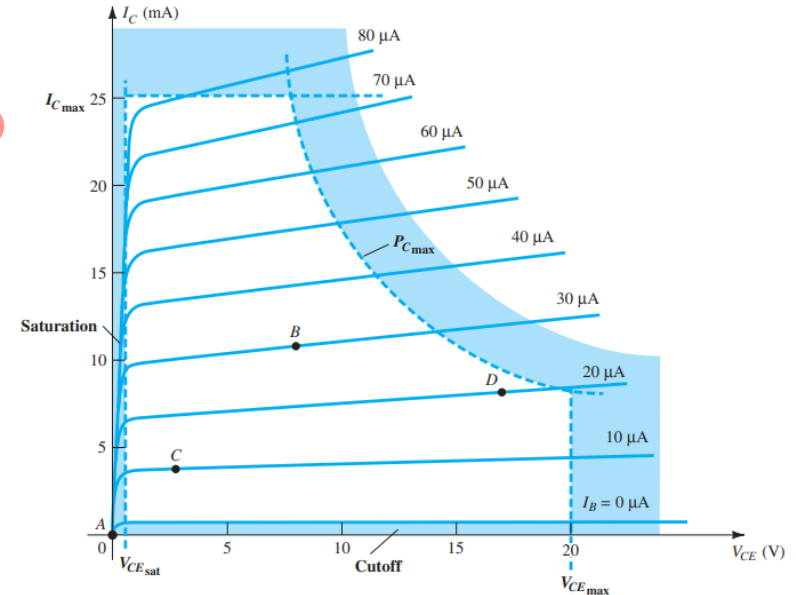
The operating point is also called as **Quiescent point** or **Q point** where **Quiescent** means 'silent' i.e., in the absence of ac signal.

Biasing circuit can be used to set the operating point within the active region. A, B, C, D are some of the possible operating points.



Various operating points within the limits of operation of a transistor.

- **Q Point at A:** If there is no bias, the device is completely off, resulting in Q point at A. This point is not suitable as complete input range can not be covered for amplification.
- **Q Point at B:** can cover a large range of input signal. It is also far from both cut-off and saturation region.
- **Q Point at C:** This is near to both cutoff and saturation region thus limiting the swing of input signal.
- **Q Point at D:** as is near to the maximum voltage and power level, limits the positive swing of input signal



Various operating points within the limits of operation of a transistor.

Operating Point

➤ Figure 4.1 shows a general output device characteristic with four operating points indicated.

The maximum ratings are indicated on the characteristics of Fig. 4.1 by

- ✓ a horizontal line for the maximum collector current I_{Cmax} and a vertical line at the maximum collector-to-emitter voltage V_{CEmax} .
- ✓ The maximum power constraint is defined by the curve P_{Cmax} .
- ✓ At the lower end of the scales are the cutoff region, defined by $I_B=0$ mA, and the saturation region, defined by $V_{CE} = V_{CEsat}$.

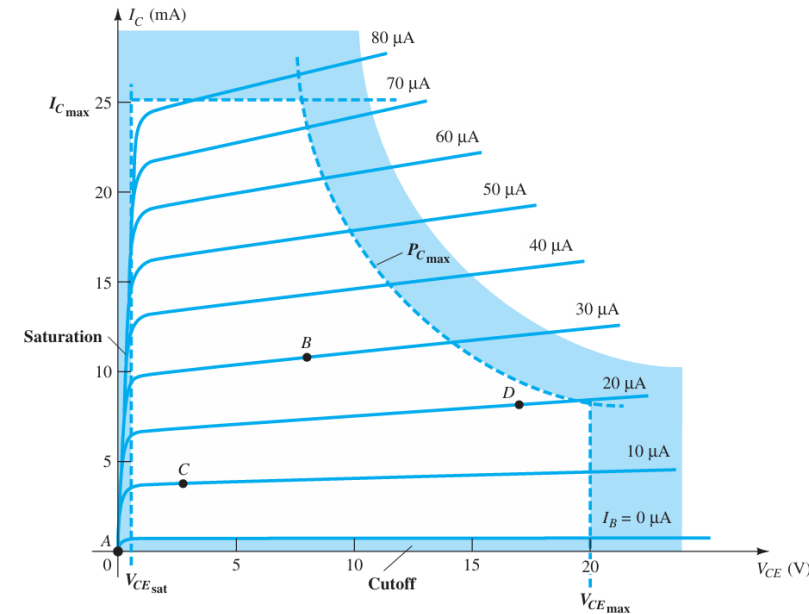


FIG. 4.1

Various operating points within the limits of operation of a transistor.

Temperature also has its effect on the Operating point.

Temperature causes the device parameters like transistor current gain (β_{ac}) and the transistor leakage current (I_{CEO}) to change.

As temperature increases, I_{CEO} increases, resulting in changing the Q point. Thus the network design must also consider temperature stability to reduce the changes in Q point.

The variation of operating point due to variation in temperature is indicated by stability factor (S). Higher the stability of the circuit, better is the circuit.

Transistor Biasing

Transistor Biasing:

Biasing is the process of providing DC voltage which helps in the functioning of the circuit. A transistor is biased in order to make the base-emitter junction forward biased and base-collector junction reverse biased, so that it maintains in active region, to work as an amplifier.

Need for DC biasing:

If a signal of very small voltage is given to the input of BJT, it cannot be amplified. Because, for a BJT, to amplify a signal, two conditions have to be met.

- The input voltage should exceed cut-in voltage for the transistor to be ON.
- The BJT should be in the linear / active region, to be operated as an amplifier.

DC biasing:

- The operating point is a fixed point on the characteristics and is also called quiescent point, denoted by Q-point. The term biasing means the application of dc voltages used to setup a fixed level of current and voltage. Called “dc basing” or “biasing analysis”
- This step determines both the region of operation and the small-signal parameters of each device

Different regions of operation of BJT

Operation in the cutoff, saturation, and linear regions of the BJT characteristic are provided as follows:

1. Linear-region (or Active region) operation:

Base-emitter junction forward-biased Base-collector junction reverse-biased

2. Cutoff-region operation:

Base-emitter junction reverse-biased Base-collector junction reverse-biased

3. Saturation-region operation:

Base-emitter junction forward-biased Base-collector junction forward-biased

Types of BJT Biasing Configuration

Objective is to **determine the terminal voltages and currents of *BJT transistor* and obtain the conditions that ensure biasing in the Active /Linear mode**

Types of Biasing (only the first 3 is included in the syllabus)

1. Fixed Bias configuration
2. Emitter- bias configuration
3. Voltage –divider bias configuration
4. Collector-feedback
5. Emitter-follower
6. Common-base

Fixed-Bias Configuration

- The fixed-bias circuit of Fig. 4.2 is the simplest transistor dc bias configuration. Hence, it is also called as simple biasing.
- For dc, $f = 0$ Hz, and $X_C = \frac{1}{j\omega C} = \frac{1}{j0C} = \infty \Omega$.

Forward Bias of Base-Emitter

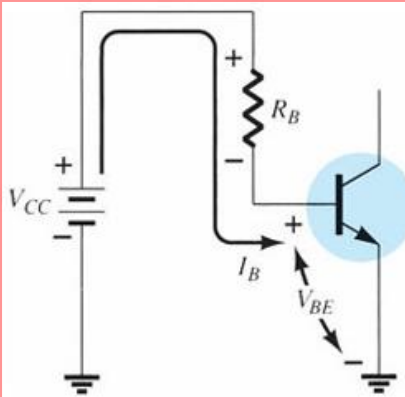
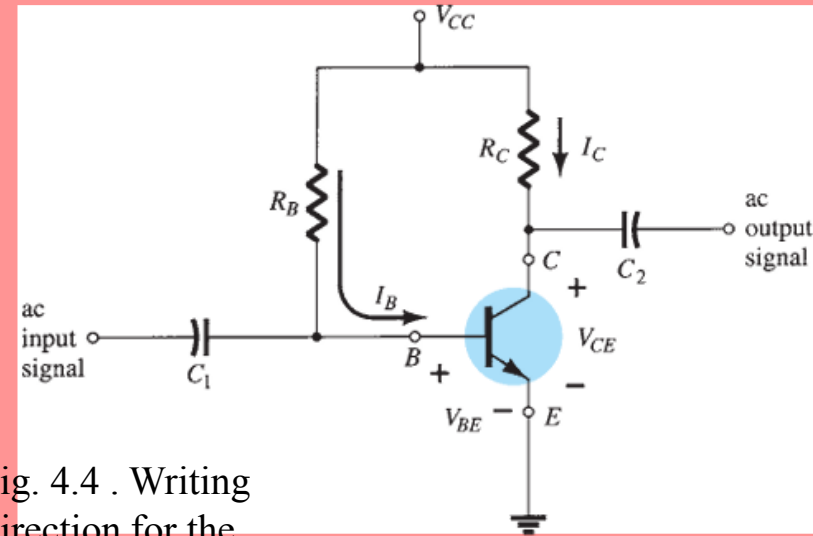


FIG. 4.4

Base-emitter loop.

Consider first the base-emitter circuit loop of Fig. 4.4 . Writing Kirchhoff's voltage equation in the clockwise direction for the loop, we obtain $+V_{CC} - I_B R_B - V_{BE} = 0$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$



Fixed-Bias Configuration 0.7

The supply voltage V_{CC} and the base-emitter voltage V_{BE} are constants, the selection of a base resistor R_B sets the level of base current for the operating point.

Collector-Emitter Loop

The collector-emitter section of the network appears in Fig. 4.5 with the indicated direction of current I_C and the resulting polarity across R_C $I_C = \beta I_B$

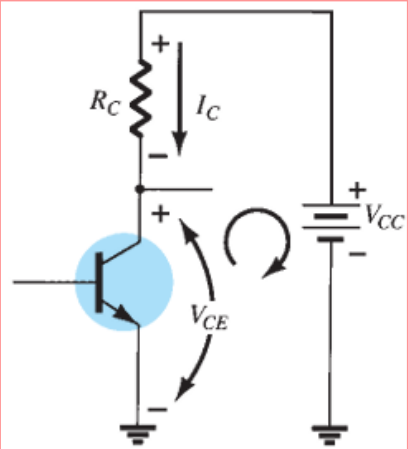


FIG. 4.5

Collector-emitter loop.

- Changing R_C to any level will not affect the level of I_B or I_C as long as the device is in the active region.
- The level of R_C will determine the magnitude of V_{CE} .

$$V_{CE} + I_C R_C - V_{CC} = 0$$

$$V_{CE} = V_{CC} - I_C R_C$$

$$V_{CE} = V_C - V_E$$

$$V_{BE} = V_B - V_E$$

$$I_E = I_B + I_C$$

$$I_E \approx I_C \quad (\text{since } I_B \ll I_C)$$

Since, emitter is ground

$$V_{CE} = V_C$$

$$V_{BE} = V_B$$

Fixed-Bias Configuration

Determine the following for the fixed-bias configuration of Fig. 4.7

- I_{BQ} and I_{CQ} .
- V_{CEQ} .
- V_B and V_C .
- V_{BC} .

$$\text{a)} \quad I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12 \text{ V} - 0.7 \text{ V}}{240 \text{ k}\Omega} = 47.08 \mu\text{A}$$

$$I_{CQ} = \beta I_{BQ} = (50)(47.08 \mu\text{A}) = 2.35 \text{ mA}$$

$$\text{b)} \quad V_{CEQ} = V_{CC} - I_C R_C \\ = 12 \text{ V} - (2.35 \text{ mA})(2.2 \text{ k}\Omega) \\ = 6.83 \text{ V}$$

$$\text{c)} \quad V_B = V_{BE} = 0.7 \text{ V} \\ V_C = V_{CE} = 6.83 \text{ V}$$

$$\text{d)} \quad V_{BC} = V_B - V_C = 0.7 \text{ V} - 6.83 \text{ V} \\ = -6.13 \text{ V}$$

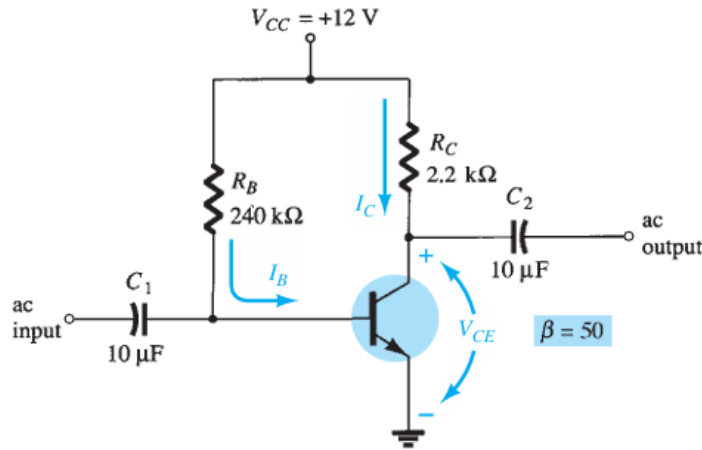


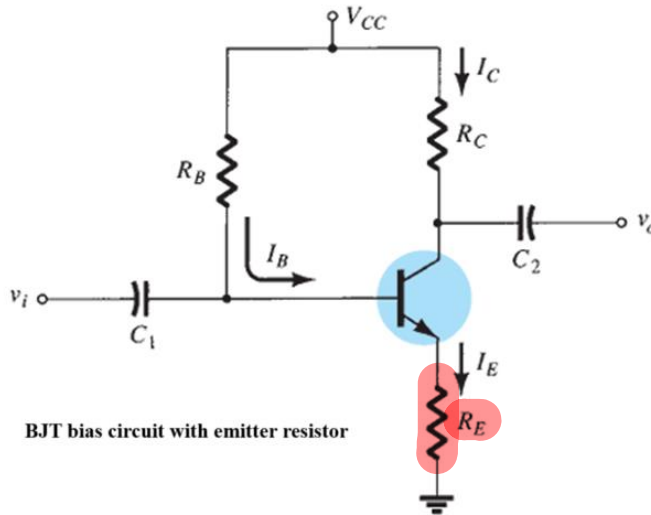
FIG. 4.7

DC fixed-bias circuit for Example 4.1.

The negative sign revealing that the junction is reversed-biased, as it should be for linear amplification.

Emitter-Bias Configuration

- The DC bias network of fig below contains an emitter resistor to improve the stability level over that of the fixed bias configuration.



Emitter-Bias Configuration

Base-Emitter Loop

Applying Kirchhoff's voltage law around the indicated loop in the clockwise direction results in the following equation:

$$+V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

$$I_E \approx (\beta + 1)I_B$$

Since $I_E = (\beta + 1)I_B$, substitute in the above equation

$$V_{CC} - I_B R_B - V_{BE} - (\beta + 1)I_B R_E = 0$$

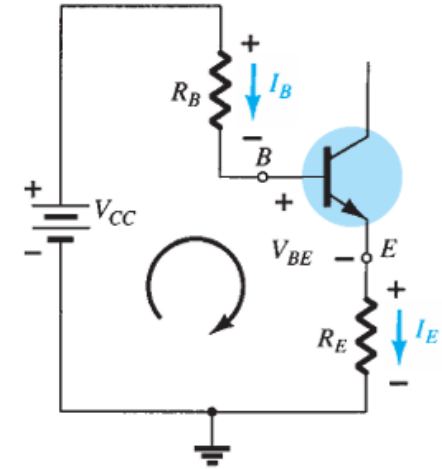
$$-I_B(R_B + (\beta + 1)R_E) + V_{CC} - V_{BE} = 0$$

Multiplying through by (-1), we have

$$I_B(R_B + (\beta + 1)R_E) - V_{CC} + V_{BE} = 0$$

$$I_B(R_B + (\beta + 1)R_E) = V_{CC} - V_{BE}$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E}$$



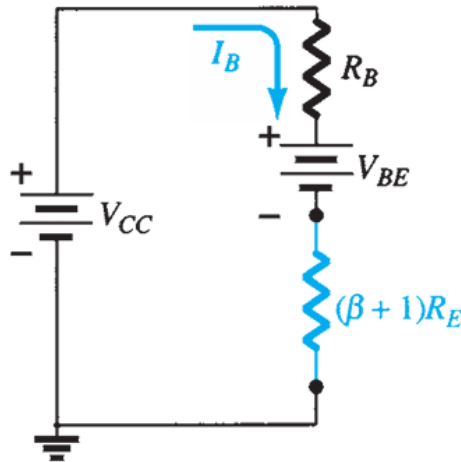
Base-emitter loop

Note: that the only difference between this equation for I_B and that obtained for the fixed bias configuration is the term $(\beta + 1)R_E$.

Emitter-Bias Configuration

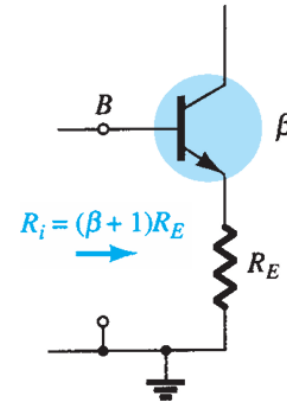
$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E}$$

the network derived from this I_B equation is



- ✓ Note that aside from the base-to-emitter voltage V_{BE} , the resistor R_E is reflected back to the input base circuit by a factor $(\beta+1)$.
- ✓ In other words, the emitter resistor, which is a part of the collector-emitter loop, “appears as” $(\beta+1)R_E$ in the base-emitter loop.
- ✓ Thus, the reflected impedance level of R_E , in the base circuit as shown in figure. Therefore, for this configuration the **input impedance is given by**

$$R_i = (\beta + 1)R_E$$



Key Points:

1. Emitter Resistor Reflection:

- The resistor R_E is physically in the **collector-emitter loop**.
- However, due to the transistor's current gain (β), R_E is "reflected" into the **base-emitter loop** with a magnified value of $(\beta + 1)R_E$.
- This happens because the base current (I_B) is smaller than the emitter current (I_E), with $I_E \approx (\beta + 1)I_B$. The impedance seen at the base includes this scaling effect.

2. Input Impedance (R_i):

- The input impedance at the base is the effective resistance seen by the input signal.
- Due to the reflection of R_E , the input impedance is given by:

$$R_i = (\beta + 1)R_E$$

3. Why It Matters:

- The reflected impedance increases the input impedance seen by the base. This is important for signal matching in amplifier circuits, as higher input impedance minimizes the loading effect on the previous stage.

Practical Explanation:

Think of the transistor's current gain (β) as a scaling factor that "amplifies" the impedance of the emitter resistor when observed from the base. This reflected impedance is part of how the transistor operates in a common emitter amplifier to maintain signal integrity and control current flow.

Emitter-Bias Configuration

Collector–Emitter Loop

The collector–emitter loop appears as shown in figure. Applying Kirchhoff's voltage law for the indicated loop in the clockwise direction results in

$$+I_E R_E + V_{CE} + I_C R_C - V_{CC} = 0$$

Substituting $I_E \cong I_C$ and grouping terms gives

$$V_{CE} - V_{CC} + I_C(R_C + R_E) = 0$$

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

Voltage from emitter to ground is given by

$$V_E = I_E R_E$$

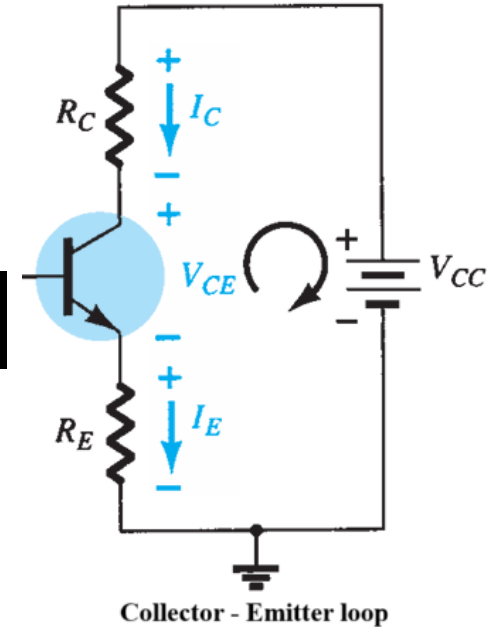
Voltage between collector and emitter is $V_{CE} = V_C - V_E$

Voltage from collector to ground is given by

$$V_C = V_{CE} + V_E$$

$$V_C = V_{CC} - I_C R_C$$

$$I_C = \beta I_B$$



Voltage at base w.r.t to ground is given by

$$V_B = V_{CC} - I_B R_B$$

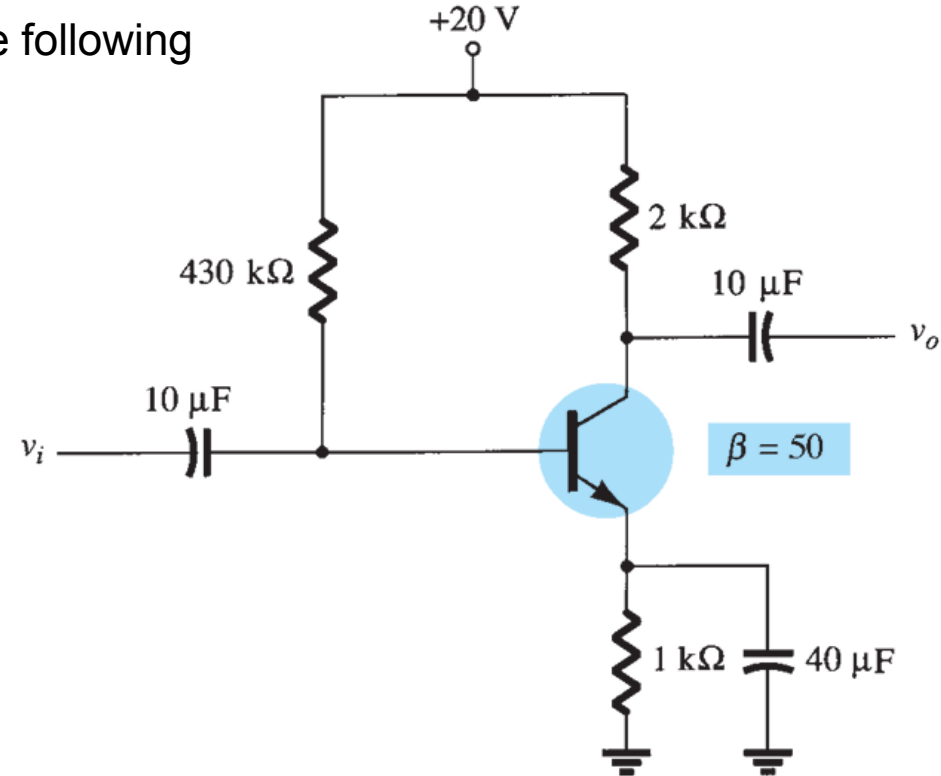
$$V_B = V_{BE} + V_E$$

Emitter-Bias Configuration

Problem

For the emitter-bias network of Fig. 4.23 , determine the following

- a. I_B .
- b. I_C .
- c. V_{CE} .
- d. V_C .
- e. V_E .
- f. V_B .
- g. V_{BC} .



Emitter-Bias Configuration

Solution

$$\begin{aligned} \text{a)} \quad I_B &= \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{20 \text{ V} - 0.7 \text{ V}}{430 \text{ k}\Omega + (51)(1 \text{ k}\Omega)} \\ &= \frac{19.3 \text{ V}}{481 \text{ k}\Omega} = \mathbf{40.1 \mu A} \end{aligned}$$

$$\begin{aligned} \text{b)} \quad I_C &= \beta I_B \\ &= (50)(40.1 \mu A) \\ &\cong \mathbf{2.01 \text{ mA}} \end{aligned}$$

$$\begin{aligned} \text{c)} \quad V_{CE} &= V_{CC} - I_C(R_C + R_E) \\ &= 20 \text{ V} - (2.01 \text{ mA})(2 \text{ k}\Omega + 1 \text{ k}\Omega) = 20 \text{ V} - 6.03 \text{ V} \\ &= \mathbf{13.97 \text{ V}} \end{aligned}$$

$$\begin{aligned} \text{d)} \quad V_C &= V_{CC} - I_C R_C \\ &= 20 \text{ V} - (2.01 \text{ mA})(2 \text{ k}\Omega) = 20 \text{ V} - 4.02 \text{ V} \\ &= \mathbf{15.98 \text{ V}} \end{aligned}$$

$$\begin{aligned} \text{e)} \quad V_E &= V_C - V_{CE} \\ &= 15.98 \text{ V} - 13.97 \text{ V} \\ &= \mathbf{2.01 \text{ V}} \end{aligned}$$

$$\begin{aligned} \text{f)} \quad V_B &= V_{BE} + V_E \\ &= 0.7 \text{ V} + 2.01 \text{ V} \\ &= \mathbf{2.71 \text{ V}} \end{aligned} \quad \text{or} \quad \begin{aligned} V_E &= I_E R_E \cong I_C R_E \\ &= (2.01 \text{ mA})(1 \text{ k}\Omega) \\ &= \mathbf{2.01 \text{ V}} \end{aligned}$$

$$\begin{aligned} \text{g)} \quad V_{BC} &= V_B - V_C \\ &= 2.71 \text{ V} - 15.98 \text{ V} \\ &= \mathbf{-13.27 \text{ V}} \text{ (reverse-biased as required)} \end{aligned}$$

Emitter-Bias Configuration

➤ Improved Bias Stability in Emitter bias configuration

The addition of the emitter resistor to the dc bias of the BJT provides improved stability, that is, the dc bias currents and voltages remain closer to where they were set by the circuit when outside conditions, such as temperature and transistor beta, change.

- The comparison table of the bias voltage and currents of the circuits of Fixed and emitter bias for the given value of $\beta = 50$ and for a new value of $\beta = 100$. The changes in I_C and V_{CE} for the same increase in β is.

Fixed bias configuration				Emitter bias configuration			
β	$I_B (\mu A)$	$I_C (mA)$	$V_{CE} (V)$	β	$I_B (\mu A)$	$I_C (mA)$	$V_{CE} (V)$
50	47.08	2.35	6.83	50	40.1	2.01	13.97
100	47.08	4.71	1.64	100	36.3	3.63	9.11

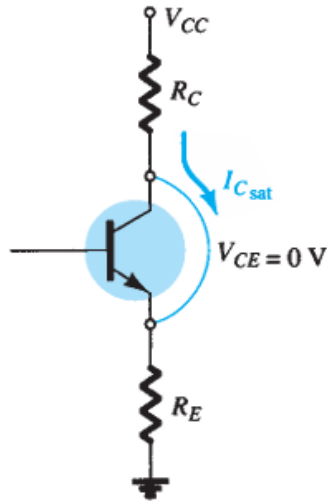
The collector current is seen to change by 100% due to the 100% change in the value of β . The value of I_B is the same, and V_{CE} decreased by 76%				The collector current increases by about 81% due to the 100% increase in β . I_B decreased, maintain the value of I_C —or at least reducing the overall change in I_C due to the change in β . The change in V_{CE} has dropped to about 35%			
--	--	--	--	--	--	--	--

- Thus, the emitter – bias configuration is therefore more stable than that of fixed bias for the same change in β .

Emitter-Bias Configuration

Saturation Level

- The collector saturation level or maximum collector current for an emitter-bias design can be determined using the same approach applied to the fixed-bias configuration
- Apply a short circuit between the collector–emitter terminals as shown in below and calculate the resulting collector current.



$$I_{C\text{ sat}} = \frac{V_{CC}}{R_C + R_E}$$

Voltage- Divider Bias Configuration

- In the previous bias configurations, the bias current I_{CQ} and voltage V_{CEQ} were a function of the current gain β of the transistor.
- Since, β is temperature sensitive, especially for silicon transistors, and the actual value of beta is usually not well defined.
- it is necessary to develop a bias circuit that is less dependent on, or independent of, the transistor beta.
- Thus, Voltage divider bias configuration is preferred.

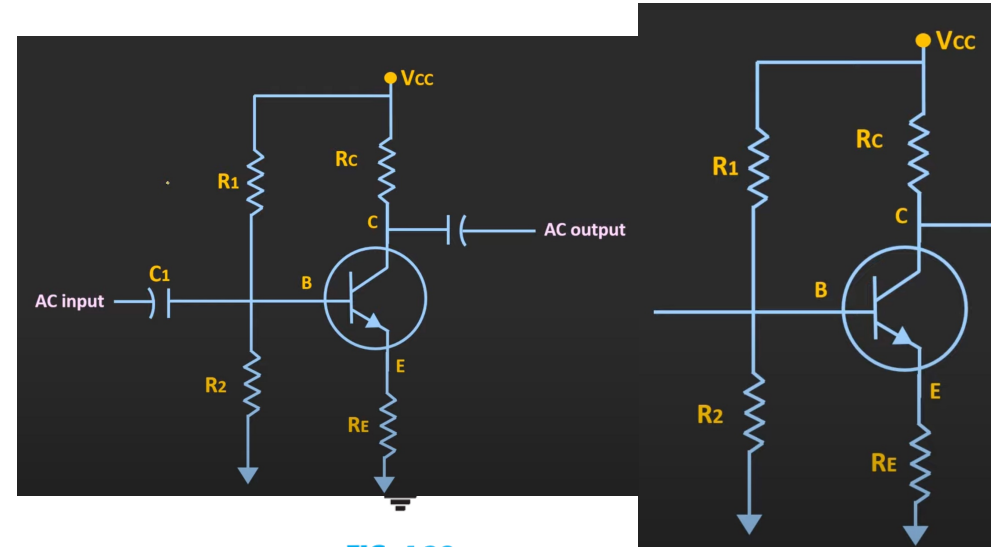


FIG. 4.28

Voltage-divider bias configuration.

Voltage- Divider Bias Configuration

- **Approximate Analysis:** can be applied only if specific conditions are satisfied
- The input section of the voltage-divider configuration can be represented by the network of Fig. 4.36
- The R_i is the resistance between base and ground for the transistor with the emitter resistance, with the reflected resistance between the base and emitter, similar to the Emitter bias configuration and is given by $R_i = (\beta + 1)R_E$
- If R_i is much larger than the resistance R_2 , the current I_B will be much smaller

than I_2 and I_2 will be approximately equal to I_1

Since, $I_B \cong 0$, the voltage drop across R_2 is given by $\frac{R_2}{R_1 + R_2} V_{CC}$

(using the voltage divider rule, hence the name of the configuration)

- **If $\beta R_E \geq 10R_2$, then the approximation method can be applied to determine the biasing current and voltage.**

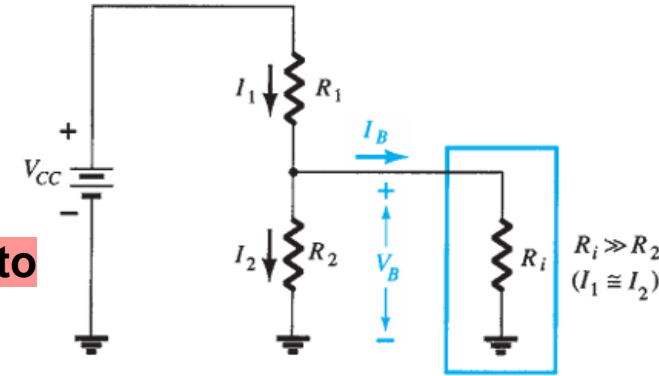


FIG. 4.36

Partial-bias circuit for calculating the approximate base voltage V_B .

Voltage- Divider Bias Configuration

- Approximate Analysis:

➤ V_E and I_E are calculated by $V_E = V_B - V_{BE}$ and $I_E = \frac{V_E}{R_E}$

➤ And $I_{CQ} \cong I_E$

➤ The collector-to-emitter voltage is determined by $V_{CE} = V_{CC} - I_C R_C - I_E R_E$

➤ Since $I_E \cong I_C$,

$$V_{CEQ} = V_{CC} - I_C(R_C + R_E)$$

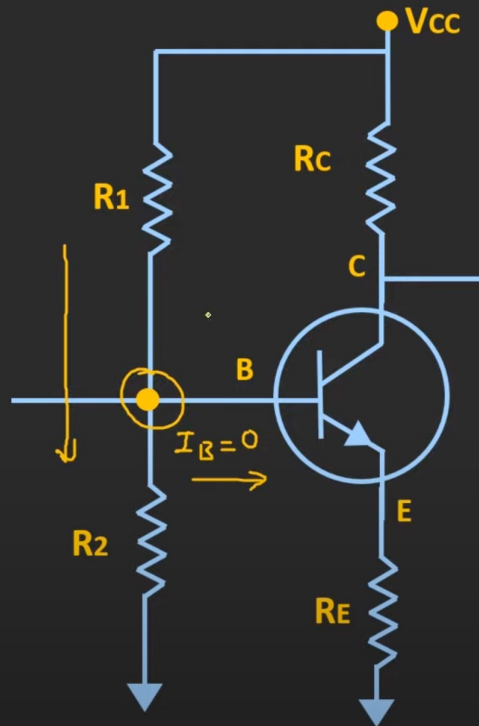
➤ In all the biasing voltage and current equation β does not appear and the Q -point (as determined by I_{CQ} and V_{CEQ}) is therefore independent of the value of β .

➤ Transistor Saturation

$$I_{C_{sat}} = I_{C_{max}} = \frac{V_{CC}}{R_C + R_E}$$

Approximate Analysis

$$I_B \approx 0$$

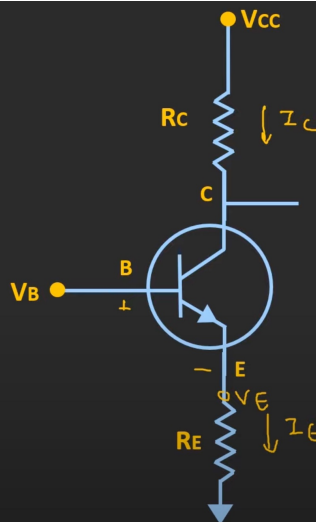


$$V_B = \frac{R_2}{R_1 + R_2} \times V_{CC}$$

Approximate Analysis

$$I_B \approx 0$$

$$I_C \approx I_E$$

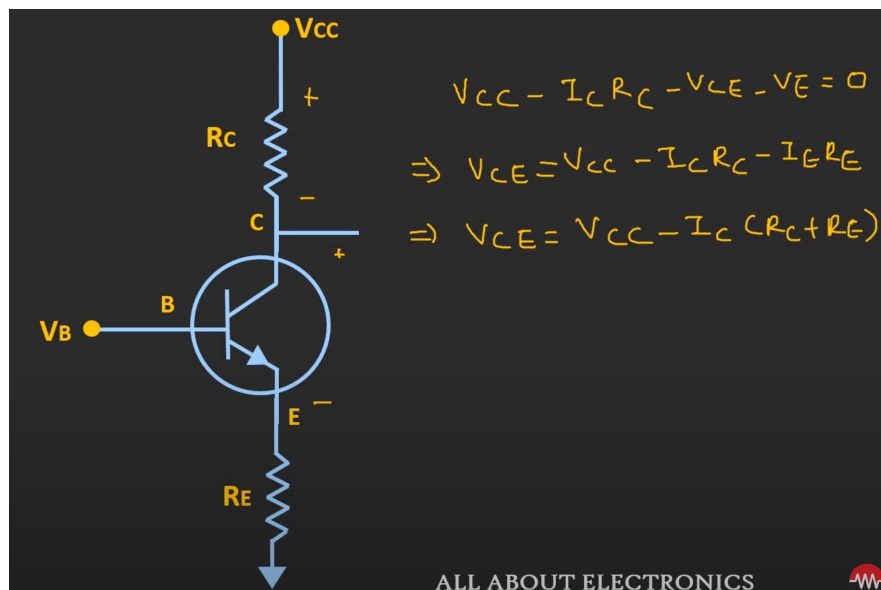


$$V_B = \frac{R_2}{R_1 + R_2} \times V_{CC}$$

$$V_E = V_B - V_{BE}$$

$$I_E = \frac{V_E}{R_E}$$

ALL ABOUT ELECTRONICS



$$V_{CC} - I_C R_C - V_{CE} - V_E = 0$$

$$\Rightarrow V_{CE} = V_{CC} - I_C R_C - I_E R_E$$

$$\Rightarrow V_{CE} = V_{CC} - I_C (R_C + R_E)$$

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Voltage- Divider Bias Configuration

Determine the dc bias voltage V_{CE} and the current I_C for the voltage divider configuration of Fig. 4.35 .

Solution: check the condition to apply approximate method

$$\beta R_E \geq 10R_2$$

$$(100)(1.5 \text{ k}\Omega) \geq 10(3.9 \text{ k}\Omega)$$

$$150 \text{ k}\Omega \geq 39 \text{ k}\Omega \text{ (satisfied)}$$

I_{CQ} calculation

$$\begin{aligned} V_B &= \frac{R_2 V_{CC}}{R_1 + R_2} \\ &= \frac{(3.9 \text{ k}\Omega)(22 \text{ V})}{39 \text{ k}\Omega + 3.9 \text{ k}\Omega} \\ &= 2 \text{ V} \end{aligned}$$

$$\begin{aligned} V_E &= V_B - V_{BE} \\ &= 2 \text{ V} - 0.7 \text{ V} \\ &= 1.3 \text{ V} \end{aligned}$$

$$I_{CQ} \cong I_E = \frac{V_E}{R_E} = \frac{1.3 \text{ V}}{1.5 \text{ k}\Omega} = 0.867 \text{ mA}$$

V_{CEQ} Calculation

$$\begin{aligned} V_{CEQ} &= V_{CC} - I_C(R_C + R_E) \\ &= 22 \text{ V} - (0.867 \text{ mA})(10 \text{ k}\Omega + 1.5 \text{ k}\Omega) \\ &= 22 \text{ V} - 9.97 \text{ V} \\ &= 12.03 \text{ V} \end{aligned}$$

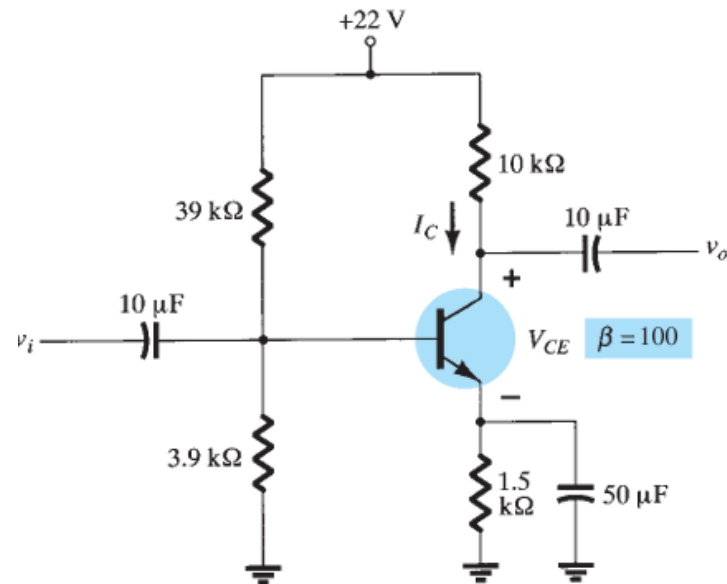
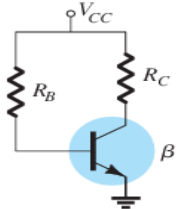
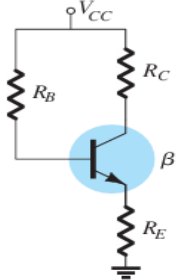
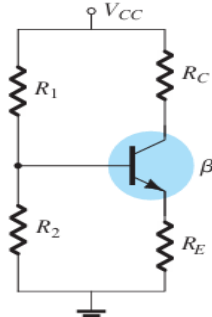


FIG. 4.35

Summary Table of biasing circuits

Type	Configuration	Pertinent Equations
Fixed-bias		$I_B = \frac{V_{CC} - V_{BE}}{R_B}$ $I_C = \beta I_B, I_E = (\beta + 1)I_B$ $V_{CE} = V_{CC} - I_C R_C$
Emitter-bias		$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E}$ $I_C = \beta I_B, I_E = (\beta + 1)I_B$ $R_i = (\beta + 1)R_E$ $V_{CE} = V_{CC} - I_C (R_C + R_E)$
Voltage-divider bias		APPROXIMATE: $\beta R_E \geq 10R_2$ $V_B = \frac{R_2 V_{CC}}{R_1 + R_2}, V_E = V_B - V_{BE}$ $I_E = \frac{V_E}{R_E}, I_B = \frac{I_E}{\beta + 1}$ $V_{CE} = V_{CC} - I_C (R_C + R_E)$

DESIGN OPERATIONS

- The design process is one where a current and/or voltage may be specified and the elements required to establish the designated levels must be determined.
- This synthesis process requires a clear understanding of the characteristics of the device, the basic equations for the network, and a firm understanding of the basic laws of circuit analysis, such as Ohm's law, Kirchhoff's voltage law, and so on.

Setting $V_E = \frac{1}{10} V_{CC}$ is a standard design practice in BJT circuits to ensure proper biasing, stabilization, and ease of calculation. It creates a stable emitter voltage that indirectly determines other resistor values in the circuit.

(a)

(b)

FIG. 4.59

Example 4.21.

DESIGN OPERATIONS

- **EXAMPLE 4.21** Given the device characteristics of Fig. 4.59a, determine V_{CC} , R_B , and R_C for the fixed-bias configuration of Fig. 4.59b.

Solution: From the load line

R_C calculation

$$V_{CC} = 20 \text{ V}$$

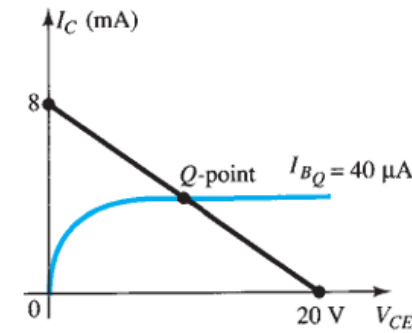
$$I_C = \frac{V_{CC}}{R_C} \Big|_{V_{CE}=0 \text{ V}}$$

$$R_C = \frac{V_{CC}}{I_C} = \frac{20 \text{ V}}{8 \text{ mA}} = 2.5 \text{ k}\Omega$$

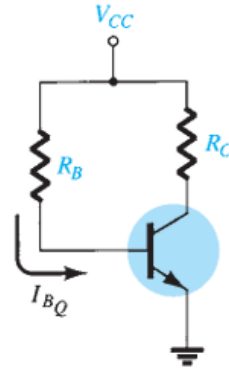
R_B calculation

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

$$\begin{aligned} R_B &= \frac{V_{CC} - V_{BE}}{I_B} \\ &= \frac{20 \text{ V} - 0.7 \text{ V}}{40 \mu\text{A}} = \frac{19.3 \text{ V}}{40 \mu\text{A}} \\ &= 482.5 \text{ k}\Omega \end{aligned}$$



(a)



(b)

Standard resistor values are

$$R_C = 2.4 \text{ k}\Omega$$

$$R_B = 470 \text{ k}\Omega$$

Using standard resistor values gives

$$I_B = 41.1 \mu\text{A}$$

which is well within 5% of the value specified.

FIG. 4.59
Example 4.21.

DESIGN OPERATIONS

- **EXAMPLE 4.22** Given that $I_{CQ} = 2 \text{ mA}$ and $V_{CEQ} = 10 \text{ V}$, determine R_1 and R_C for the network of Fig. 4.60

Solution:

R_1 calculation

$$V_E = I_E R_E \cong I_C R_E$$

$$= (2 \text{ mA})(1.2 \text{ k}\Omega) = 2.4 \text{ V}$$

$$V_B = V_{BE} + V_E = 0.7 \text{ V} + 2.4 \text{ V} = 3.1 \text{ V}$$

$$V_B = \frac{R_2 V_{CC}}{R_1 + R_2} = 3.1 \text{ V}$$

$$\frac{(18 \text{ k}\Omega)(18 \text{ V})}{R_1 + 18 \text{ k}\Omega} = 3.1 \text{ V}$$

$$324 \text{ k}\Omega = 3.1 R_1 + 55.8 \text{ k}\Omega$$

$$3.1 R_1 = 268.2 \text{ k}\Omega$$

$$R_1 = \frac{268.2 \text{ k}\Omega}{3.1} = \mathbf{86.52 \text{ k}\Omega}$$

R_C Calculation

$$R_C = \frac{V_{R_C}}{I_C} = \frac{V_{CC} - V_C}{I_C}$$

$$V_C = V_{CE} + V_E = 10 \text{ V} + 2.4 \text{ V} = 12.4 \text{ V}$$

$$R_C = \frac{18 \text{ V} - 12.4 \text{ V}}{2 \text{ mA}}$$

$$= \mathbf{2.8 \text{ k}\Omega}$$

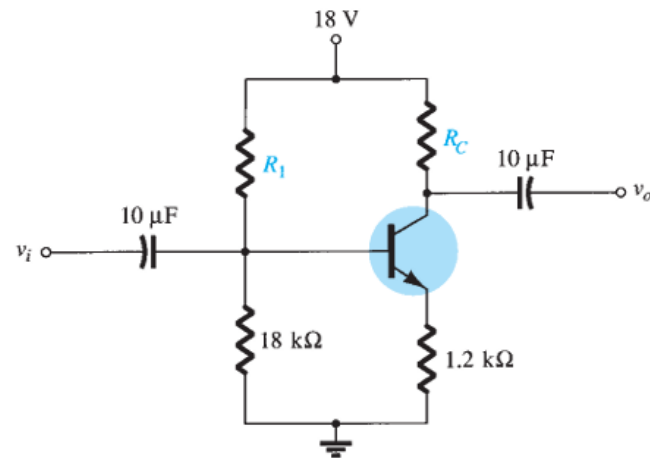


FIG. 4.60

Example 4.22.

DESIGN OPERATIONS

- **EXAMPLE 4.23** The emitter-bias configuration of Fig. 4.61 has the following specifications: $I_{CQ} = 1/2 I_{C\text{sat}}$, $I_{C\text{sat}} = 8 \text{ mA}$, $V_C = 18 \text{ V}$, and $\beta = 110$. Determine R_C , R_E , and R_B .

➤ **Solution:**

R_C calculation

$$I_{CQ} = \frac{1}{2} I_{C\text{sat}} = 4 \text{ mA}$$

$$R_C = \frac{V_{R_C}}{I_{CQ}} = \frac{V_{CC} - V_C}{I_{CQ}}$$

$$= \frac{28 \text{ V} - 18 \text{ V}}{4 \text{ mA}} = \mathbf{2.5 \text{ k}\Omega}$$

R_E calculation

$$I_{C\text{sat}} = \frac{V_{CC}}{R_C + R_E}$$

$$R_C + R_E = \frac{V_{CC}}{I_{C\text{sat}}} = \frac{28 \text{ V}}{8 \text{ mA}} = 3.5 \text{ k}\Omega$$

$$R_E = 3.5 \text{ k}\Omega - R_C$$

$$= 3.5 \text{ k}\Omega - 2.5 \text{ k}\Omega$$

$$= \mathbf{1 \text{ k}\Omega}$$

R_B calculation

$$I_{BQ} = \frac{I_{CQ}}{\beta} = \frac{4 \text{ mA}}{110} = 36.36 \mu\text{A}$$

$$I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E}$$

$$R_B + (\beta + 1)R_E = \frac{V_{CC} - V_{BE}}{I_{BQ}}$$

$$R_B = \frac{V_{CC} - V_{BE}}{I_{BQ}} - (\beta + 1)R_E$$

$$= \frac{28 \text{ V} - 0.7 \text{ V}}{36.36 \mu\text{A}} - (111)(1 \text{ k}\Omega)$$

$$= \frac{27.3 \text{ V}}{36.36 \mu\text{A}} - 111 \text{ k}\Omega$$

$$= \mathbf{639.8 \text{ k}\Omega}$$

For standard values,

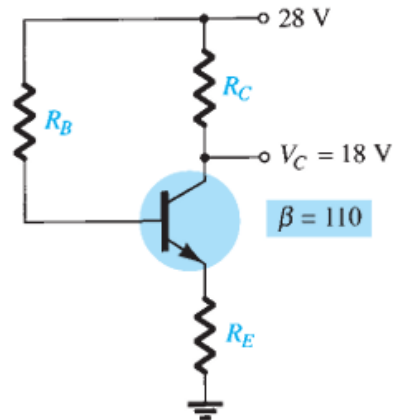


FIG. 4.61
Example 4.23.

$$R_C = 2.4 \text{ k}\Omega$$

$$R_E = 1 \text{ k}\Omega$$

$$R_B = 620 \text{ k}\Omega$$

DESIGN OPERATIONS

- **EXAMPLE 4.24** Determine the resistor values for the network of Fig. 4.62 for the indicated operating point and supply voltage.

➤ **Solution:**

$$V_E = \frac{1}{10} V_{CC} = \frac{1}{10} (20 \text{ V}) = 2 \text{ V}$$

$$R_E = \frac{V_E}{I_E} \cong \frac{V_E}{I_C} = \frac{2 \text{ V}}{2 \text{ mA}} = 1 \text{ k}\Omega$$

$$R_C = \frac{V_{R_C}}{I_C} = \frac{V_{CC} - V_{CE} - V_E}{I_C} = \frac{20 \text{ V} - 10 \text{ V} - 2 \text{ V}}{2 \text{ mA}} = \frac{8 \text{ V}}{2 \text{ mA}} = 4 \text{ k}\Omega$$

$$I_B = \frac{I_C}{\beta} = \frac{2 \text{ mA}}{150} = 13.33 \mu\text{A}$$

$$R_B = \frac{V_{R_B}}{I_B} = \frac{V_{CC} - V_{BE} - V_E}{I_B} = \frac{20 \text{ V} - 0.7 \text{ V} - 2 \text{ V}}{13.33 \mu\text{A}} \cong 1.3 \text{ M}\Omega$$

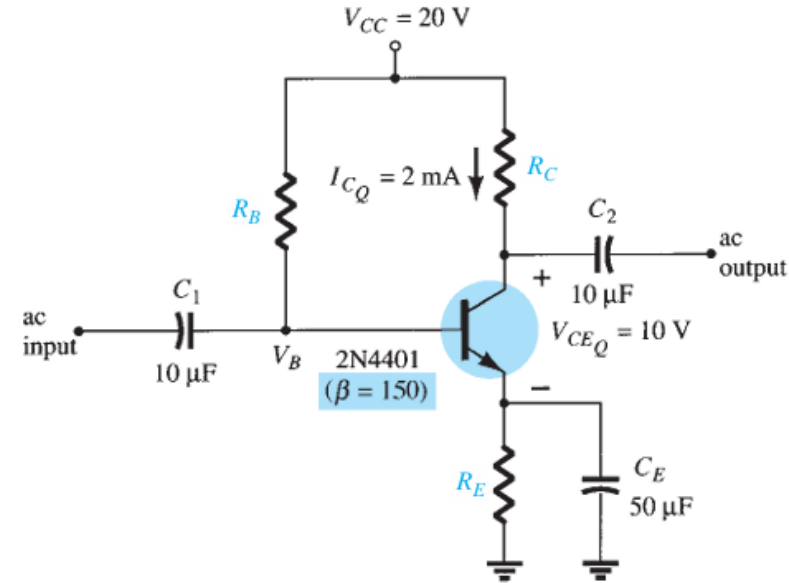


FIG. 4.62

Emitter-stabilized bias circuit for design consideration.

DESIGN OPERATIONS

- **EXAMPLE 4.25** Determine the levels of R_C , R_E , R_1 , and R_2 for the network of Fig. 4.63 for the operating point indicated.

➤ **Solution:**

$$V_E = \frac{1}{10}V_{CC} = \frac{1}{10}(20 \text{ V}) = 2 \text{ V}$$

$$R_E = \frac{V_E}{I_E} \cong \frac{V_E}{I_C} = \frac{2 \text{ V}}{10 \text{ mA}} = \mathbf{200 \text{ } \Omega}$$

$$R_C = \frac{V_{R_C}}{I_C} = \frac{V_{CC} - V_{CE} - V_E}{I_C} = \frac{20 \text{ V} - 8 \text{ V} - 2 \text{ V}}{10 \text{ mA}} = \frac{10 \text{ V}}{10 \text{ mA}} = \mathbf{1 \text{ k}\Omega}$$

$$V_B = V_{BE} + V_E = 0.7 \text{ V} + 2 \text{ V} = 2.7 \text{ V}$$

- **Using $\beta R_E \geq 10R_2$,**

R_2 calculation

$$R_2 \leq \frac{1}{10}\beta R_E$$

$$V_B = \frac{R_2}{R_1 + R_2} V_{CC}$$

$$R_2 \leq \frac{1}{10}(80)(0.2 \text{ k}\Omega) = \mathbf{1.6 \text{ k}\Omega}$$

R_1 calculation

$$V_B = 2.7 \text{ V} = \frac{(1.6 \text{ k}\Omega)(20 \text{ V})}{R_1 + 1.6 \text{ k}\Omega}$$

$$2.7R_1 + 4.32 \text{ k}\Omega = 32 \text{ k}\Omega$$

$$2.7R_1 = 27.68 \text{ k}\Omega$$

$$R_1 = \mathbf{10.25 \text{ k}\Omega} \quad (\text{use } 10 \text{ k}\Omega)$$

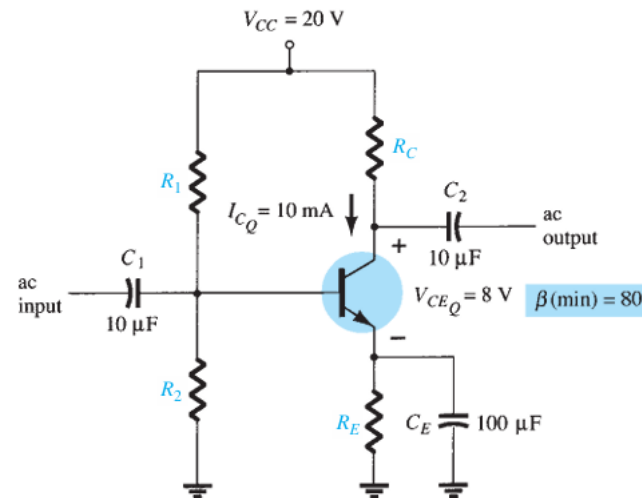


FIG. 4.63

Current-gain-stabilized circuit for design considerations.

Single stage CE Amplifier

- Fig shows a single stage CE amplifier. The different circuit elements and their functions are described as follows.

(i) Biasing circuit :

The resistances R_1 , R_2 and R_E form the biasing and stabilization circuit.

(ii) Input capacitance C_{in} :

This is used to couple the signal to the base of the transistor.

If this is not used, the signal source resistance will come across

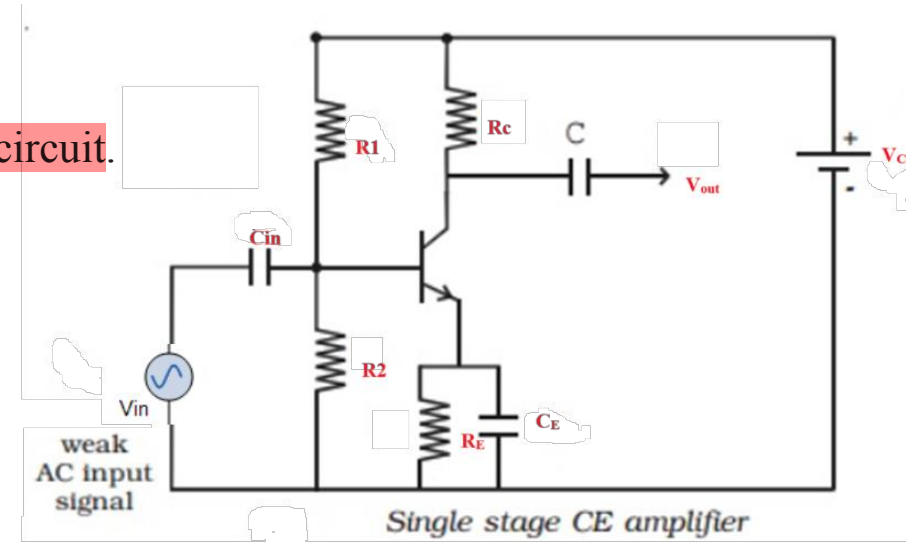
R_2 and thus changes the bias condition of the transistor.

The capacitor C_{in} allows only a.c. signal to flow.

(iii) Emitter bypass capacitor C_E :

This is connected in parallel with R_E to provide a low reactance path to the amplified a.c. signal.

If it is not used, then amplified a.c. signal flowing through R_E will cause a voltage drop across it, thereby shifting the output voltage.



Slide 1: Circuit Elements and Their Functions

The diagram depicts a single-stage CE amplifier, a fundamental transistor-based amplifier used to amplify weak AC signals.

1. Biasing Circuit (R_1 , R_2 , and R_E):

- These resistors form a voltage divider network to establish a stable operating point for the transistor.
- The biasing ensures the transistor operates in the **active region**, essential for proper amplification.

2. Input Capacitance (C_{in}):

- **Function:** Couples the input AC signal to the base of the transistor while blocking any DC component of the signal.
- **Why it's necessary:** If C_{in} is not used, the source resistance (e.g., a signal generator) could directly affect the bias conditions of the transistor, leading to improper operation.

3. Emitter Bypass Capacitor (C_E):

- **Function:** Provides a low-reactance path for amplified AC signals, bypassing the emitter resistor (R_E).
- **Why it's necessary:** If C_E is absent, the AC signal will cause a voltage drop across R_E , reducing the overall voltage gain.

Single stage CE Amplifier

(iv) Coupling capacitor C :

This is used to couple the amplified signal to the output device. This capacitor C allows only a.c. signal to flow.

(v) Working

- When a weak input a.c. signal is applied to the base of the transistor, a small base current flows.
- Due to transistor action, a much larger a.c. current flows through collector load R_C , a large voltage appears across R_C and hence at the output.
- Therefore, a weak signal applied to the base appears in amplified form at the collector of circuit.
- Voltage gain (A_v) of the amplifier is the ratio of the amplified output voltage to the input voltage.

Slide 2: Additional Components and Working

4. Coupling Capacitor (C):

- **Function:** Couples the amplified AC output signal to the next stage (e.g., a load or another amplifier) while blocking the DC component.
- **Why it's necessary:** Prevents any DC offset from affecting the connected circuit or device.

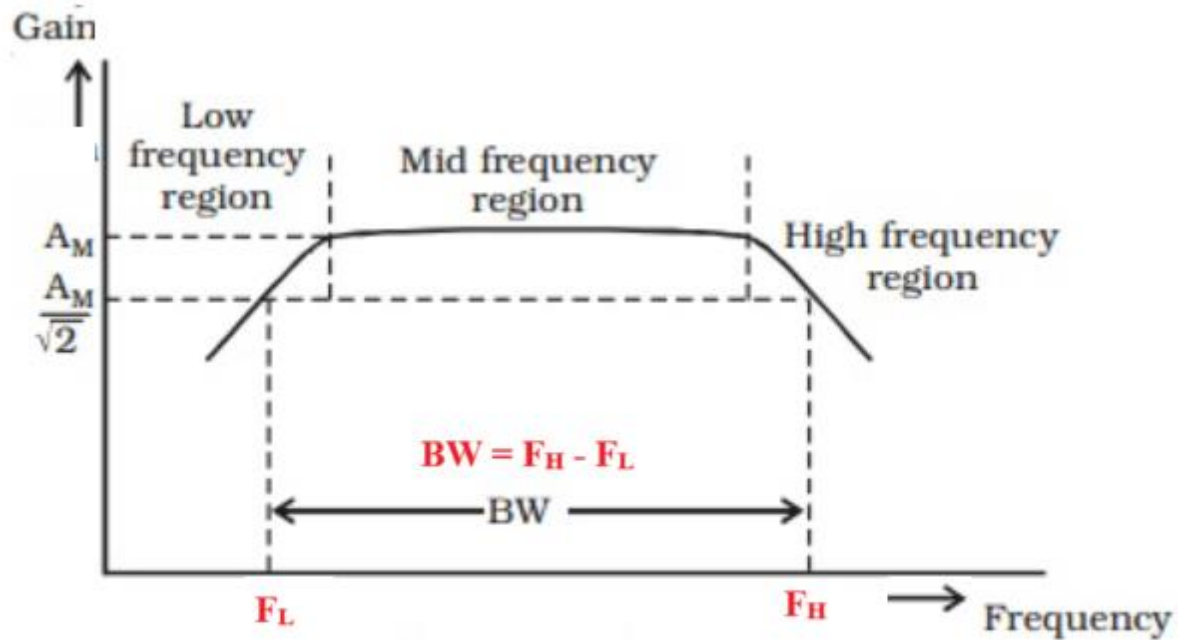
5. Working:

- **Input Signal:** A weak AC signal is applied to the base via C_{in} .
- **Amplification Process:**
 - The base current is small due to transistor action.
 - A much larger AC current flows through the collector load resistor (R_C) due to the transistor's amplification.
 - This causes a significant voltage drop across R_C , producing an amplified output signal.
- **Voltage Gain (A_V):**

$$A_V = \frac{\text{Amplified Output Voltage}}{\text{Input Voltage}}$$

Single stage CE Amplifier

(vi) *Frequency response and bandwidth*



Frequency response curve

Slide 3: Frequency Response and Bandwidth

The frequency response curve demonstrates how the amplifier's gain varies with the frequency of the input signal.

6. Frequency Response Regions:

- **Low-Frequency Region:**
 - The gain is lower due to the reactance of coupling and bypass capacitors being significant.
- **Mid-Frequency Region:**
 - The gain remains constant, as the capacitors provide negligible reactance, and the transistor operates efficiently.
- **High-Frequency Region:**
 - The gain decreases due to parasitic capacitances (e.g., junction capacitances) and other high-frequency effects in the transistor.

7. Bandwidth (BW):

- **Definition:** The range of frequencies over which the amplifier provides consistent gain.
- **Formula:**

$$BW = f_H - f_L$$

- f_H : Upper cutoff frequency where the gain starts to drop significantly at high frequencies.
- f_L : Lower cutoff frequency where the gain starts to drop at low frequencies.



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Unit 3 –Transistors and Operational Amplifiers

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playlist?
list=PLwjK_iyK4LLDDBB1
E9MFbxGCEnmMMOAX
OH](https://www.youtube.com/playlist?list=PLwjK_iyK4LLDDBB1E9MFbxGCEnmMMOAXOH)

Op-Amp yt Playlist

Operational Amplifier - Introduction

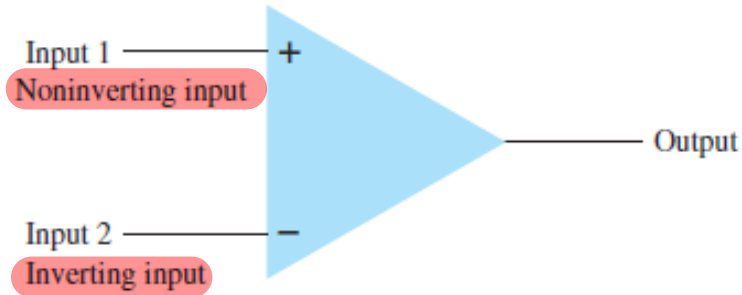


FIG. 10.1
Basic op-amp.

- ❖ Operational Amplifier or op-amp is a very high gain differential amplifier.
- ❖ It has very high input impedance. Ideally its value is infinity
- ❖ It has a low output impedance. Ideally its value is zero
- ❖ It finds applications in filters and oscillators

ELECTRONIC PRINCIPLES AND DEVICES

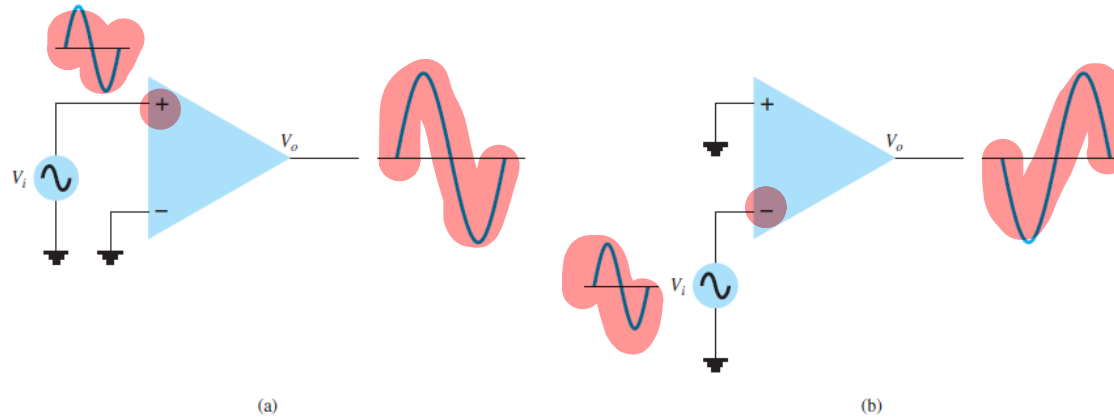


FIG. 10.2
Single-ended operation.

- ❖ **Single-ended input operation** results when the input signal is connected to one input with the other input connected to the ground.
- ❖ When the input signal is applied to the inverting input, the output is phase-shifted by 180°

ELECTRONIC PRINCIPLES AND DEVICES

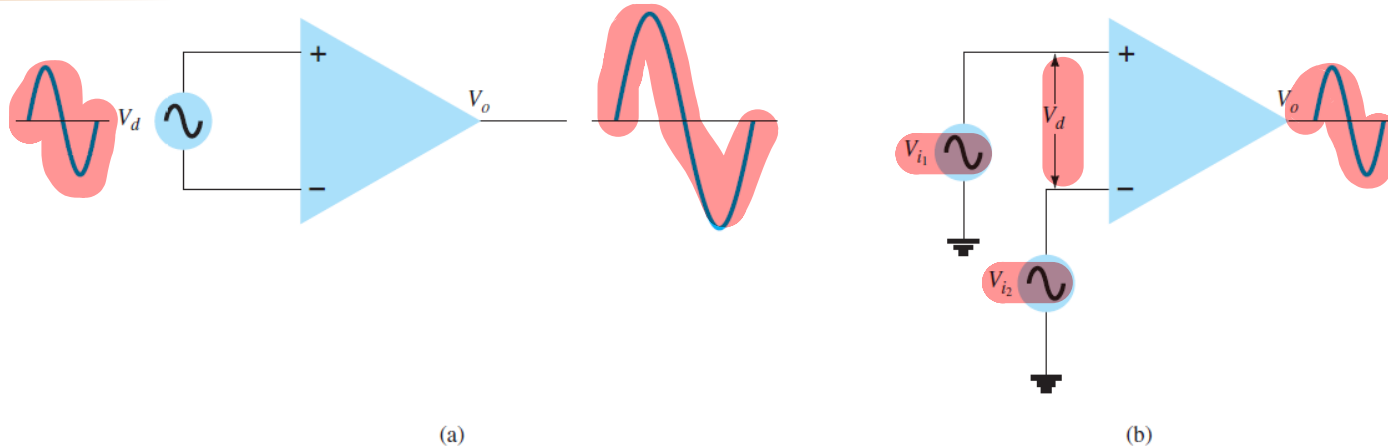


FIG. 10.3

Double-ended (differential) operation.

- ❖ Double-ended operation is one where an input, V_d is applied between the two input terminals (recall that neither input is at ground), with the resulting amplified output in phase with that applied between the inputs.
- ❖ Figure (b) shows the same action resulting when two separate signals are applied to the inputs, the difference signal being $V_{i1} - V_{i2}$

ELECTRONIC PRINCIPLES AND DEVICES

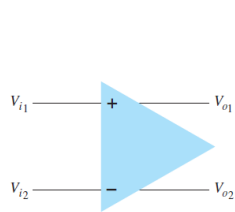


FIG. 10.4

Double-ended input with double-ended output.

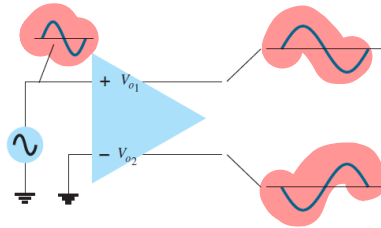


FIG. 10.5

Single-ended input with double-ended output.

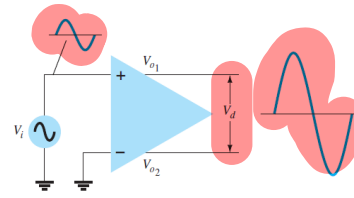


FIG. 10.6

Differential output.

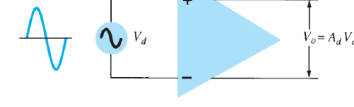


FIG. 10.7

Differential input, differential-output operation.

- ❖ **Double Ended Output:** An input applied to either input will result in outputs from both output terminals, these outputs always being opposite in polarity.
- ❖ Figure 10.5 shows a single-ended input with a double-ended output. As shown, the signal applied to the plus input results in two amplified outputs of opposite polarity.
- ❖ Figure 10.6 shows the same operation with a single output measured between output terminals. This difference output signal is $V_{o1} - V_{o2}$. The difference output is also referred to as a *floating signal* since neither output terminal is the ground (reference) terminal.
- ❖ Figure 10.7 shows a differential input, differential output operation. The input is applied between the two input terminals, and the output is taken from between the two output terminals. This is a fully differential operation.

Common Mode Operation

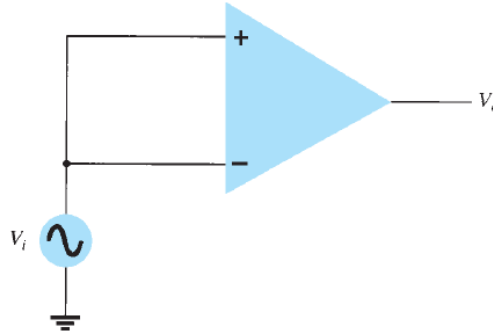


FIG. 10.8
Common-mode operation.

When the same input signal is applied to both inputs, common-mode operation results, as shown in Fig. Ideally, the two inputs are equally amplified, and since they result in opposite-polarity signals at the output, these signals cancel, resulting in 0-V output. Practically, a small output signal will result.

Basic Op-Amp

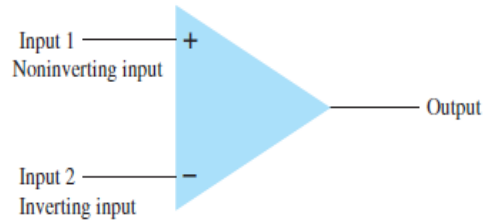
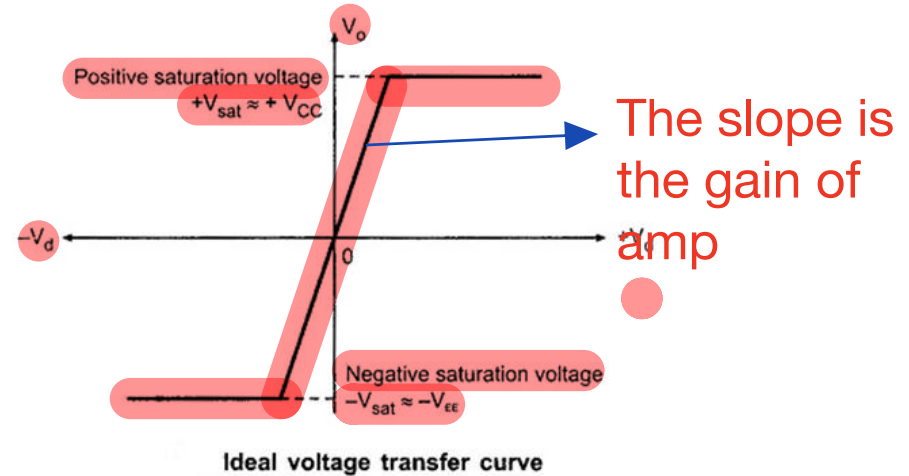


FIG. 10.1
Basic op-amp.

- ❖ Op-Amp has two inputs as shown in the figure
- ❖ Input1=Noninverting input
- ❖ Input2= Inverting input

Ideal Voltage Transfer Curve



- ❖ V_d = Differential Voltage
- ❖ $+V_{sat}$ = +Ve Saturation Voltage
- ❖ $-V_{sat}$ = - Ve Saturation Voltage
- ❖ It rises linearly between $+V_{sat}$ and $-V_{sat}$

ELECTRONIC PRINCIPLES AND DEVICES

Op-Amp Parameters

Input Offset Voltage: It is the differential DC voltage that must be applied between the input terminals (inverting and non-inverting) of the op-amp to make the output voltage zero. Ideally its value is 0v. Its value is 1 mv to 6mv for IC 741.

Output offset voltage: This refers to the small DC voltage that appears at its output even when the input terminals are shorted together and no input signal is applied. This offset is caused by internal mismatches in the op-amp, such as differences in the transistor parameters or imbalances in the internal circuitry. Ideally its value is 0v.

Input Resistance: The input resistance of an op-amp refers to the effective resistance seen at its input terminals. It is generally measured in open-loop condition. Ideally its value is infinity. Its value is $2\text{M}\Omega$ for IC 741 under open loop condition.

Output resistance: The output resistance of the op-amp is the resistance seen at the output terminal when the op-amp is configured in an open-loop condition. Ideally, its value is 0. Its value is 75Ω IC 741.

Op-Amp Parameters

Gain Bandwidth: The frequency at which the gain drops by 3 dB is known as the cutoff frequency f_c of the Op-Amp. The unity-gain frequency f_1 and cutoff frequency are related by

$$f_1 = A_{VD} f_c$$

Unity-gain frequency may also be called the gain–bandwidth product of the op-amp. Ideally its value is infinity. Its value is 1MHz for IC741

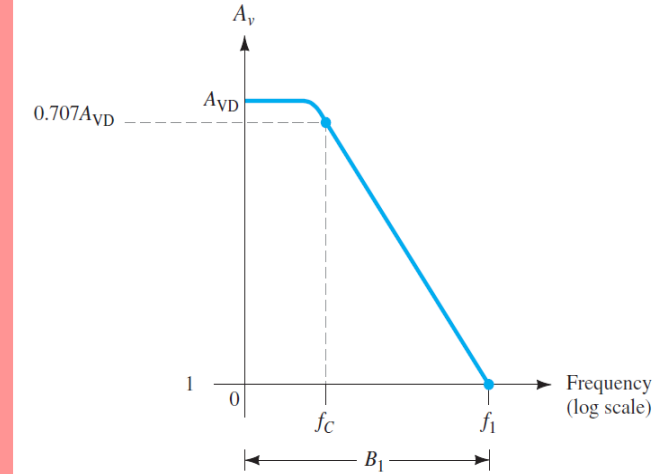


FIG. 10.47

Gain versus frequency plot.

1. Gain Bandwidth

- **Definition:** The product of the amplifier's bandwidth (f_c) and the gain (A_{VD}) at a particular point.
- **Significance:** Indicates the frequency range over which the op-amp can operate effectively at a specific gain.
- For a constant gain-bandwidth product, as the gain increases, the bandwidth decreases, and vice versa.

2. Cutoff Frequency (f_c)

- **Definition:** The frequency at which the gain of the amplifier drops by 3 dB from its maximum value.
- **3 dB Drop:** Corresponds to a gain value of approximately $0.707A_{VD}$.
- **Importance:** Marks the upper limit of the op-amp's effective operating range for the given gain.

3. Unity-Gain Frequency (f_1)

- **Definition:** The frequency at which the gain of the op-amp becomes unity ($A_V = 1$).
- **Relation:** The unity-gain frequency and cutoff frequency are related by the formula:

$$f_1 = A_{VD}f_c$$

- **Also Called:** Gain-bandwidth product.
- **Practical Value:** For the IC 741 op-amp, f_1 is typically 1 MHz.

5. Bandwidth (B_1)

- **Definition:** The range of frequencies over which the op-amp can provide amplification.
- **In this Image:** B_1 is shown as the range extending from low frequencies up to the cutoff frequency (f_c).

6. Gain-Frequency Plot (Log Scale)

- **X-Axis:** Frequency (in logarithmic scale).
- **Y-Axis:** Gain (A_V).
- **Key Features:**
 - Starts at a high gain value (A_{VD}).
 - Drops gradually with increasing frequency.
 - Hits the cutoff frequency (f_c) at $0.707A_{VD}$.
 - Continues dropping until it reaches unity gain at f_1 .

4. Gain (A_{VD})

- **Definition:** The ratio of the output voltage to the input voltage at a given frequency.
- **In this context:** A_{VD} is the amplifier's voltage gain at the cutoff frequency.

Op-Amp Parameters

Slew rate: The maximum rate at which amplifier output can change in volts per microsecond is known as **Slew rate**. The slew rate provides a parameter specifying the maximum rate of change of the output voltage when driven by a large step-input signal. Ideal value of slew rate is infinity. It is 0.5 V/ μ s for IC741

$$SR = \frac{\Delta V_o}{\Delta t} \text{ V}/\mu\text{s} \quad \text{with } t \text{ in } \mu\text{s}$$

Common-Mode Gain (A_{CM})

- **Definition:** The gain of an amplifier when the same input signal is applied to both input terminals (in-phase signals).
- Mathematically:

$$A_{CM} = \frac{V_{out}}{V_{CM}}$$

where V_{CM} is the common-mode input voltage (same for both terminals).

- **Ideal Case:** In an ideal differential amplifier, $A_{CM} = 0$, meaning no output should result from common-mode signals.
- **Practical Case:** Due to non-idealities like mismatched components, there is a small A_{CM} , but it should be minimized.

Differential-Mode Gain (A_{DM})

- **Definition:** The gain of an amplifier when the two input terminals receive opposite signals (differential input signals).
- Mathematically:

$$A_{DM} = \frac{V_{out}}{V_{DM}}$$

where $V_{DM} = V_1 - V_2$ is the difference between the two input voltages.

- **Ideal Case:** In an ideal differential amplifier, the output depends entirely on V_{DM} , and A_{DM} is maximized.
- **Practical Case:** A_{DM} is significantly larger than A_{CM} , making the amplifier suitable for differential signals.

Op-Amp Parameters

Common Mode Rejection Ratio: It is the ratio of differential gain to the common mode gain of the amplifier. Ideally, CMRR is infinity. Typically, its value is 90 dB for IC741.

$$CMRR = \frac{A_d}{A_c}$$

A_d = Differential gain of the amplifier

A_c = Common mode gain of the amplifier

The value of CMRR can also be expressed in logarithmic terms

$$CMRR(log) = 20 \log_{10} \left\{ \frac{A_d}{A_c} \right\} \text{ dB}$$

. Open Loop gain

Open loop gain is the gain of the Op Amp without a positive or negative feedback. An ideal OP Amp should have an infinite open loop gain but typically it range between 20,000 and 2, 00000.

Feedback in Operational Amplifiers (Op-Amps)

Feedback refers to the process of feeding a portion of the output signal back to the input of the op-amp. This feedback can influence the behavior and performance of the op-amp in a circuit, such as controlling gain, improving stability, or altering the input and output impedance.

Types of Feedback

1. Positive Feedback (+ Feedback):

- A portion of the output signal is fed back to the non-inverting input terminal (+).
- **Effect:**
 - Increases the overall gain.
 - Can lead to **oscillations** or instability in the circuit.
 - Often used in **oscillators** and circuits requiring signal generation.
- **Example Applications:**
 - Comparators
 - Schmitt triggers
 - Oscillators (e.g., sine wave or square wave generators)

2. Negative Feedback (- Feedback):

- A portion of the output signal is fed back to the inverting input terminal (-).
- **Effect:**
 - Reduces the overall gain but improves stability and accuracy.
 - Reduces non-linear distortions and enhances bandwidth.
 - Controls the closed-loop gain of the amplifier.
- **Example Applications:**
 - Amplifiers (voltage, current, and power amplifiers)
 - Filters
 - Voltage followers (buffer circuits)

Negative feedback in Op-amp:

Op-amps are normally used with negative feedback for the following reasons:

1. The open-loop gain of the op-amp is very high. It is typically 2×10^5 for IC 741. Such large gain is unsuitable for linear applications. Negative feedback decreases the gain and makes it suitable for linear applications such as an amplifier.
2. The open-loop gain is not stable and varies with temperature, supply voltage and frequency whereas the gain with feedback is stable.
3. The open-loop BW is very small making it unsuitable for any practical application. With negative BW increases.

ELECTRONIC PRINCIPLES AND DEVICES

Virtual Ground Concept

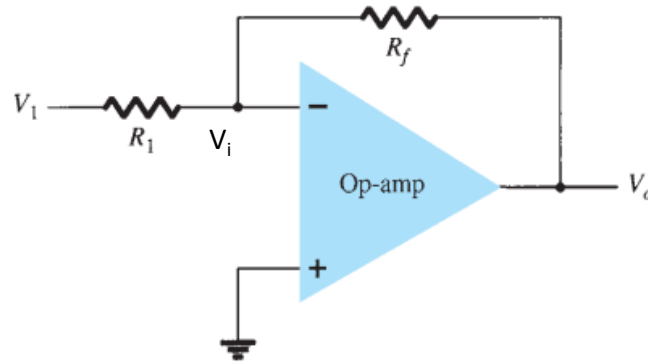
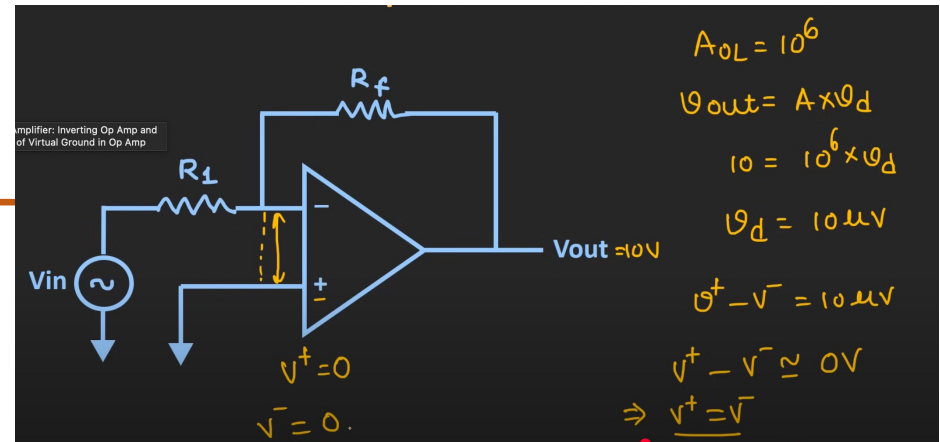


Fig. Basic Op-amp Connection

- The output voltage of an Op-amp is limited by the supply voltage V_1 as shown in the figure and the voltage gains are very high.
- For example, if $V_o = -10\text{ V}$ and gain $A_v = 20,000$, then the input voltage $V_i = -\frac{V_o}{A_v} = -\frac{10}{20,000} = 0.5\text{ mV}$.
- If the circuit has an overall gain of 1, then value of V_1 is 10V.
- Compared to all other input and output voltages, the value of V_i is very small and can be considered as 0V.
- The fact that $V_i \approx 0\text{ V}$ leads to the concept that at the amplifier input there exists a virtual short-circuit or virtual ground.



Virtual Ground Concept

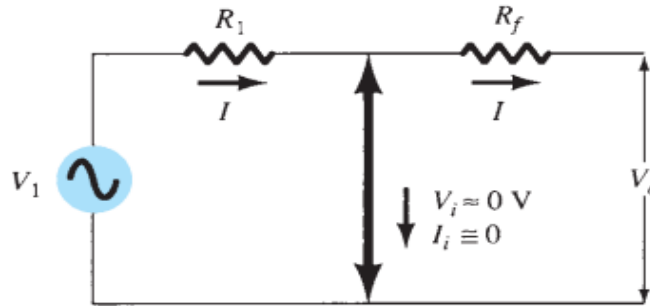


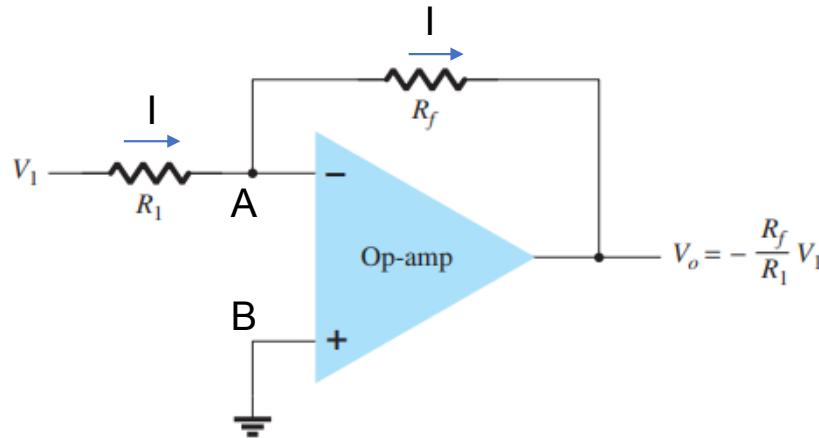
Fig. Virtual ground in Op-amp

- From above figure, Virtual short or Virtual ground implies that although the input voltage V_i is nearly 0V, there is no current through the amplifier input to ground.
- The heavy line in the figure is used to indicate that a short exists with $V_i \approx 0V$.
- current will not flow through short and ground and flows only through resistors R_1 and R_f as shown.
- Using the virtual ground concept, we can write equations for the current I as follows:

$$I = \frac{V_1}{R_1} = -\frac{V_0}{R_f}$$

Therefore, $\frac{V_0}{V_1} = -\frac{R_f}{R_1}$

1. INVERTING AMPLIFIER:



$V_A = V_B = 0$ (Virtual ground)

At Input, $I = (V_1 - V_A)/R_1 = V_1/R_1$ ----(1)

At Output, $I = (V_A - V_O)/R_f = -(V_O/R_f)$ ----(2)

Equating 1 and 2

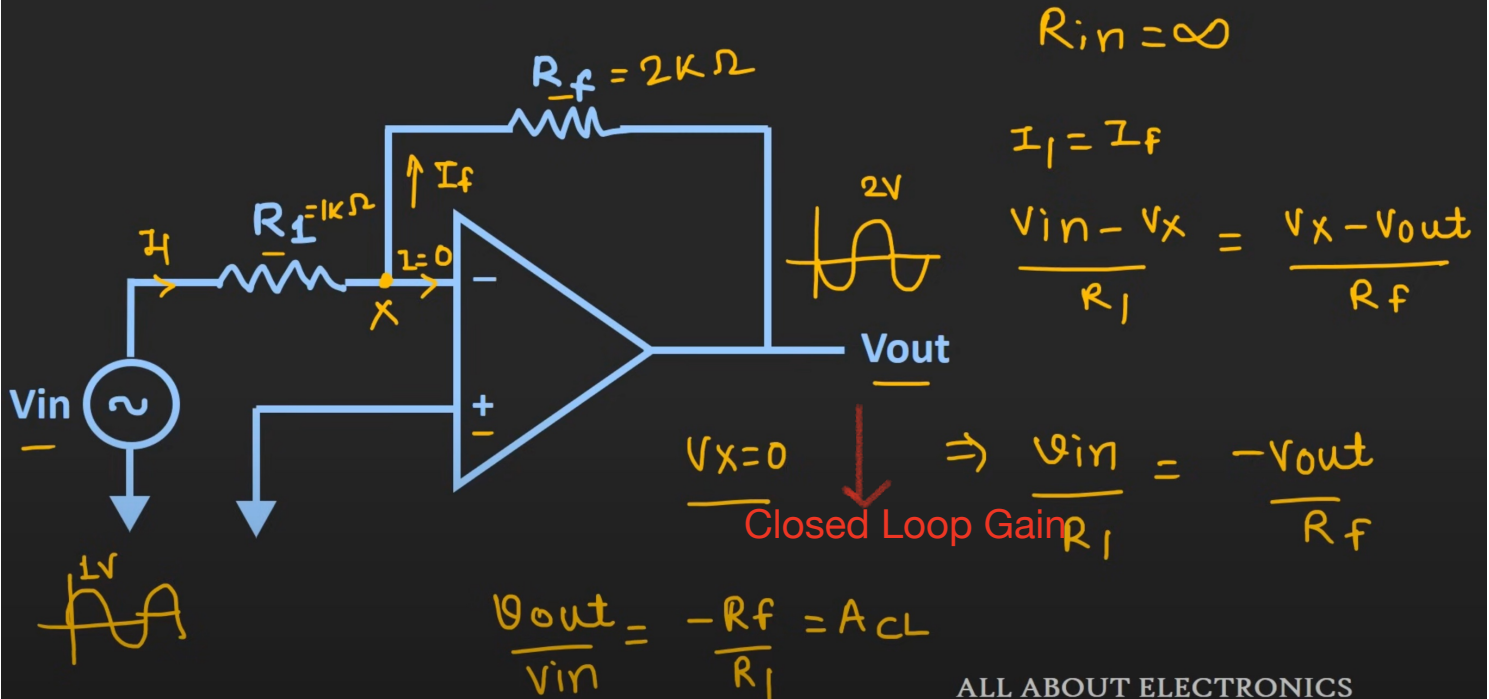
Output Voltage

$$V_O = -(R_f/R_1) * V_1$$

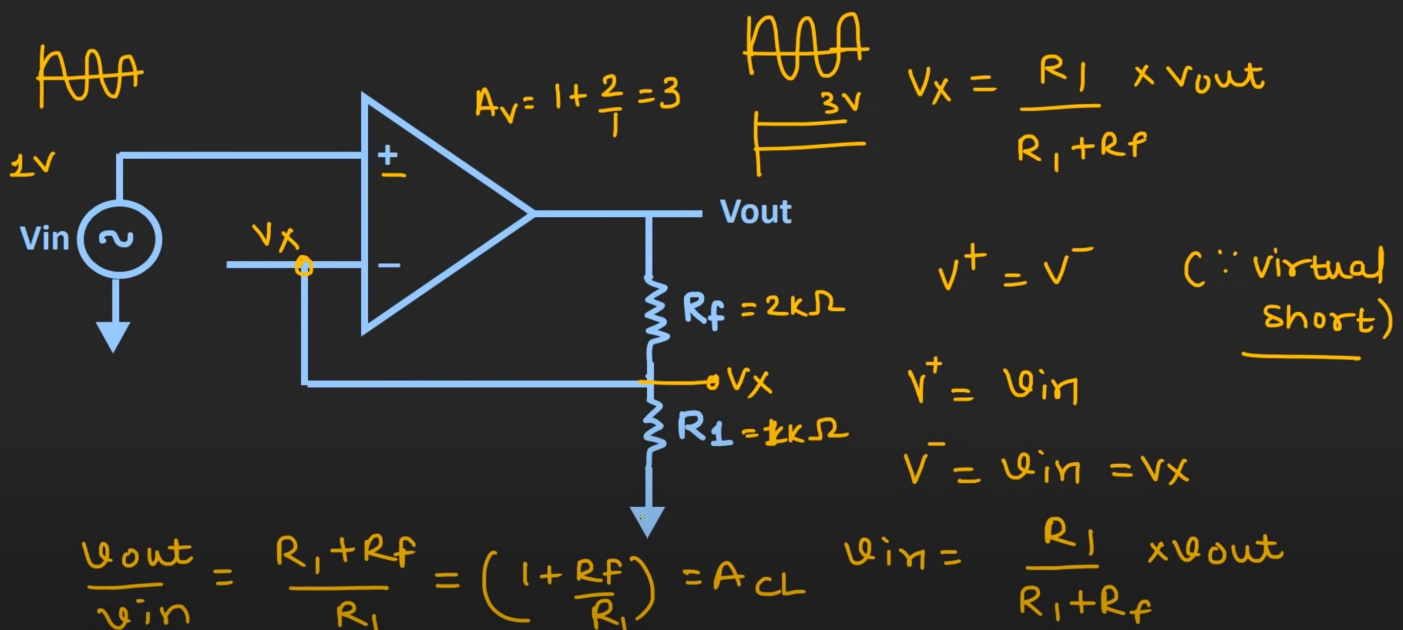
Where Gain is $-(R_f/R_1)$

- Input is given to inverting terminal of the amplifier.
- Inverting amplifier is the most widely used constant-gain amplifier circuit.
- The output is obtained by multiplying the input by a fixed or constant gain, set by the input resistor R_1 and feedback resistor R_f and output is being inverted from the input.
- Output Voltage of Inverting amplifier is $V_O = -\frac{R_f}{R_1} * V_1$ (from virtual ground concept)

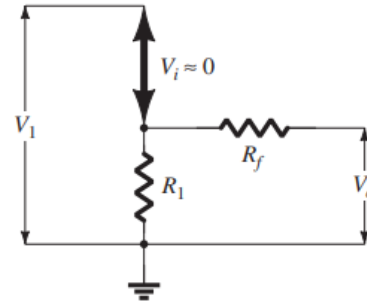
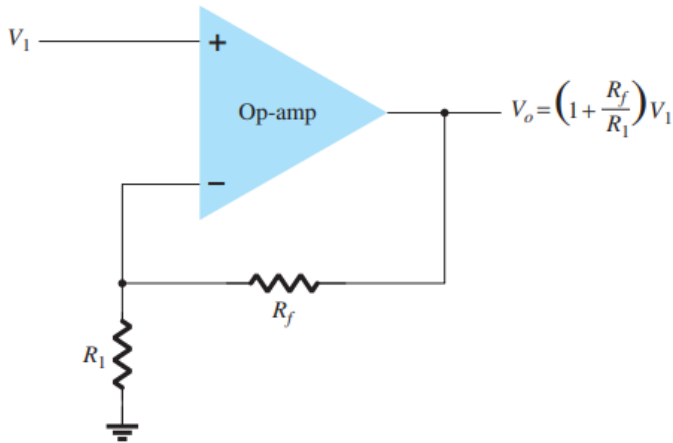
Inverting Operational Amplifier



Non-Inverting Operational Amplifier



2. NONINVERTING AMPLIFIER:

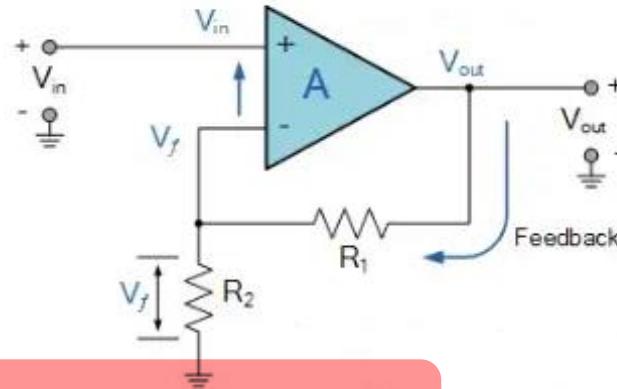


- Input is given to noninverting terminal of the op-amp.
- voltage across R_1 is V_1 (since $V_i = 0V$) and is equal to the output voltage, through a voltage divider of R_1 and R_f and is given by

$$V_1 = \frac{R_1}{R_1 + R_f} V_o \text{ or}$$

$$\frac{V_o}{V_1} = \frac{R_1 + R_f}{R_1} = 1 + \frac{R_f}{R_1}$$

Ideal OP-Amp Negative Feedback



$$\text{Closed loop Gain} = \frac{V_{out}}{V_{in}} = 1 + \frac{R_1}{R_2}$$

$$\text{Feed back } \beta = \frac{V_f}{V_{out}} = \frac{R_2}{R_1 + R_2}$$

3. UNITY FOLLOWER OR VOLTAGE FOLLOWER:

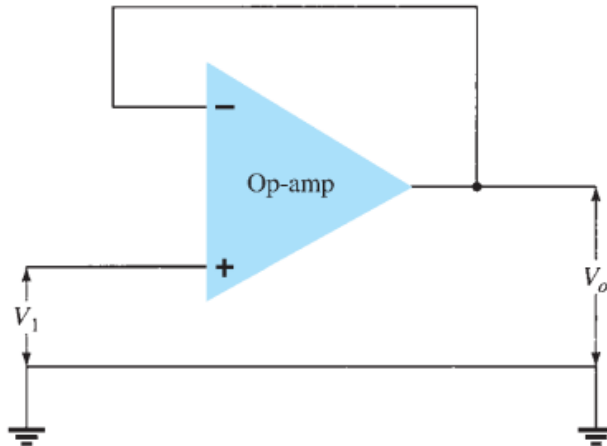


Fig a

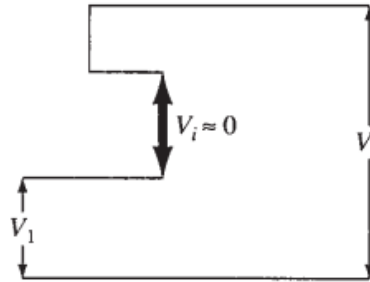


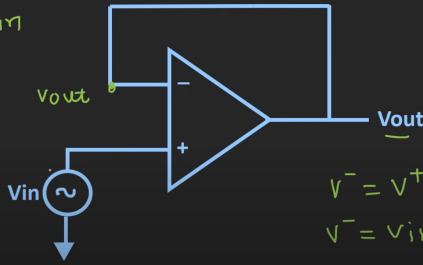
Fig b

- The unity-follower circuit provides a gain of unity with no polarity or phase reversal as shown in fig a.
- From the equivalent circuit as in fig b, $V_o = V_1$.
- It has a very high input resistance and a very low output resistance. It is used as a buffer circuit for impedance matching purposes.

Op-Amp as Buffer

$$Z_{in} = \infty$$

$$V_{out} = V_{in}$$



$$R_f = 0$$

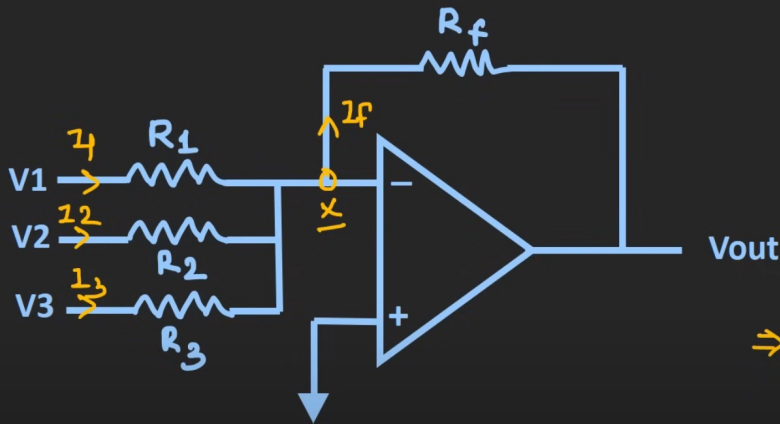
$$R_i = \infty$$

$$V^- = V^+$$

$$V^- = V_{in} = V_{out}$$

High input impedance in an op-amp is a desirable characteristic that ensures accurate amplification by minimizing loading effects and preserving the signal source's characteristics. This is particularly important when dealing with high-impedance signal sources or when precise signal amplification is required.

Inverting Summing Amplifier



Applying KCL

$$I_1 + I_2 + I_3 = I_f$$

$$\frac{V_1 - 0}{R_1} + \frac{V_2 - 0}{R_2} + \frac{V_3 - 0}{R_3} = \frac{0 - V_{out}}{R_f}$$

$$\Rightarrow \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = -\frac{V_{out}}{R_f}$$

$$\Rightarrow V_{out} = - \left[\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 \right]$$

$$R_1 \neq R_2 \neq R_3$$

Vout

$$\frac{R_f}{R_1} \neq \frac{R_f}{R_2} \neq \frac{R_f}{R_3}$$

$$V_{out} = - [A V_1 + B V_2 + C V_3]$$

4. SUMMING AMPLIFIER:

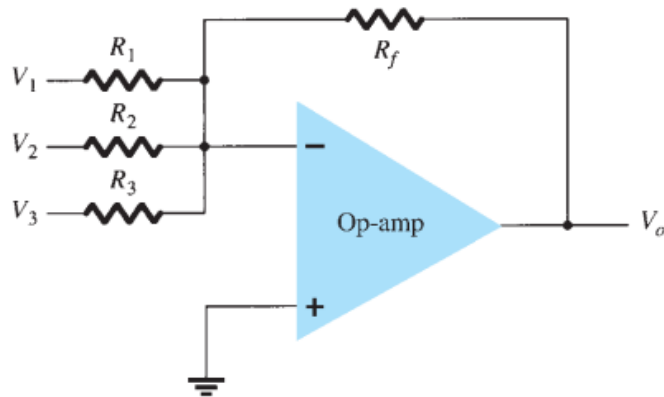


Fig a

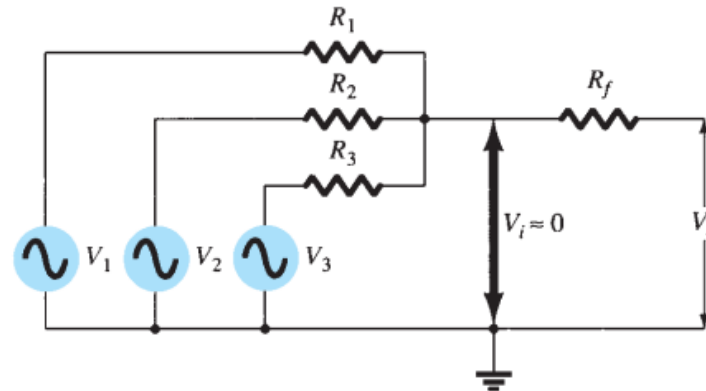
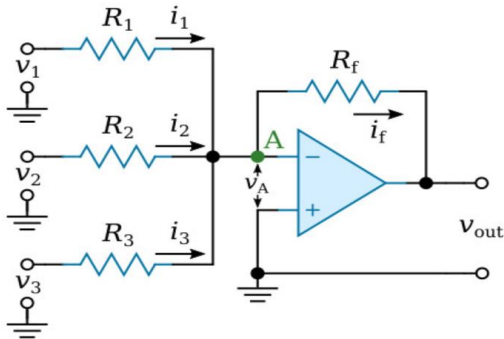


Fig b

- Above figure shows a three-input inverting summing amplifier circuit, which algebraically sums(adds) three voltages, each multiplied by a constant-gain(through resistors).
- If more inputs are used, they each add an additional component to the output.

OUTPUT OF A SUMMING AMPLIFIER:



Applying KCL at node A,

$$i_1 + i_2 + i_3 = i_f \quad \text{---(1)}$$

Where,

$$\begin{aligned} i_1 &= \frac{v_1 - v_A}{R_1} & i_3 &= \frac{v_3 - v_A}{R_3} \\ i_2 &= \frac{v_2 - v_A}{R_2} & i_f &= \frac{v_A - v_{out}}{R_f} \end{aligned} \quad \text{----(2)}$$

Substituting eq (2) in (1) and considering $v_A = 0$ (Virtual ground),

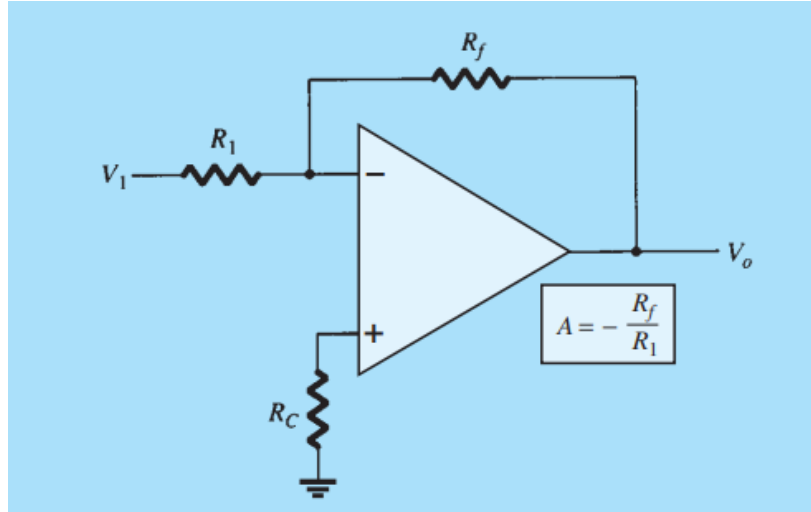
$$\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} = \frac{-v_{out}}{R_f} \quad \text{or}$$

$$v_{out} = -R_f \left(\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} \right)$$

5. CONSTANT GAIN/FIXED GAIN AMPLIFIER:

a. Inverting Constant gain Multiplier: Provides a precise gain of

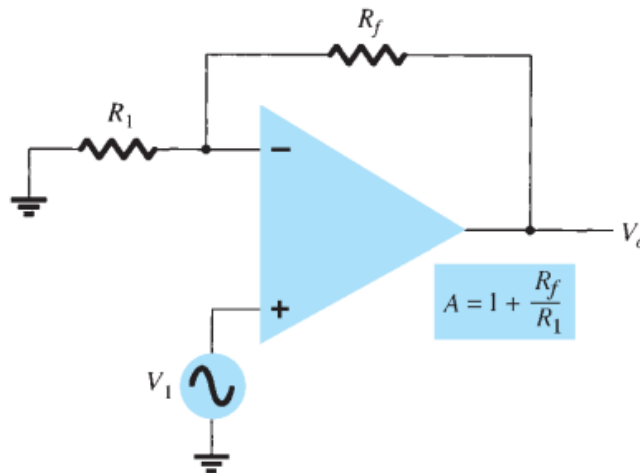
$$A = -\frac{R_f}{R_1}$$



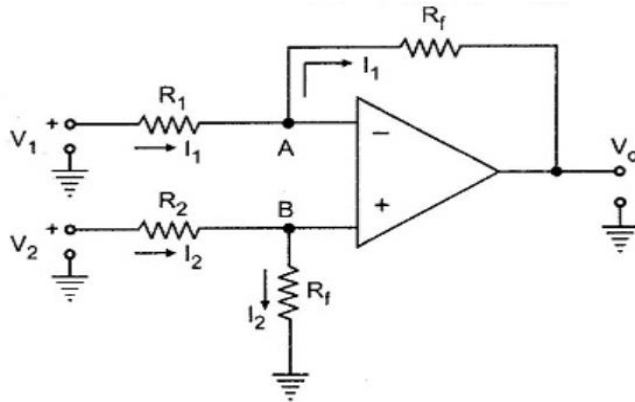
5. CONSTANT GAIN/FIXED AMPLIFIER:

b. Non-Inverting Constant gain Multiplier: Provides a precise gain of

$$A = 1 + \frac{R_f}{R_1}$$



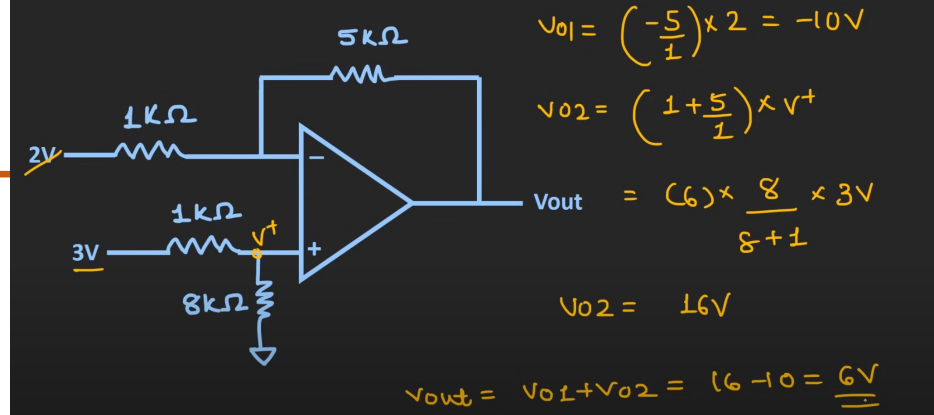
6. VOLTAGE SUBTRACTION:



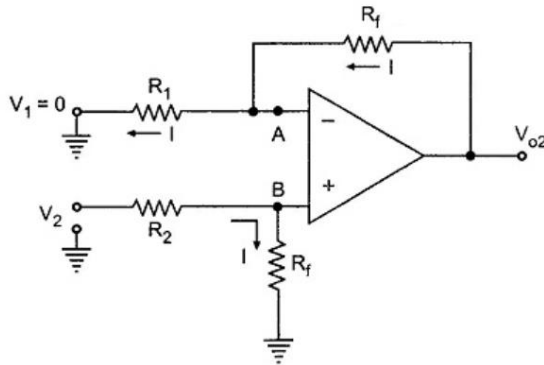
- Superposition principle is used to find the relation between output and input.
- Let V_{o1} be the output, with input V_1 , assuming V_2 to be zero. And V_{o2} be the output, with input V_2 , assuming V_1 to be zero.
- With V_2 zero, the circuit acts as an inverting amplifier and the output equation is

$$V_{o1} = -\frac{R_f}{R_1} V_1 \quad \text{-----(1)}$$

Example



While with V_1 as zero, the circuit reduces to as shown.



Let potential of node B be V_B . The potential of node A is same as B i.e. $V_A = V_B$ (Virtual short). Applying voltage divider rule to the input V_2 loop,

$$V_B = \frac{R_f}{R_2 + R_f} V_2 \quad \text{-----(2)}$$

$$I = \frac{V_A}{R_1} = \frac{V_B}{R_1} \quad \text{-----(3)}$$

$$I = \frac{V_{o2} - V_A}{R_f} = \frac{V_{o2} - V_B}{R_f} \quad \text{-----(4)}$$

Equating the equations (3) and (4),

Practical Op-amp Circuits

$$\frac{V_B}{R_1} = \frac{V_{o2} - V_B}{R_f}$$

$$V_{o2} = \frac{R_1 + R_f}{R_1} V_B$$

$$V_{o2} = \left[1 + \frac{R_f}{R_1} \right] V_B \quad \text{-----(5)}$$

Substituting V_B from (2) in (5) we get,

$$V_{o2} = \left[1 + \frac{R_f}{R_1} \right] \left[\frac{R_f}{R_2 + R_f} \right] V_2$$

Hence using Superposition principle,

$$V_o = V_{o1} + V_{o2}$$

$$= -\frac{R_f}{R_1} V_1 + \left[1 + \frac{R_f}{R_1} \right] \left[\frac{R_f}{R_2 + R_f} \right] V_2$$

If $R_1 = R_2$,

$$V_o = -\frac{R_f}{R_1} V_1 + \left[1 + \frac{R_f}{R_1}\right] \left[\frac{R_f}{R_1 + R_f}\right] V_2$$

$$= -\frac{R_f}{R_1} V_1 + \frac{R_f}{R_1} V_2$$

$$V_o = +\frac{R_f}{R_1} (V_2 - V_1)$$

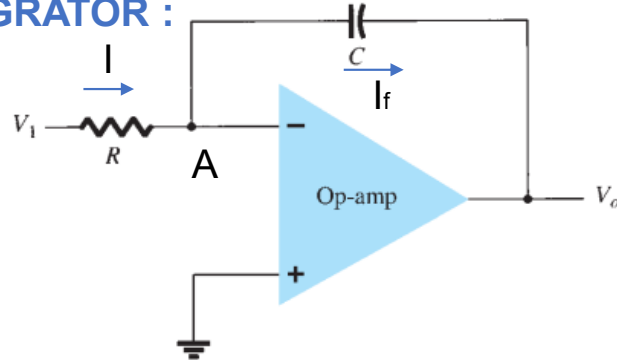
- The output voltage is proportional to the difference between the two input voltages. Thus it acts as a Subtractor using Op Amp circuit or difference amplifier.

If $R_1 = R_2 = R_f$ is selected, then $V_o = V_2 - V_1$

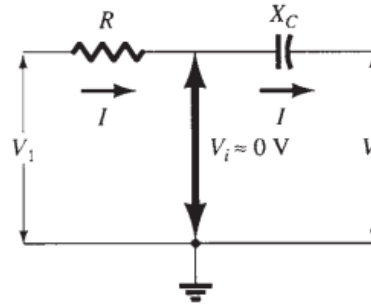
ELECTRONIC PRINCIPLES AND DEVICES

Practical Op-amp Circuits

7. INTEGRATOR :



Output



Applying KCL at node A, $I = I_f$ ----(1)

Where $I = (V_1 - V_A)/R$ and $I_f = C \cdot d(V_A - V_o)/dt$ ----(2)

($V_A = 0$ due to virtual ground property)

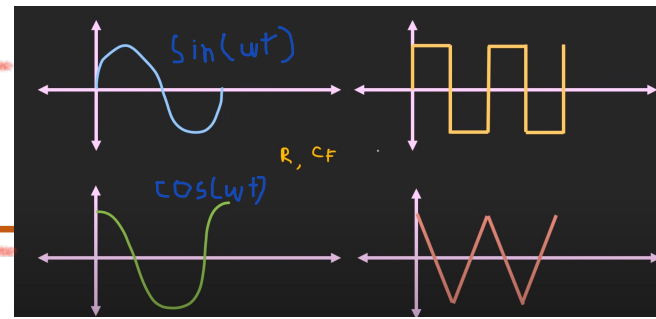
Substituting 2 in 1,

$$\frac{V_1 - 0}{R} = C \frac{d(0 - V_o)}{dt} \quad \text{or} \quad \frac{dV_o}{dt} = -\frac{1}{RC} V_1$$

Slope

$$V_{out}(t) = -\frac{1}{RC} \int_0^t V_{in}(t) dt$$

Where Scale factor is $1/RC$



- Integrator has capacitor as a feedback element.
- Integration is summing the area under a waveform or a curve over a period of time.
- If a fixed voltage is applied in the form of a square wave as an input to an integrator circuit, the output voltage grows over a period of time, resulting in a ramp voltage.

Scaling Factor:

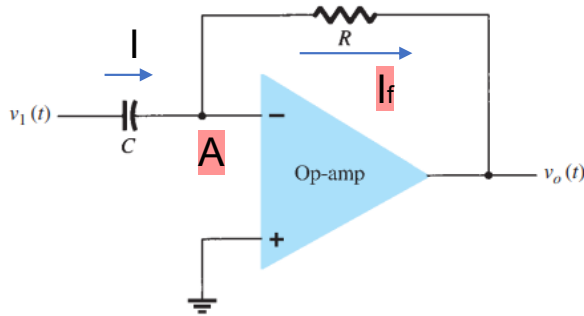
The scaling factor of the integrator circuit is:

$$\frac{1}{RC}$$

This factor determines:

1. **Integration Speed:** A larger RC product results in a slower response (smaller scaling factor).
2. **Amplitude of V_{out} :** The scaling factor directly affects how much the integral is amplified or attenuated.

8. DIFFERENTIATOR :



- A differentiator op amp circuit produces an output signal proportional to the input signal's rate of change.
- Involves an inverting amplifier with a capacitor at the input terminal.

Applying KCL at node A, $I = I_f$ ----(1)

Where $I = C \cdot d(V_i - V_A)/dt$ and $I_f = (V_A - V_0)/R$ ----(2)

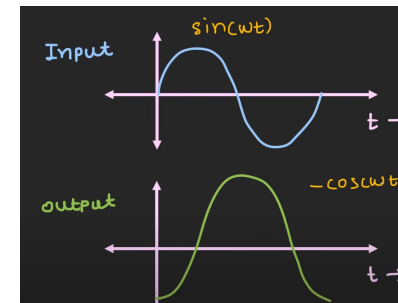
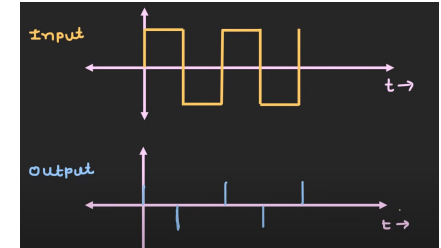
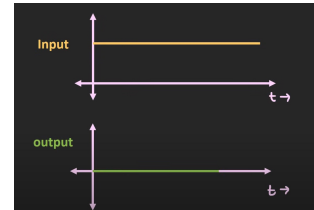
($V_A = 0$ due to virtual ground property)

Substituting 2 in 1,

$$C \frac{dv_i}{dt} = -\frac{v_0}{R}$$

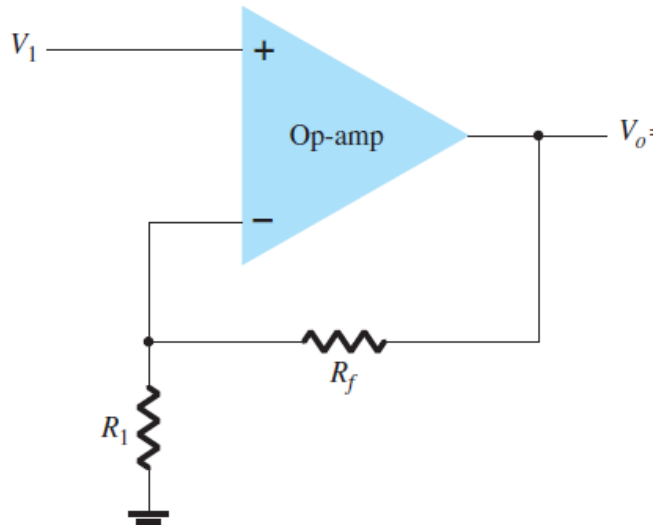
$$v_0 = -RC \frac{dv_i}{dt}$$

Where the scale factor is $-RC$



Numerical-1

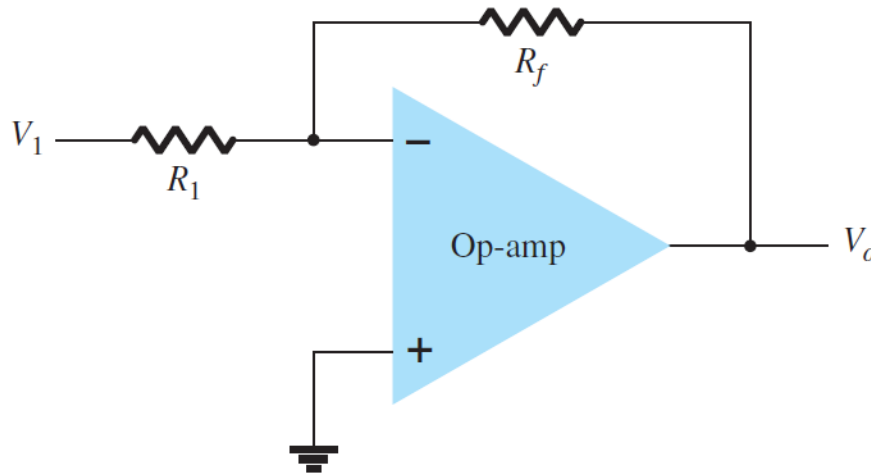
Calculate the output voltage of a noninverting amplifier shown in figure



$$V_o = \left(1 + \frac{R_f}{R_1}\right)V_1 = \left(1 + \frac{500 \text{ k}\Omega}{100 \text{ k}\Omega}\right)(2 \text{ V}) = 6(2 \text{ V}) = +12 \text{ V}$$

Numerical-2

If the circuit of the figure has $R_1=100\text{K Ohms}$ and $R_f=500\text{K Ohms}$,
What is the output voltage if $V_1= 2\text{V}$?



$$V_o = -\frac{R_f}{R_1} V_1 = -\frac{500 \text{ k}\Omega}{100 \text{ k}\Omega} (2 \text{ V}) = -10 \text{ V}$$

Numerical-3

3. What is the Output voltage in the Circuit shown in Figure

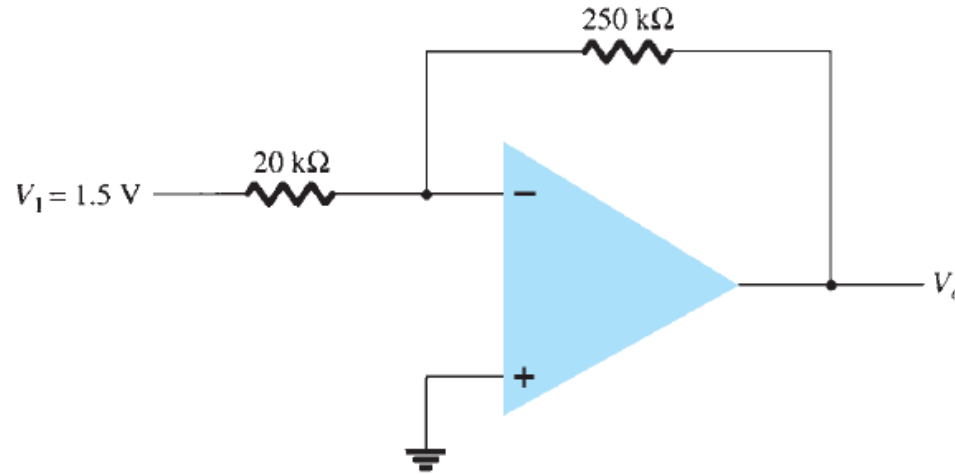


FIG. 10.62

$$V_o = -\frac{R_F}{R_1} V_1 = -\frac{250 \text{ k}\Omega}{20 \text{ k}\Omega} (1.5 \text{ V}) = -18.75 \text{ V}$$

ELECTRONIC PRINCIPLES AND DEVICES

Numerical-4

What is the range of voltage gain adjustment in the circuit of the figure shown

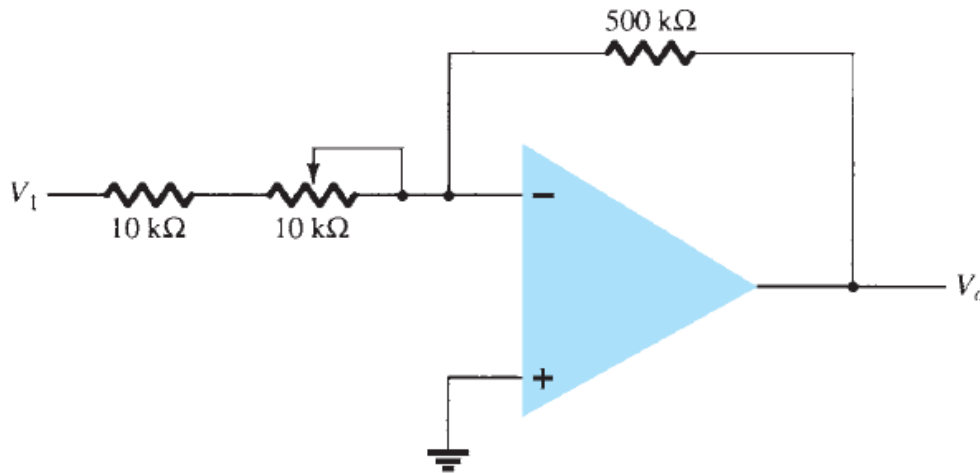


FIG. 10.63

$$A_v = \frac{V_o}{V_i} = -\frac{R_F}{R_1}$$

For $R_1 = 10\text{ k}\Omega$:

$$A_v = -\frac{500\text{ k}\Omega}{10\text{ k}\Omega} = -50$$

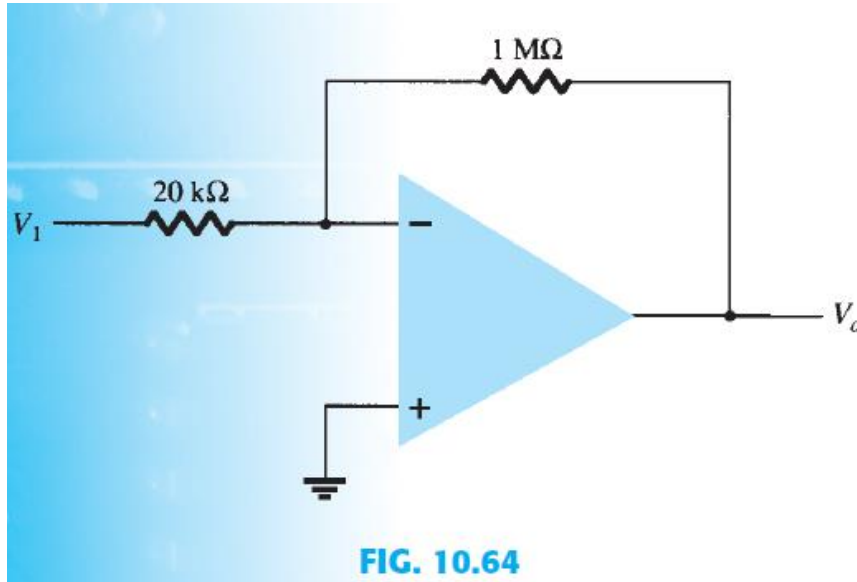
For $R_1 = 20\text{ k}\Omega$:

$$A_v = -\frac{500\text{ k}\Omega}{20\text{ k}\Omega} = -25$$

ELECTRONIC PRINCIPLES AND DEVICES

Numerical-5

What is the input voltage that results in an output of 2V in circuit shown in Figure



$$V_o = -\frac{R_f}{R_1} V_1 = -\left(\frac{1 \text{ M}\Omega}{20 \text{ k}\Omega}\right) V_1 = 2 \text{ V}$$
$$V_1 = \frac{2 \text{ V}}{-50} = -40 \text{ mV}$$

ELECTRONIC PRINCIPLES AND DEVICES

Numerical-6

What is the range of output voltage in the circuit shown in the figure if the input can vary from 0.1 to 0.5 V?

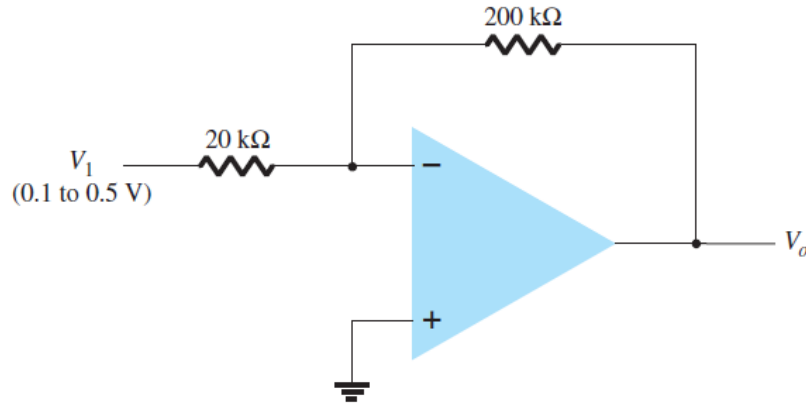


FIG. 10.65

$$V_o = -\frac{R_F}{R_1} V_1 = -\frac{200 \text{ k}\Omega}{20 \text{ k}\Omega} V_1 = -10 V_1$$

For $V_1 = 0.1 \text{ V}$:

$$V_o = -10(0.1 \text{ V}) = -1 \text{ V}$$

For $V_1 = 0.5 \text{ V}$:

$$V_o = -10(0.5 \text{ V}) = -5 \text{ V}$$

} V_o ranges from -1 V to -5 V

ELECTRONIC PRINCIPLES AND DEVICES

Numerical-7

What is the range of output voltage developed in the circuit shown in the figure

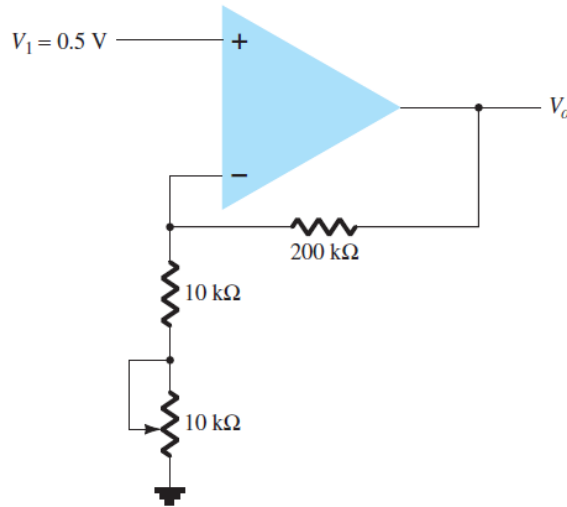


FIG. 10.67
Problem 7.

$$V_o = \left(1 + \frac{R_F}{R_1}\right) V_1$$

For $R_1 = 10\text{ k}\Omega$:

$$V_o = \left(1 + \frac{200\text{ k}\Omega}{10\text{ k}\Omega}\right) (0.5\text{ V}) = 21(0.5\text{ V}) = \mathbf{10.5\text{ V}}$$

For $R_1 = 20\text{ k}\Omega$:

$$V_o = \left(1 + \frac{200\text{ k}\Omega}{20\text{ k}\Omega}\right) (0.5\text{ V}) = 11(0.5\text{ V}) = \mathbf{5.5\text{ V}}$$

V_o ranges from 5.5 V to 10.5 V.

Numerical-8

What is the results in the circuit shown if the input for an input of $V_1 = -0.3 \text{ V}$?

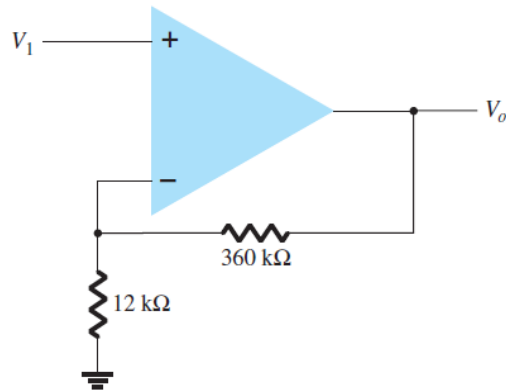


FIG. 10.66

$$\begin{aligned} V_o &= \left(1 + \frac{R_F}{R_1} \right) V_1 = \left(1 + \frac{360 \text{ k}\Omega}{12 \text{ k}\Omega} \right) (-0.3 \text{ V}) \\ &= 31(-0.3 \text{ V}) = -9.3 \text{ V} \end{aligned}$$

ELECTRONIC PRINCIPLES AND DEVICES

Numerical-9

What input must be applied to the input of the circuit shown in the figure to result in an output of 2.4 V?

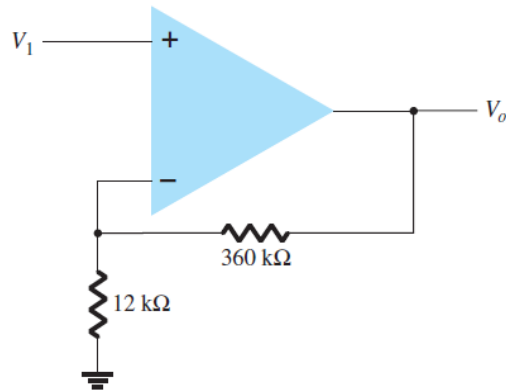


FIG. 10.66

$$V_o = \left(1 + \frac{R_F}{R_1}\right) V_1 = \left(1 + \frac{360 \text{ k}\Omega}{12 \text{ k}\Omega}\right) V_1 = 2.4 \text{ V}$$
$$V_1 = \frac{2.4 \text{ V}}{31} = 77.42 \text{ mV}$$

ELECTRONIC PRINCIPLES AND DEVICES

Numerical-10

What output voltage results in the circuit shown in the figure for $V_1 = +0.5\text{V}$?

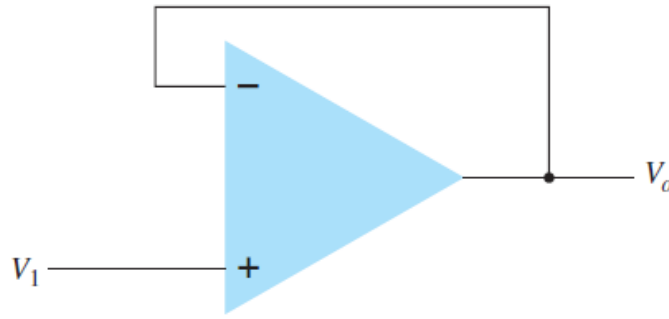


FIG. 10.70

$$V_o = V_1 = +0.5 \text{ V}$$

Practical Op-amp Circuits

Numerical-11

Calculate the output voltage of a noninverting amplifier for values of $V_1 = 2\text{V}$, $R_f = 500\text{ k}\Omega$, and $R_1 = 100\text{ k}\Omega$.

$$\text{Soln: } V_o = \left(1 + \frac{R_f}{R_1}\right)V_1 = \left(1 + \frac{500\text{ k}\Omega}{100\text{ k}\Omega}\right)(2\text{ V}) = 6(2\text{ V}) = +12\text{ V}$$

Numerical-12

Calculate the output voltage of an op-amp summing amplifier for the following sets of voltages and resistors. Use $R_f = 1\text{ M}$ in all cases.

a. $V_1 = +1\text{ V}$, $V_2 = +2\text{ V}$, $V_3 = +3\text{ V}$, $R_1 = 500\text{ k}$, $R_2 = 1\text{ M}$, $R_3 = 1\text{ M}$.

b. $V_1 = -2\text{ V}$, $V_2 = +3\text{ V}$, $V_3 = +1\text{ V}$, $R_1 = 200\text{ k}$, $R_2 = 500\text{ k}$, $R_3 = 1\text{ M}$.

Soln: Using Summing Amplifier Equation,

$$\text{a. } V_o = - \left[\frac{1000\text{ k}\Omega}{500\text{ k}\Omega} (+1\text{ V}) + \frac{1000\text{ k}\Omega}{1000\text{ k}\Omega} (+2\text{ V}) + \frac{1000\text{ k}\Omega}{1000\text{ k}\Omega} (+3\text{ V}) \right]$$

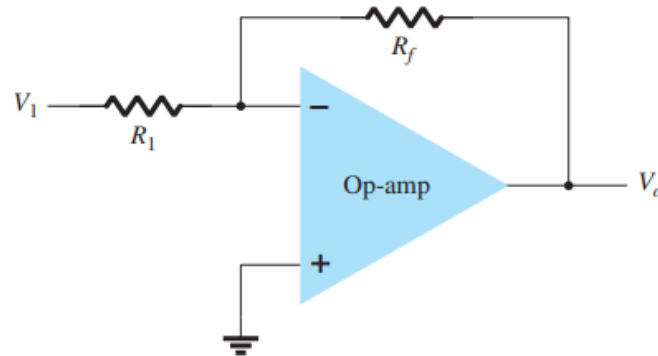
$$= -[2(1\text{ V}) + 1(2\text{ V}) + 1(3\text{ V})] = -7\text{ V}$$

$$\text{b. } V_o = - \left[\frac{1000\text{ k}\Omega}{200\text{ k}\Omega} (-2\text{ V}) + \frac{1000\text{ k}\Omega}{500\text{ k}\Omega} (+3\text{ V}) + \frac{1000\text{ k}\Omega}{1000\text{ k}\Omega} (+1\text{ V}) \right]$$

$$= -[5(-2\text{ V}) + 2(3\text{ V}) + 1(1\text{ V})] = +3\text{ V}$$

Numerical-13

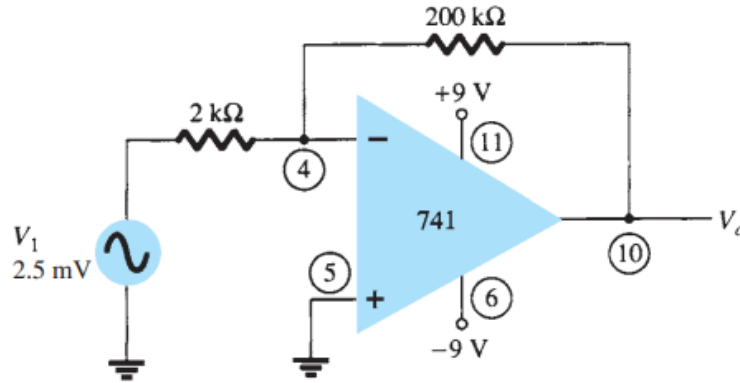
For the circuit shown, $R_1 = 100 \text{ k}$ and $R_f = 500 \text{ k}$, what output voltage results for an input of $V_1 = 2 \text{ V}$?



Soln:
$$V_o = -\frac{R_f}{R_1} V_1 = -\frac{500 \text{ k}\Omega}{100 \text{ k}\Omega} (2 \text{ V}) = -10 \text{ V}$$

Numerical-14

Determine the output voltage for the circuit of Fig with a sinusoidal input of 2.5 mV.



Soln:

$$\text{Gain } A = -\frac{R_f}{R_1} = -\frac{200 \text{ k}\Omega}{2 \text{ k}\Omega} = -100$$

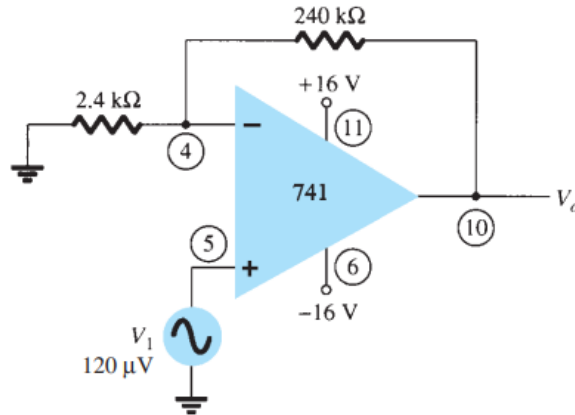
The output voltage is then

$$V_o = AV_i = -100(2.5 \text{ mV}) = -250 \text{ mV} = -0.25 \text{ V}$$

Practical Op-amp Circuits

Numerical-15

Calculate the output voltage from the circuit of Fig for an input of $120 \mu\text{V}$.



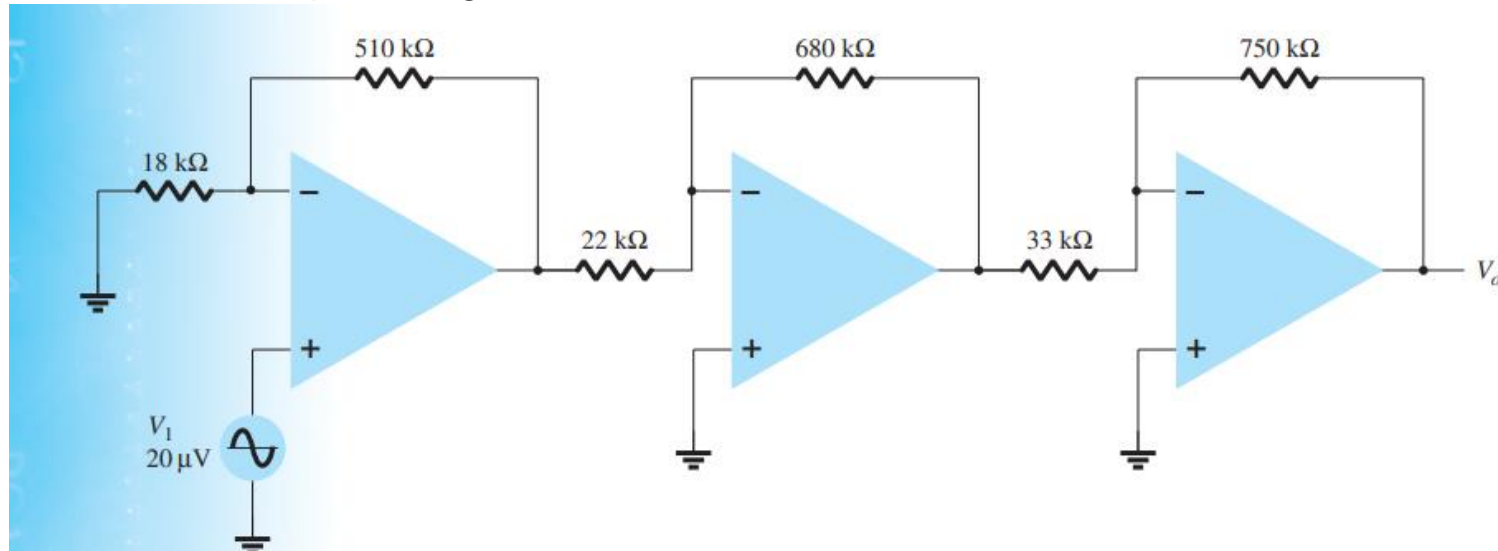
Soln: Gain $A = 1 + \frac{R_f}{R_1} = 1 + \frac{240 \text{ k}\Omega}{2.4 \text{ k}\Omega} = 1 + 100 = 101$

The output voltage is then

$$V_o = AV_i = 101(120 \mu\text{V}) = \mathbf{12.12 \text{ mV}}$$

Numerical-16

Calculate the output voltage in the circuit.

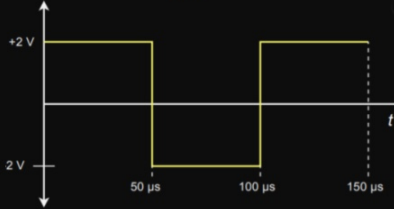


Soln:

$$\begin{aligned} V_o &= \left(1 + \frac{510 \text{ k}\Omega}{18 \text{ k}\Omega} \right) (20 \text{ }\mu\text{V}) \left[-\frac{680 \text{ k}\Omega}{22 \text{ k}\Omega} \right] \left[-\frac{750 \text{ k}\Omega}{33 \text{ k}\Omega} \right] \\ &= (29.33)(-30.91)(-22.73)(20 \text{ }\mu\text{V}) \\ &= \mathbf{412 \text{ mV}} \end{aligned}$$

Numerical-17

Sketch the output of the integrator circuit if the input signal is a 10 kHz, 2 V peak square wave. Assume $R=10\text{K}\Omega$ and $C=10\text{nF}$.



Soln:

$$V_{in}(t) = 2 \text{ from } t = 0, \text{ to } t = 50\mu\text{s}$$

$$V_{in}(t) = -2 \text{ from } t = 50\mu\text{s}, \text{ to } t = 100\mu\text{s}$$

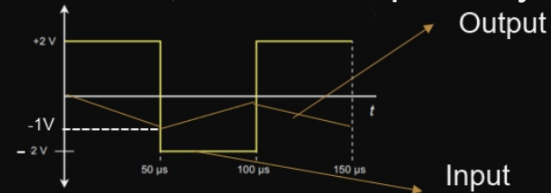
$$V_{out}(t) = -\frac{1}{RC} \int_0^t V_{in}(t) dt + C \quad \text{where } C \text{ is constant of integration-----(1)}$$

Substituting $R=10\text{K}\Omega$, $C=10\text{nF}$ and $V_{in}(t)=2$ in eq (1), $V_{out}(t) = -20000t + C_1$. hence At $t=0$, $V_{out}=C_1=0$ and at $t=50\mu\text{s}$, $V_{out}=-1\text{V}$

Similarly Substituting $R=10\text{K}\Omega$, $C=10\text{nF}$ and $V_{in}(t)=-2$ in eq (1) from limit $t=50\mu\text{s}$ to t , at $t=50\mu\text{s}$, $V_{out}=20000(t-50\mu\text{s}) + C_2 = V_{out}=20000(50\mu\text{s}-50\mu\text{s}) + (-1) = -1$ and at $t=100\mu\text{s}$, $V_{out}=20000(100\mu\text{s}-50\mu\text{s}) + C_3 = 0\text{V}$.

Note:

C_1, C_2 and C_3 are constant of integration and their initial values are 0, -1 and -1 respectively



2. For $t = 0 \mu s$ to $t = 50 \mu s$:

- $V_{in}(t) = 2 \text{ V}$.
- Substituting into the integration formula:

$$V_{out}(t) = -\frac{1}{RC} \int_0^t 2 dt + C_1$$

- Evaluate the integral:

$$V_{out}(t) = -\frac{1}{10,000 \cdot 10 \times 10^{-9}} \cdot (2 \cdot t) + C_1$$

$$V_{out}(t) = -20,000 \cdot t + C_1$$

- At $t = 0$, assume the initial condition $V_{out}(0) = 0$, so $C_1 = 0$.
- At $t = 50 \mu s$:

$$V_{out}(50 \mu s) = -20,000 \cdot 50 \cdot 10^{-6} + 0$$

$$V_{out}(50 \mu s) = -1 \text{ V}.$$

3. For $t = 50 \mu s$ to $t = 100 \mu s$:

- $V_{in}(t) = -2 \text{ V}$.
- Substituting into the integration formula:

$$V_{out}(t) = -\frac{1}{RC} \int_{50 \mu s}^t (-2) dt + C_2$$

- Evaluate the integral:

$$V_{out}(t) = -(-20,000) \cdot (t - 50 \mu s) + C_2$$

$$V_{out}(t) = 20,000 \cdot (t - 50 \mu s) + C_2$$

- At $t = 50 \mu s$, the previous output is $V_{out}(50 \mu s) = -1 \text{ V}$, so:

$$-1 = 20,000 \cdot (50 \mu s - 50 \mu s) + C_2$$

$$C_2 = -1 \text{ V}.$$

- At $t = 100 \mu s$:

$$V_{out}(100 \mu s) = 20,000 \cdot (100 \cdot 10^{-6} - 50 \cdot 10^{-6}) - 1$$

$$V_{out}(100 \mu s) = 20,000 \cdot 50 \cdot 10^{-6} - 1$$

$$V_{out}(100 \mu s) = 0 \text{ V}.$$

4. Output Waveform:

The output waveform alternates between a triangular shape:

- Decreasing from 0 V to -1 V (for $t = 0 \mu s$ to $t = 50 \mu s$).
- Increasing from -1 V to 0 V (for $t = 50 \mu s$ to $t = 100 \mu s$).

Numerical-18

Sketch the output waveform for the differentiator circuit if the input is 3 volt peak triangle wave at 4 kHz. Assume $R_f = 5k\Omega$ and $C = 10nF$

Soln:

$$T = \frac{1}{4kHz}$$
$$T = 250\mu s$$

The slope (considering p-p) is

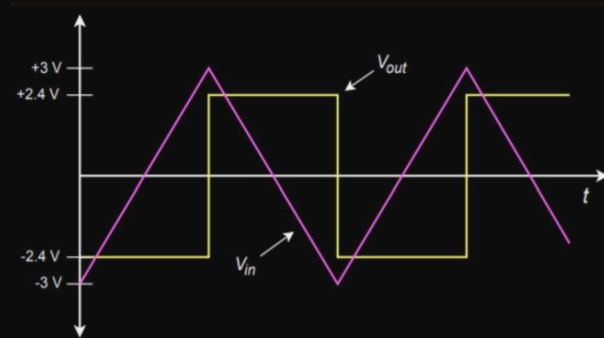
$$\text{Slope} = \frac{6V}{125\mu s}$$
$$\text{Slope} = 48000V/s$$

$$V_{in}(t) = 48000t$$

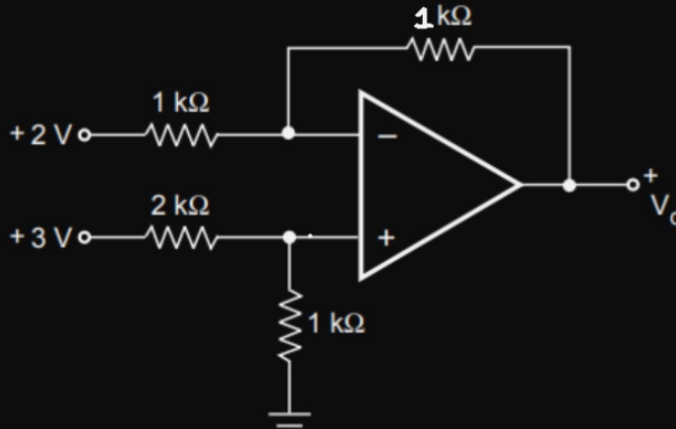
$$V_{out}(t) = -R_f C \frac{dV_{in}(t)}{dt}$$

$$V_{out}(t) = -5k \times 10nF \frac{d48000t}{dt}$$

$$V_{out}(t) = -2.4V \text{ Peak}$$



Numerical19: What is the output voltage of a subtractor circuit shown.



Soln: From Superposition theorem.

$$V_o = V_{o1} + V_{o2}$$

$$= -\frac{R_f}{R_1} V_1 + \left[1 + \frac{R_f}{R_1}\right] \left[\frac{R_f}{R_2 + R_f}\right] V_2$$

$$= -2 + (2/3) \cdot 3$$

$$= 0V$$

Where $R_1=1k, R_2=2K, R_f=1K$

The Slides Below are deleted from
Pesu Slides, so out of syllabus

Electronic Principles and Devices

r. Ananda M

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ommunication Engineering

ELECTRONIC PRINCIPLES AND DEVICES

Operational Amplifier

Op-Amp parameters and its values for Op-Amp 741 – Input and output offset voltages, Input and output resistances, GBW, SR, CMRR (Definitions and significance only), Ideal Op-amp, Negative feedback.

Dr. Ananda M

Department of Electronics and Communication.

12/12/2024

The 8 pin DIP package of IC 741 is shown in the Fig. 1.59.



Fig. 1.59 8 Pin diagram

1.26.3 Ideal Vs Practical Characteristics of IC 741 Op-Amp

The Table 1.7 lists the ideal op-amp characteristics and the typical characteristics of 741 IC, a popular general purpose op-amp IC.

Sr. No	Parameter	Symbol	Ideal	Typical for 741 IC
1	Open loop voltage gain	A_{OL}	∞	2×10^5
2	Output impedance	Z_{out}	0	75Ω
3	Input impedance	Z_{in}	∞	2 M Ω
4	Input offset current	I_{ios}	0	20 nA
5	Input offset voltage	V_{ios}	0	1 mV
6	Bandwidth	B.W	∞	1 MHz
7	CMRR	ρ	∞	90 dB
8	Slew rate	S	∞	0.5 V/ μ sec
9	Input bias current	I_b	0	80 nA
10	PSRR	PSRR	0	30 μ V/V

Characteristics and performance parameters of Op-amp

Input offset Voltage

Input offset current

Input bias current

Output Offset voltage

Input and output impedances and Differential input resistance

CMRR

Output voltage swing

PSRR

Slew rate

Gain Bandwidth product

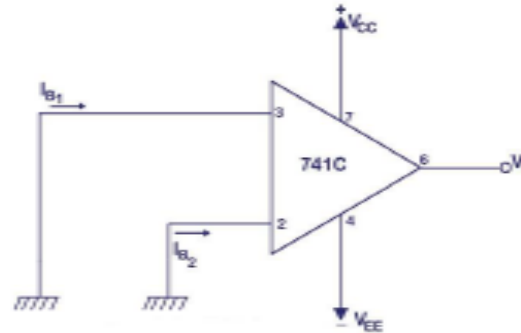
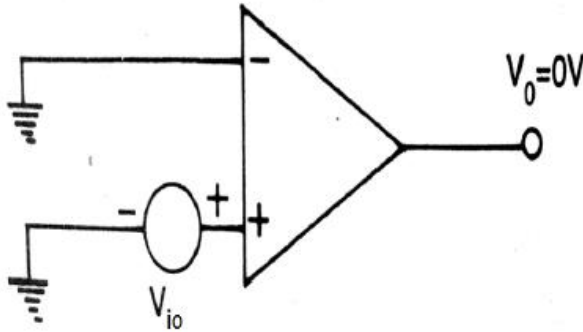
Negative feedback

1. Input offset Voltage :

The differential voltage that must be applied between the two input terminals of an op-amp, to make the output voltage zero.

It is denoted as V_{ios}

For op-amp 741C the input offset voltage is 6mV



2. Input offset current :

The algebraic difference between the currents flowing into the two input terminals of the op-amp

It is denoted as $I_{ios} = |I_{b1} - I_{b2}|$

For op-amp 741C the input offset current is 200nA

Electronic Principles and Devices

Operational Amplifier -Parameters

3. Input bias current :

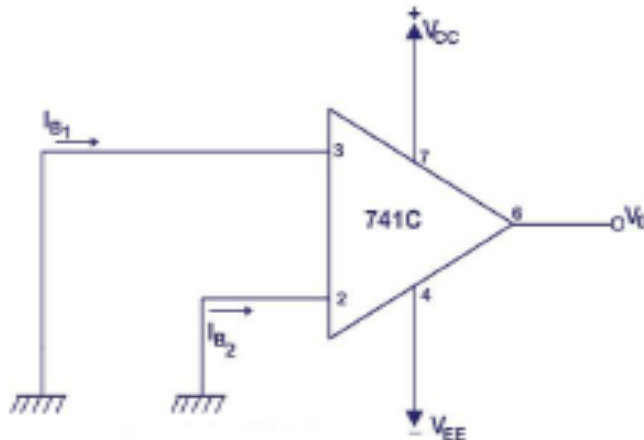
The average value of the two currents flowing into the op-amp input terminals
It is expressed mathematically as

$$\frac{I_{b1} + I_{b2}}{2}$$

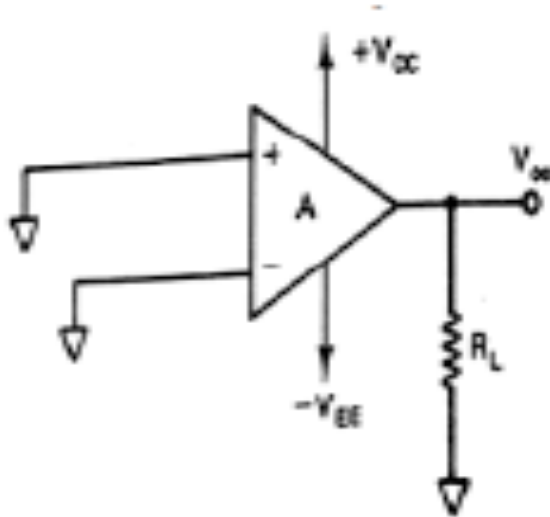
For 741C the maximum value of I_b is 500nA

Where I_{b1} = DC bias current flowing into the non-inverting input

I_{b2} = DC bias current flowing into the inverting input



4. Output Offset voltage : The output offset voltage is the dc voltage present at the output terminals when both the input terminals are grounded. It is denoted as V_{oo}



Electronic Principles and Devices

Operational Amplifier -Parameters



5. Input and output impedances

Ideal Op-Amp Input impedance of op-amp is ∞ – No current flow in or out of input terminals Output impedance of op-amp (with respect to ground) is '0'.

Input impedance : It is the ratio of the input voltage to input current. It should be infinite without any leakage of current from the supply to the inputs. But there will be a few Pico ampere current leakages in most Op Amps.

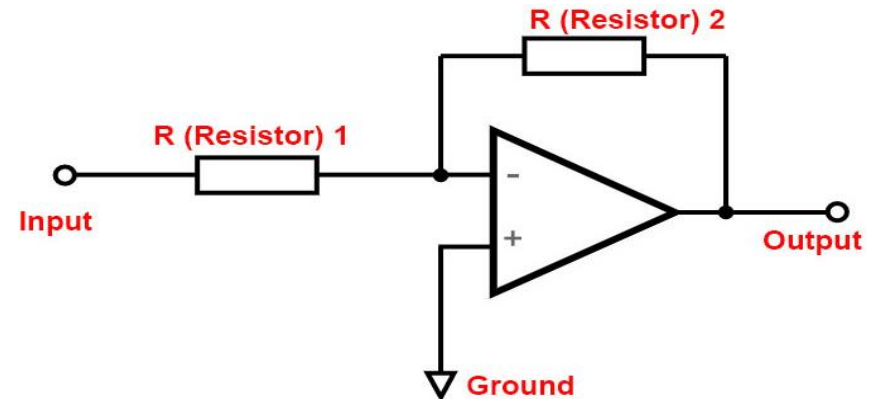
Output impedance :The ideal Op Amp should have zero output impedance without any internal resistance. So that it can supply full current to the load connected to the output.

6. Differential Input Resistance

It is the equivalent resistance measured at either the inverting or non inverting input terminal with the other input terminal grounded.

It is denoted as R_i

For 741C it is of the order of $2M\Omega$



Electronic Principles and Devices

Operational Amplifier -Parameters

7. CMRR

It is the ratio of differential voltage gain A_d to common mode voltage gain A_c

$$\text{CMRR} = A_d / A_c$$

A_d is open loop voltage gain A_{OL} and $A_c = V_{OC} / V_c$

For op-amp 741C CMRR is 90 dB

Common-Mode Operation

When the same input signals are applied to both inputs, common-mode operation results, as shown in Fig. 10.8. Ideally, the two inputs are equally amplified, and since they result in opposite-polarity signals at the output, these signals cancel, resulting in 0-V output. Practically, a small output signal will result

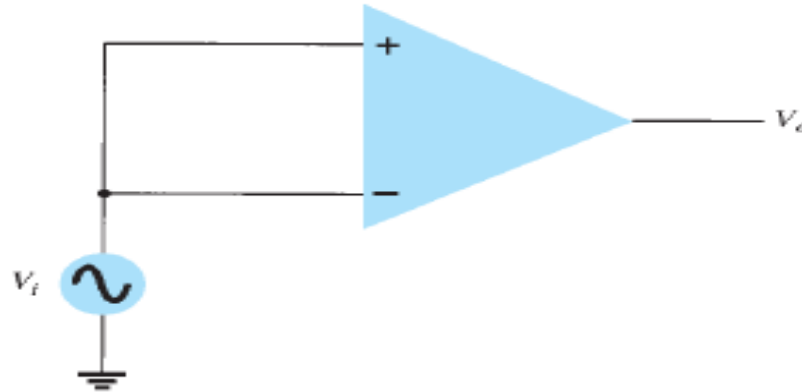


FIG. 10.8

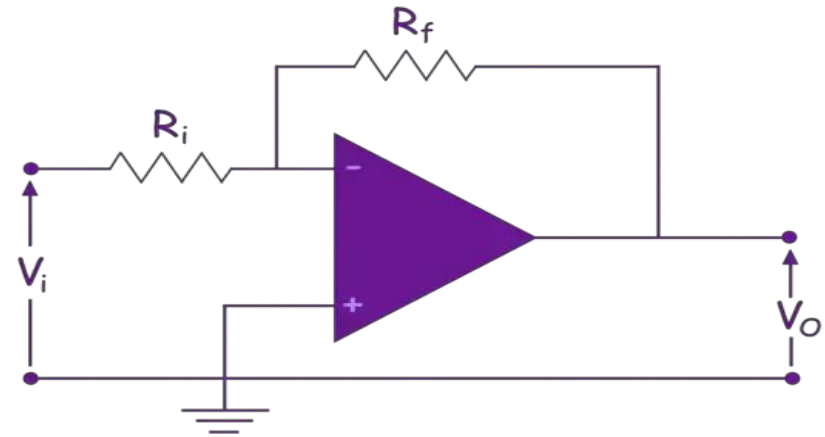
Common-mode operation.

Open loop gain

- The open-loop gain of an electronic amplifier is the gain obtained when no overall feedback is used in the circuit.
- An ideal operational amplifier (op-amp) has infinite open-loop gain.
- Typically an op-amp may have a maximal open-loop gain of around 10^5 , or 100 dB.
- An op-amp with a large open-loop gain offers high precision when used as an inverting amplifier. The definition of open-loop gain (at a fixed frequency) is

$$A_{OL} = \frac{V_{out}}{V^+ - V^-}$$

Where $V^+ - V^-$ is the input voltage difference that is being amplified. (The dependence on frequency is not displayed here.)

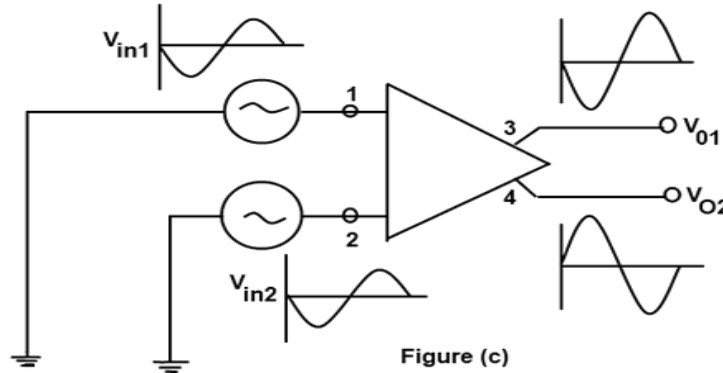


Electronic Principles and Devices

Operational Amplifier -Parameters

Difference Mode

In this mode two opposite polarity (out of phase) signals are applied to the inputs as shown in Figure (c).



Common-Mode Rejection

A significant feature of a differential connection is that the signals that are opposite at the inputs are highly amplified, whereas those that are common to the two inputs are only slightly amplified—the overall operation being to amplify the difference signal while rejecting the common signal at the two inputs.

Since noise (any unwanted input signal) is generally common to both inputs, the differential connection tends to provide attenuation of this unwanted input while providing an amplified output of the difference signal applied to the inputs. This operating feature is referred to as **common-mode rejection**.

8. Output Voltage swing

The opamp output voltage gets saturated at $+V_{CC}$ and $-V_{EE}$ and it cannot produce output voltage more than $+V_{CC}$ and $-V_{EE}$. Practically voltages $+V_{sat}$ and $-V_{sat}$ are slightly less than $+V_{CC}$ and $-V_{EE}$

For op amp 741C the saturation voltages are $+13V$ for supply voltages of $15V$

9. Power supply rejection ratio

PSRR is defined as the ratio of the change in input offset voltage due to the change in supply voltage producing it, keeping the other power supply voltage constant. It is also called as power supply sensitivity (PSV)

$$PSRR = (\Delta v_{ios} / \Delta V_{CC}) |_{\text{constant } V_{EE}}$$

$$PSRR = (\Delta v_{ios} / \Delta V_{EE}) |_{\text{constant } V_{CC}}$$

The typical value of PSRR for op-amp 741C is $30\mu V/V_{64}$

9. Slew rate

It is defined as the maximum rate of change of output voltage with time. The slew rate is specified in V/ μ sec

$$\text{Slew rate} = S = dV_o / dt \text{ max}$$

It is specified by the op amp in unity gain condition.

The slew rate is caused due to limited charging rate of the compensation capacitor and current limiting and saturation of the internal stages of op amp, when a high frequency large amplitude signal is applied.

10. Gain –Bandwidth product

It is the bandwidth of op-amp when voltage gain is unity (1). It is denoted as GB.

The GB is also called unity gain bandwidth (UGB) or closed loop bandwidth

It is about 1MHz for op-amp 741C68

Negative Feedback :

Negative feedback in an op amp is defined as connecting the output to the inverting input through a resistor to control the gain

We obtain Negative feedback in an op amp by connecting output terminal of an op amp to its inverting input terminal through a suitable resistance as shown Fig.

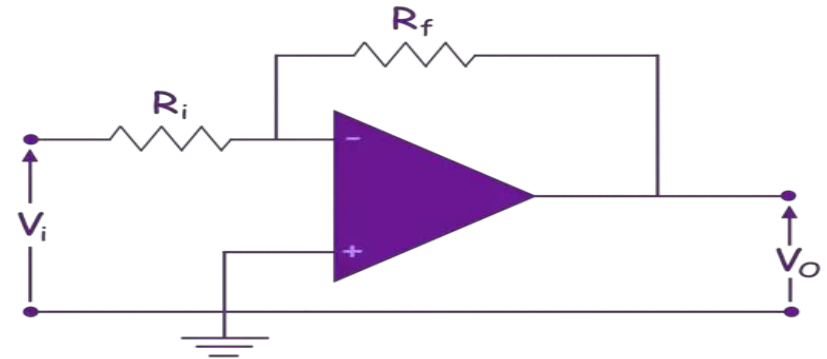
The gain of an op amp with negative feedback is called closed loop gain.

Closed Loop Gain of Op Amp

When we connect a feedback resistor and a resistance in series with the inverting input terminal of an op amp, the system's gain becomes the negative ratio of the feedback resistance to the input resistance.

The op amp has its own very large gain, ideally infinite. By choosing the appropriate values of series input resistance (R_i) and feedback resistance (R_f), we can set the system's gain regardless of the op amp's own gain (open loop gain).

$$- \frac{R_f}{R_i}$$

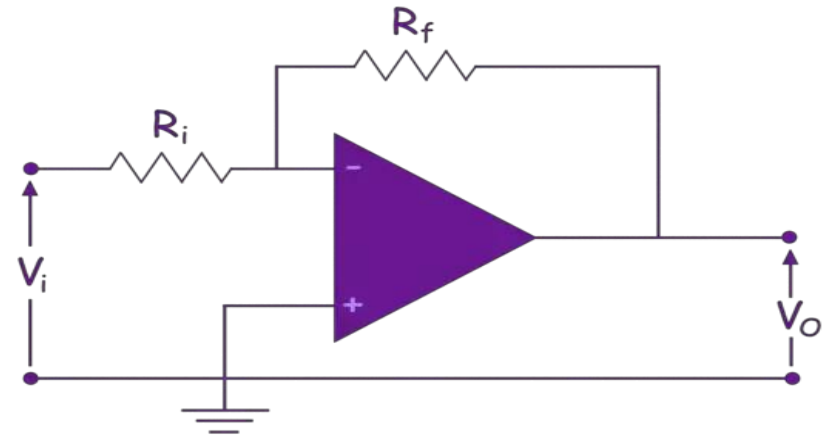


Open loop gain

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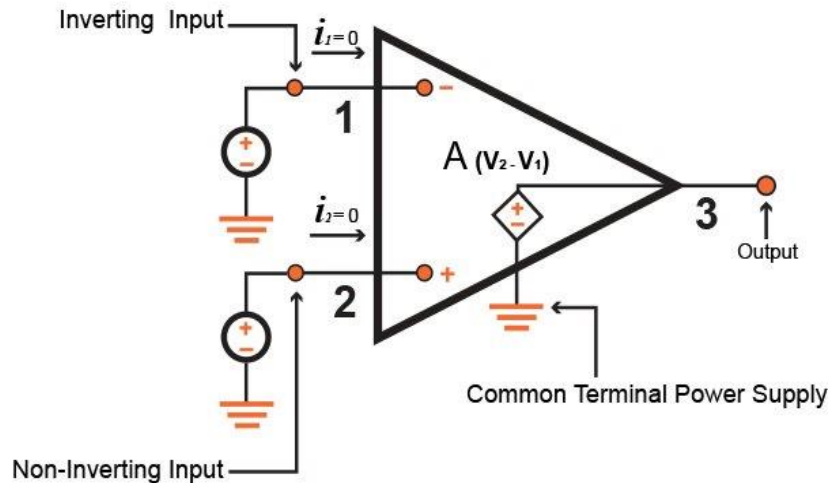
$$A_{OL} = \frac{V_{out}}{V^+ - V^-}$$

Where $V^+ - V^-$ is the input voltage difference that is being amplified. (The dependence on frequency is not displayed here.)



Operational Amplifier - Op Amp Equivalent Circuit

The output of the Op Amp should be zero when the voltage difference between the inputs is zero. But in most Op Amps, the output will not be zero when off but there will be a minute voltage from it



Circuit model of ideal op-amp

Electronic Principles and Devices

Operational Amplifier – Virtual Ground

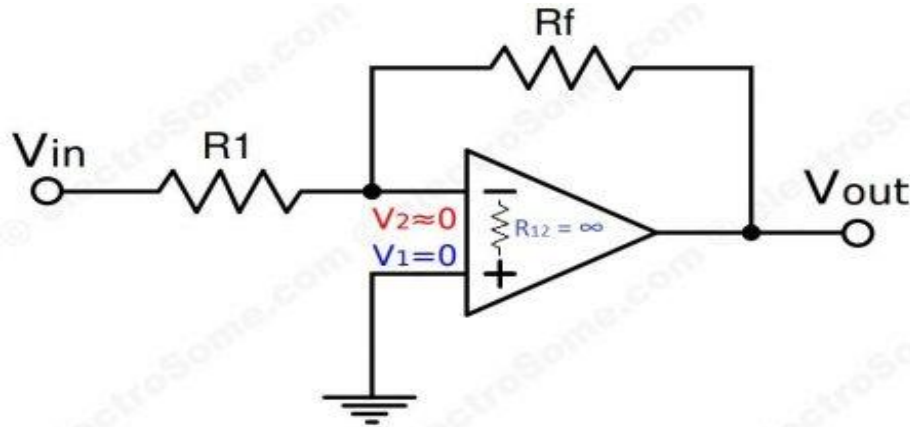
As the name indicates it is virtual, not real ground.

For some purposes we can consider it as equivalent to ground.

In op-amp the term virtual ground means that the voltage at that particular node is almost equal to ground voltage (0V).

It is not physically connected to ground.

This concept is very useful in analysis of op-amp circuits and it will make a lot of calculations very simple.



Electronic Principles and Devices

Operational Amplifier – Virtual Ground

Virtual Ground – Opamp

Lets see how the virtual ground concept is employed in inverting amplifier.

Virtual Ground – Inverting Amplifier using Op-amp

We can explain this concept in two different ways using two different characteristics of an ideal op-amp.

Using Infinite Voltage Gain

We already know that an ideal op-amp will provide infinite voltage gain. For real op-amp also the gain will be very high such that we can consider it as infinite for calculation purposes.

- Gain = V_o/V_{in}

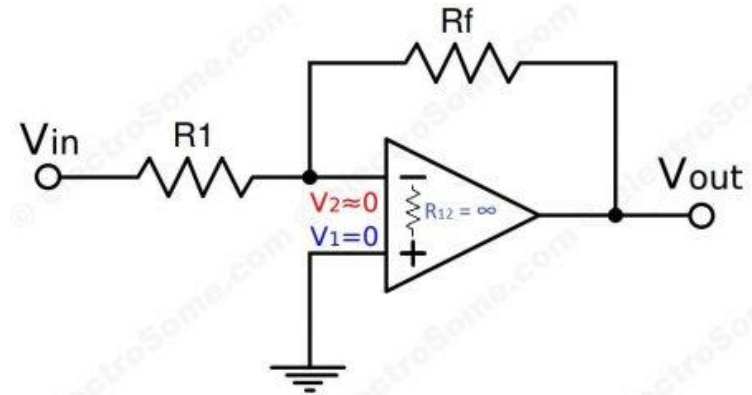
As gain is infinite, $V_{in} = 0$

$$V_{in} = V_2 - V_1$$

In the above circuit V_1 is connected to ground, so $V_1 = 0$. Thus

V_2 also will be at ground potential.

$$V_2 = 0$$



Electronic Principles and Devices

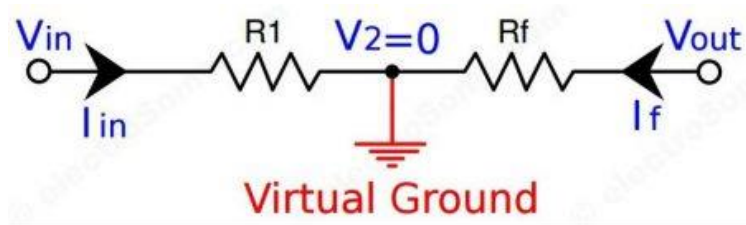
Operational Amplifier – Virtual Ground

Virtual Ground – Inverting Amplifier using Opamp

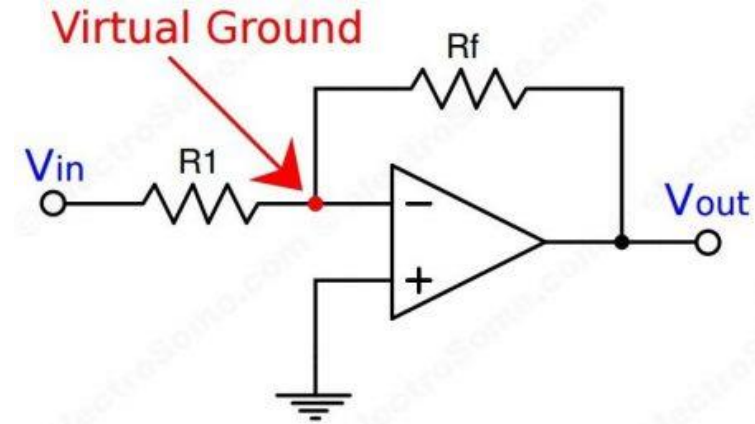
Why we need Virtual Ground ?

Virtual Ground concept is very useful in analysis of an opamp when negative feedback is employed.

It will simplify a lot of calculations and derivations.



Equivalent Circuit – Inverting Amplifier – Virtual Ground – Opamp





THANK YOU

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